



Transfer Learning

Lena Voita

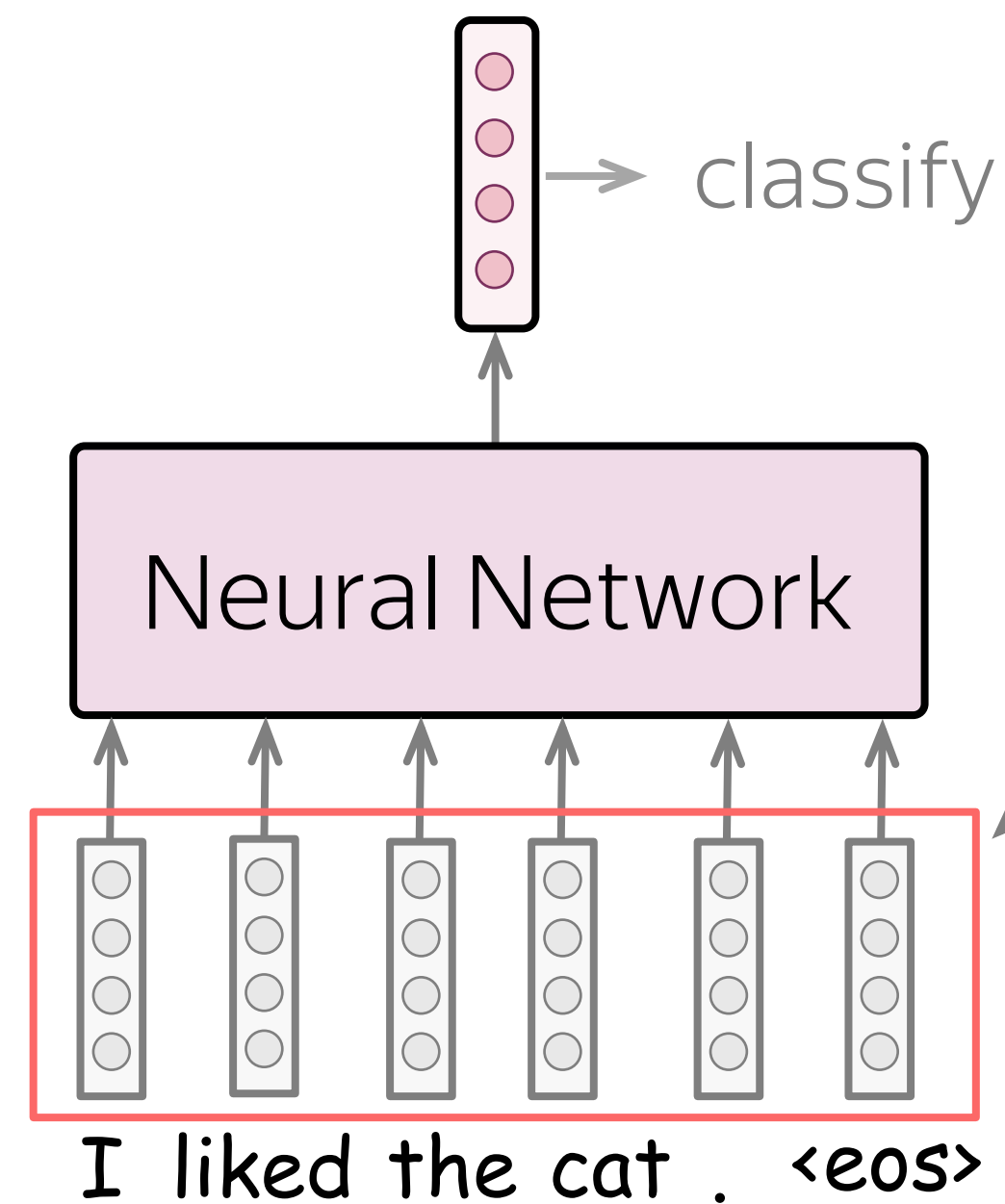
What is going to happen:

- Transfer Learning Idea
 - (recap) Word Embeddings
- Pretrained Models
 - Great Idea 1: Words $\mapsto$ Words in Context
 - Great Idea 2: Task-Specific $\mapsto$ Unified
-  Analysis and Interpretability

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Recap from Text Classification: Word Embeddings

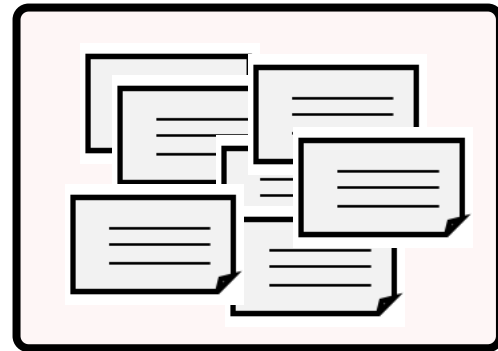


Input word embeddings:

- Train from scratch
- Take pretrained (Word2Vec, GloVe)
- Initialize with pretrained, then fine-tune

Which data do we have?

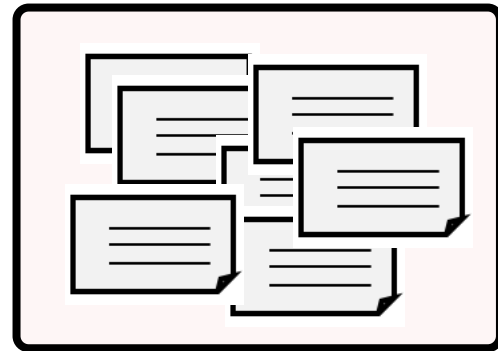
Training data for text
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- Not huge, or not diverse, or both
- Domain: task-specific

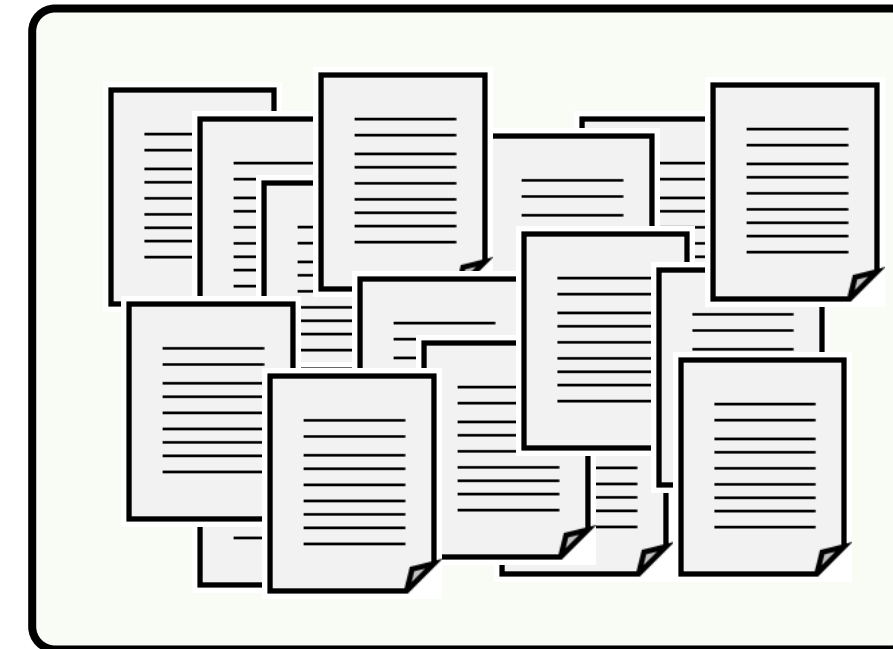
Which data do we have?

Training data for text
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- Not huge, or not diverse, or both
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Training data for word
embeddings (unlabeled)



- Huge diverse corpus (e.g., Wikipedia)
- Domain: general

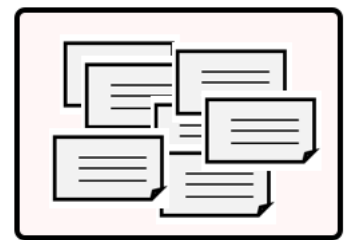
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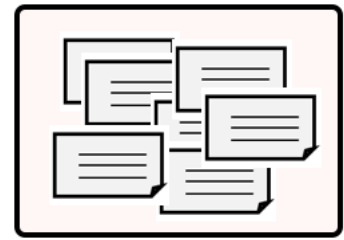


May be not enough
to learn relationships
between words

Recap from Text Classification: Word Embeddings

- Train from scratch

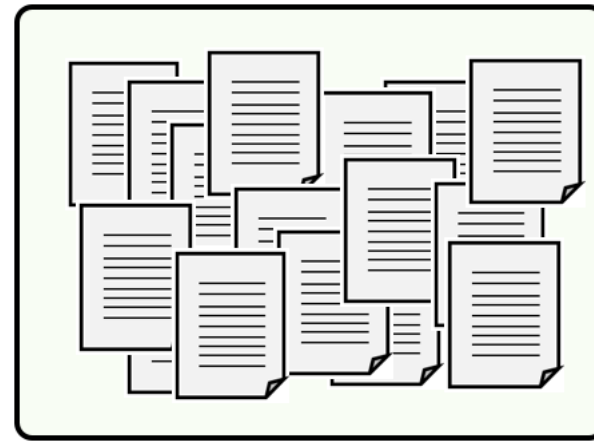
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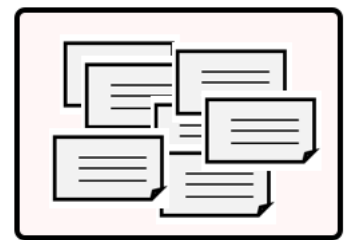
Know relationships between words,
but are **not** specific to the task

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Recap from Text Classification: Word Embeddings

- Train from scratch

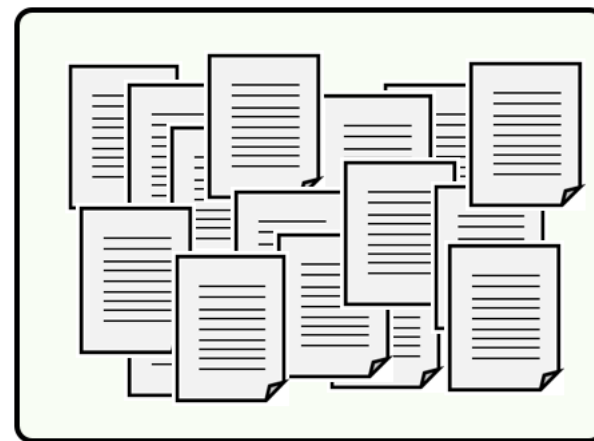
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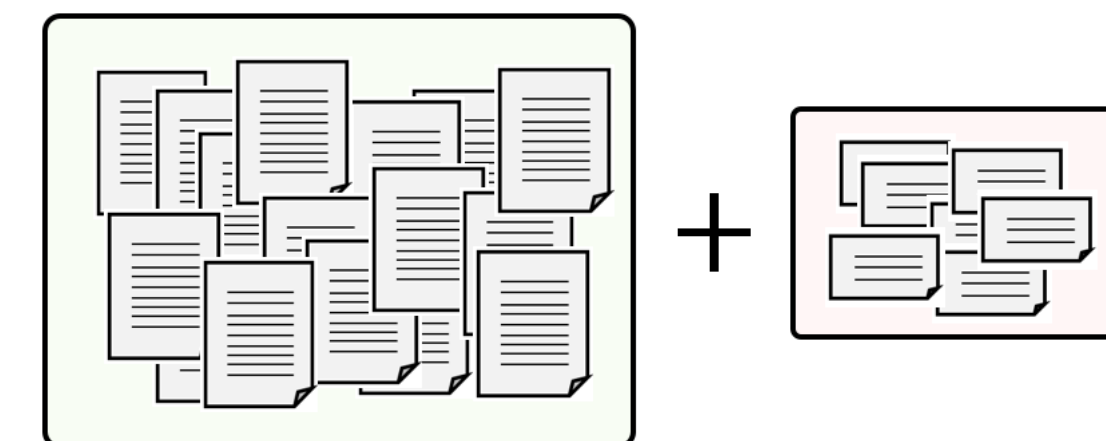
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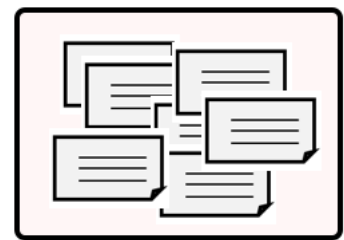


Know relationships between
words and adapted for the task

Recap from Text Classification: Word Embeddings

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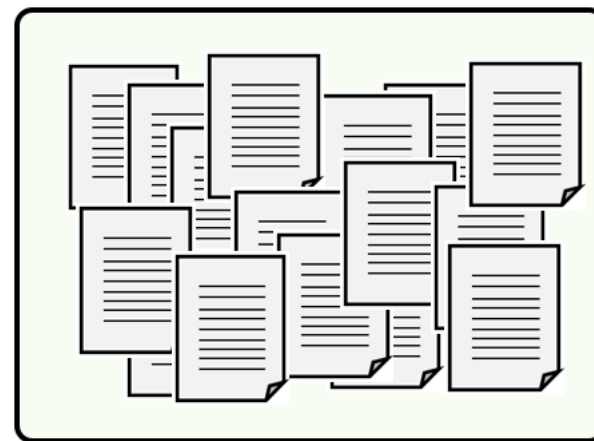
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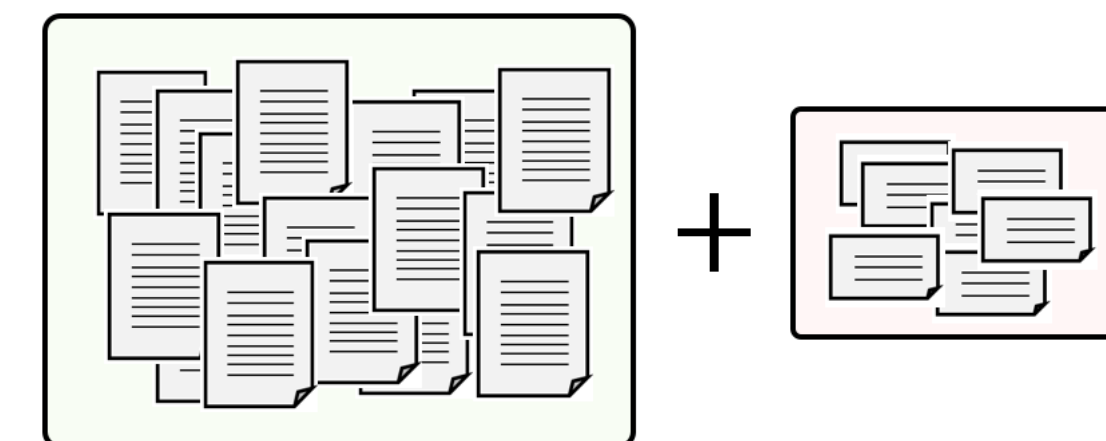
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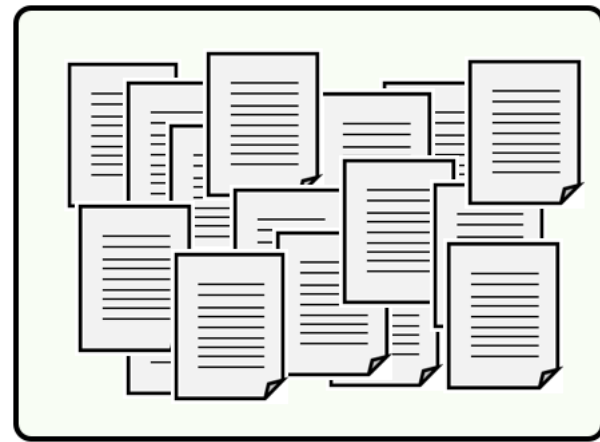
Know relationships between
words and adapted for the task

“Transfer” knowledge from a huge unlabeled
corpus to your task-specific model

We’ll learn more about this later in the course!

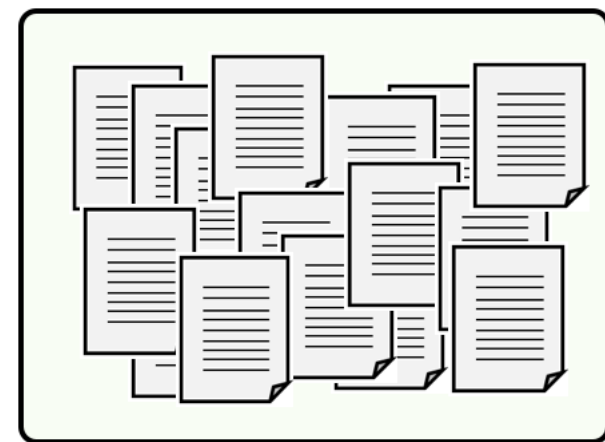
Transfer Learning Idea

Source task

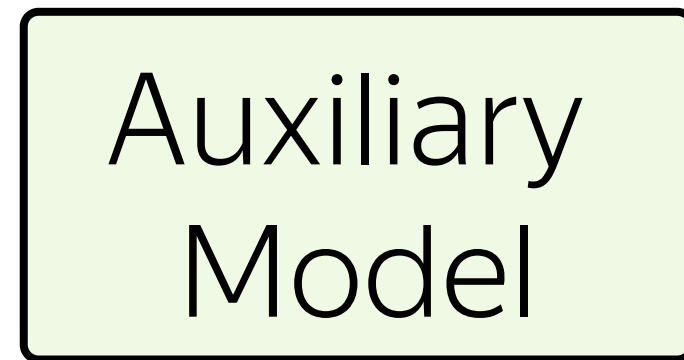


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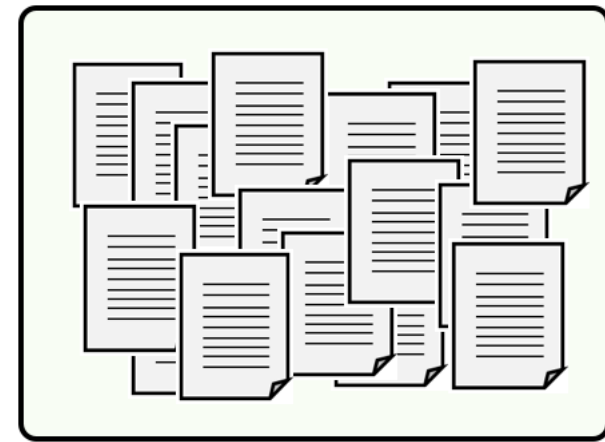


Auxiliary
Model



Transfer Learning Idea

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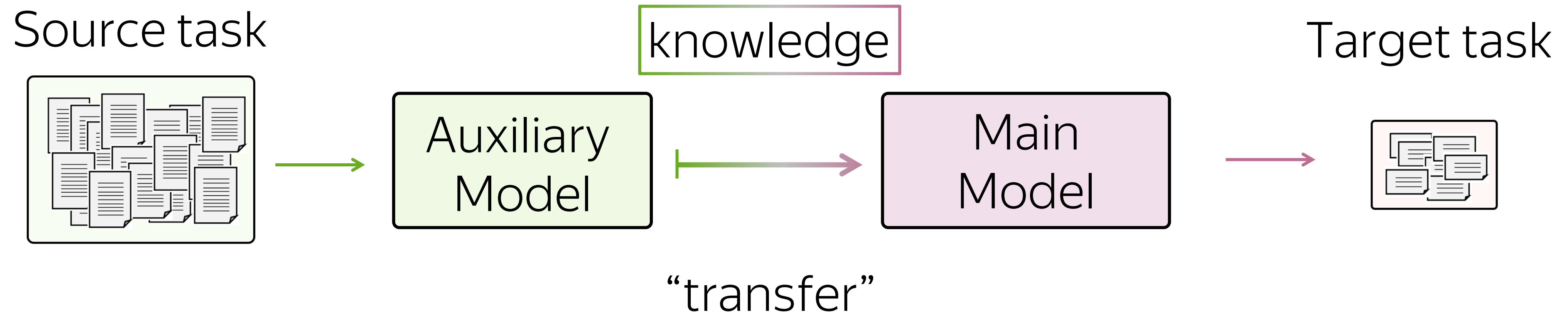
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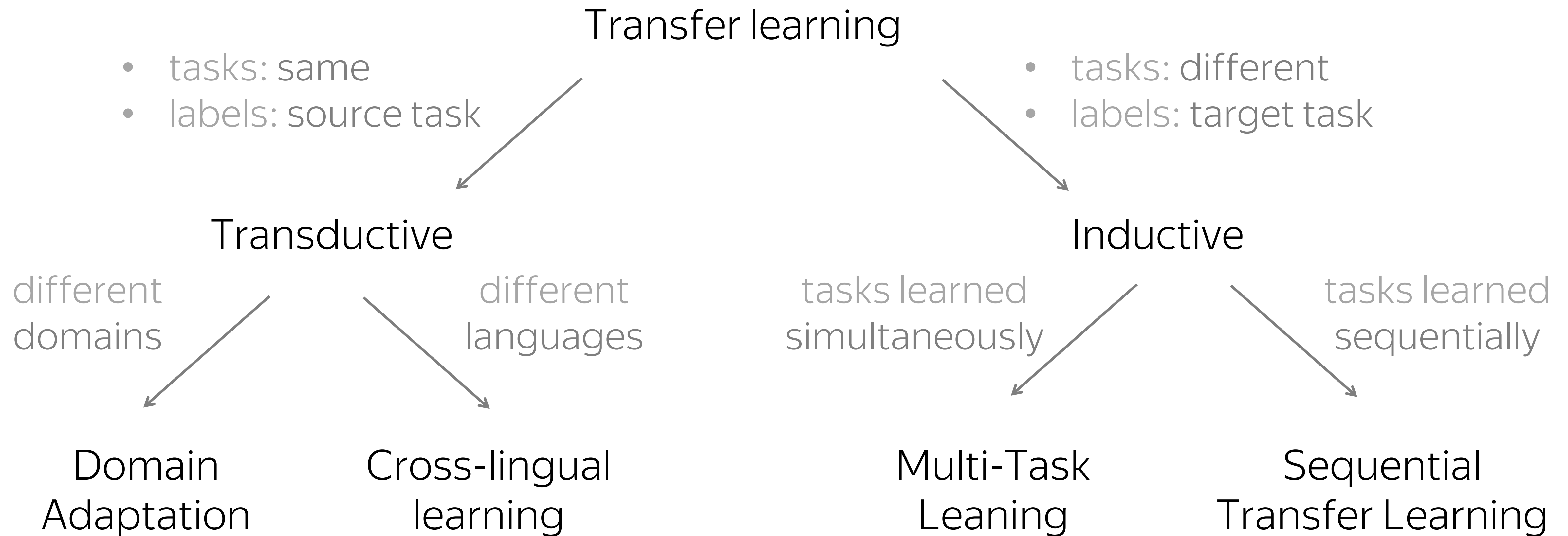


Main
Model

Transfer Learning Idea

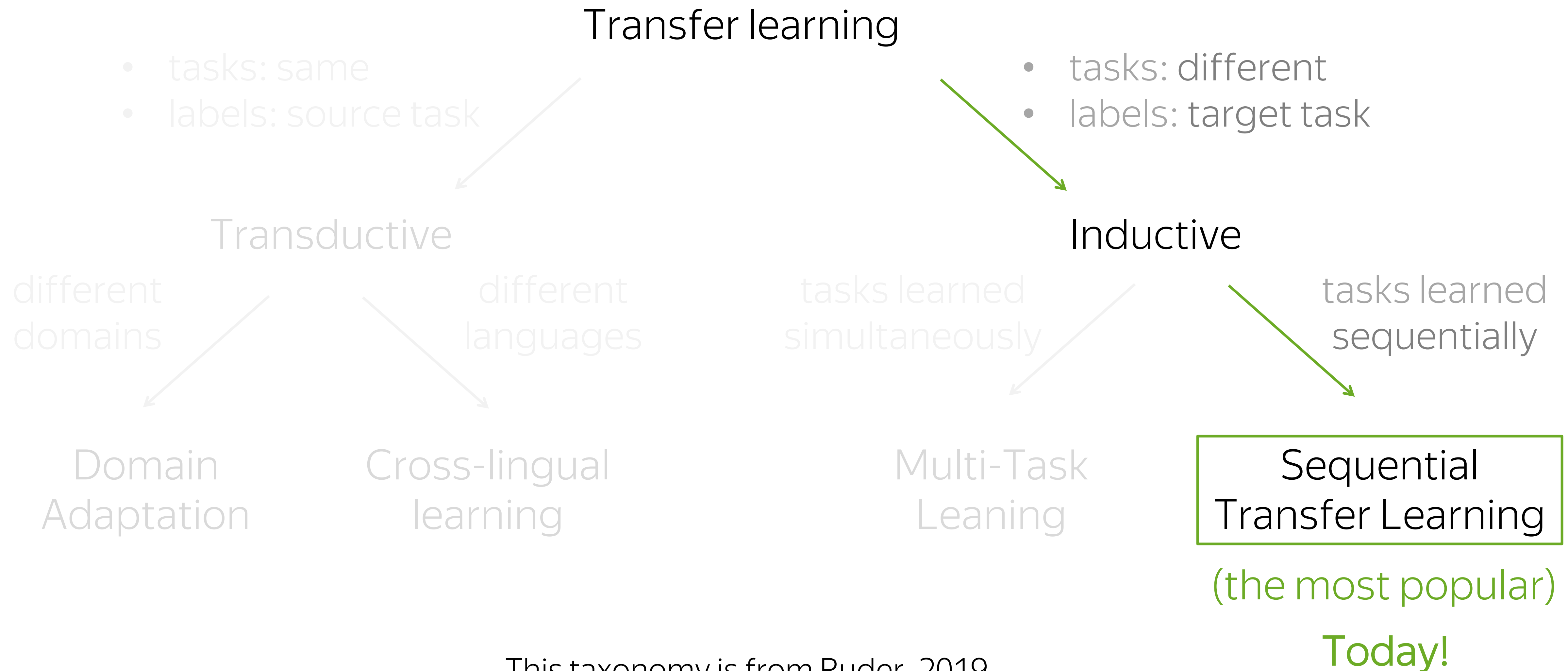


A Taxonomy of Transfer Learning in NLP



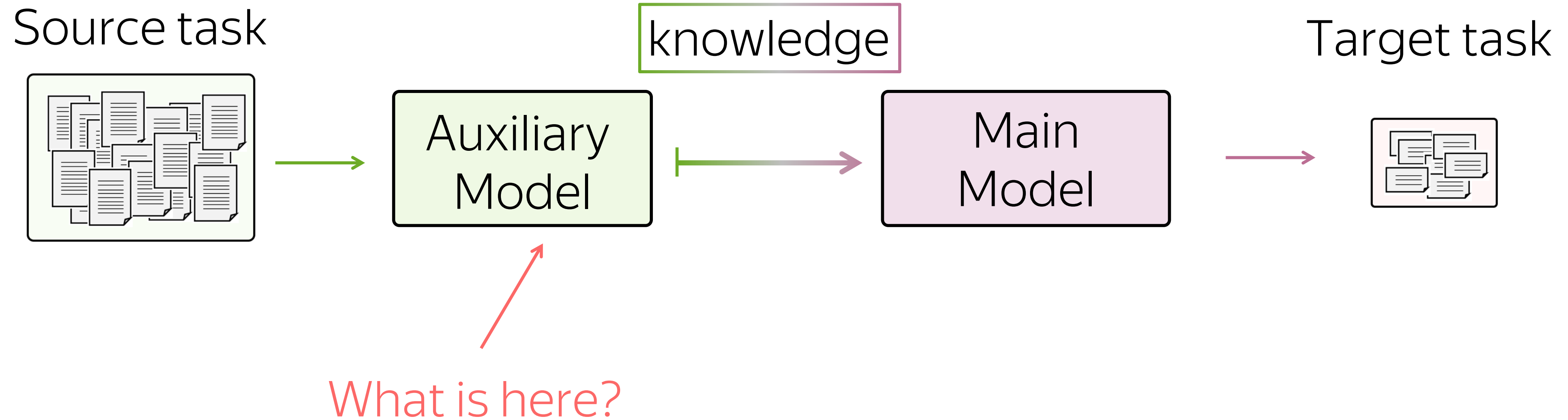
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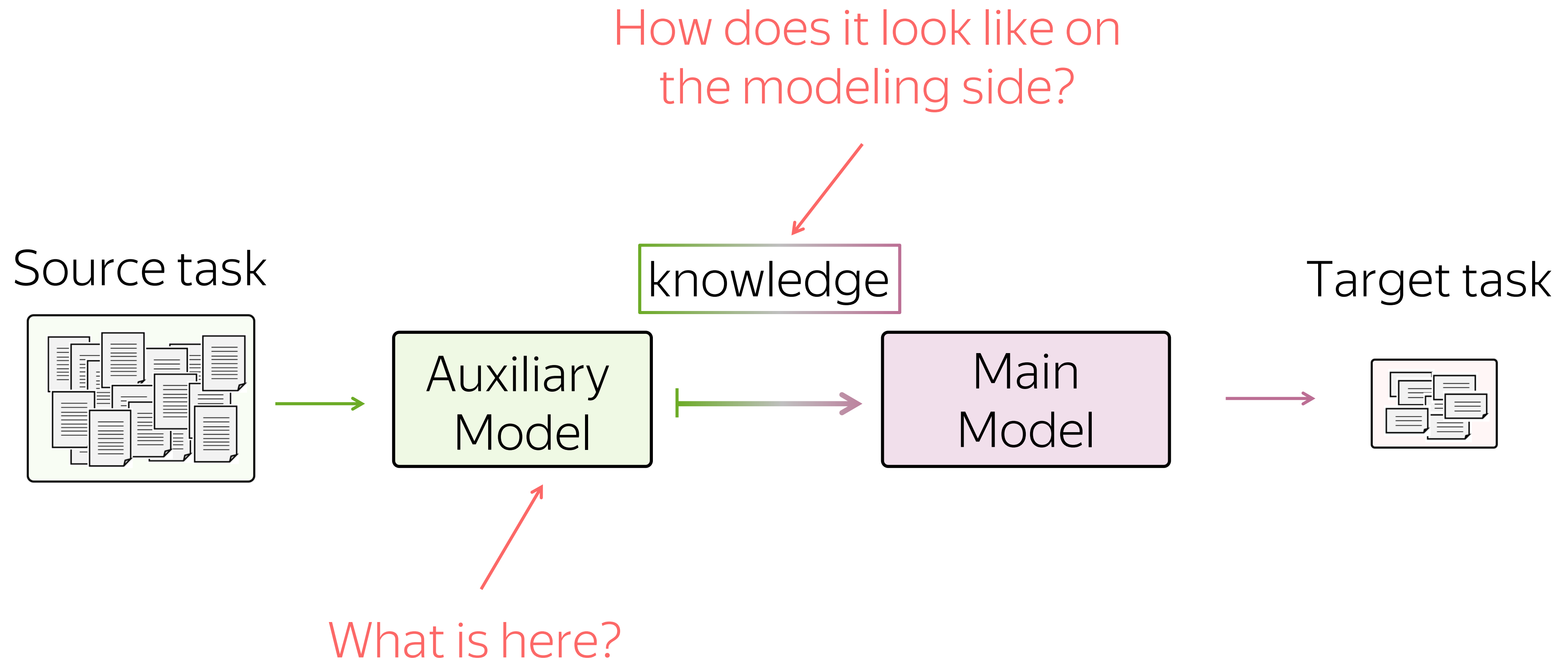


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Transfer Learning Idea



Transfer Learning Idea



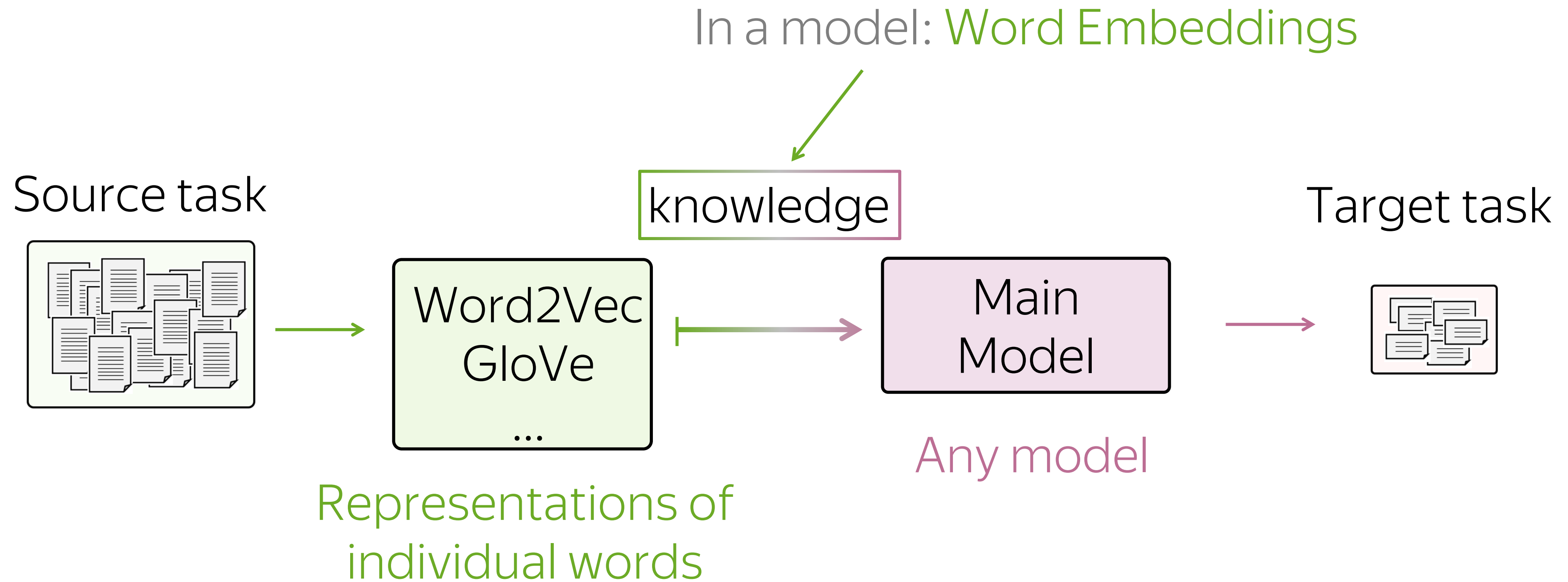
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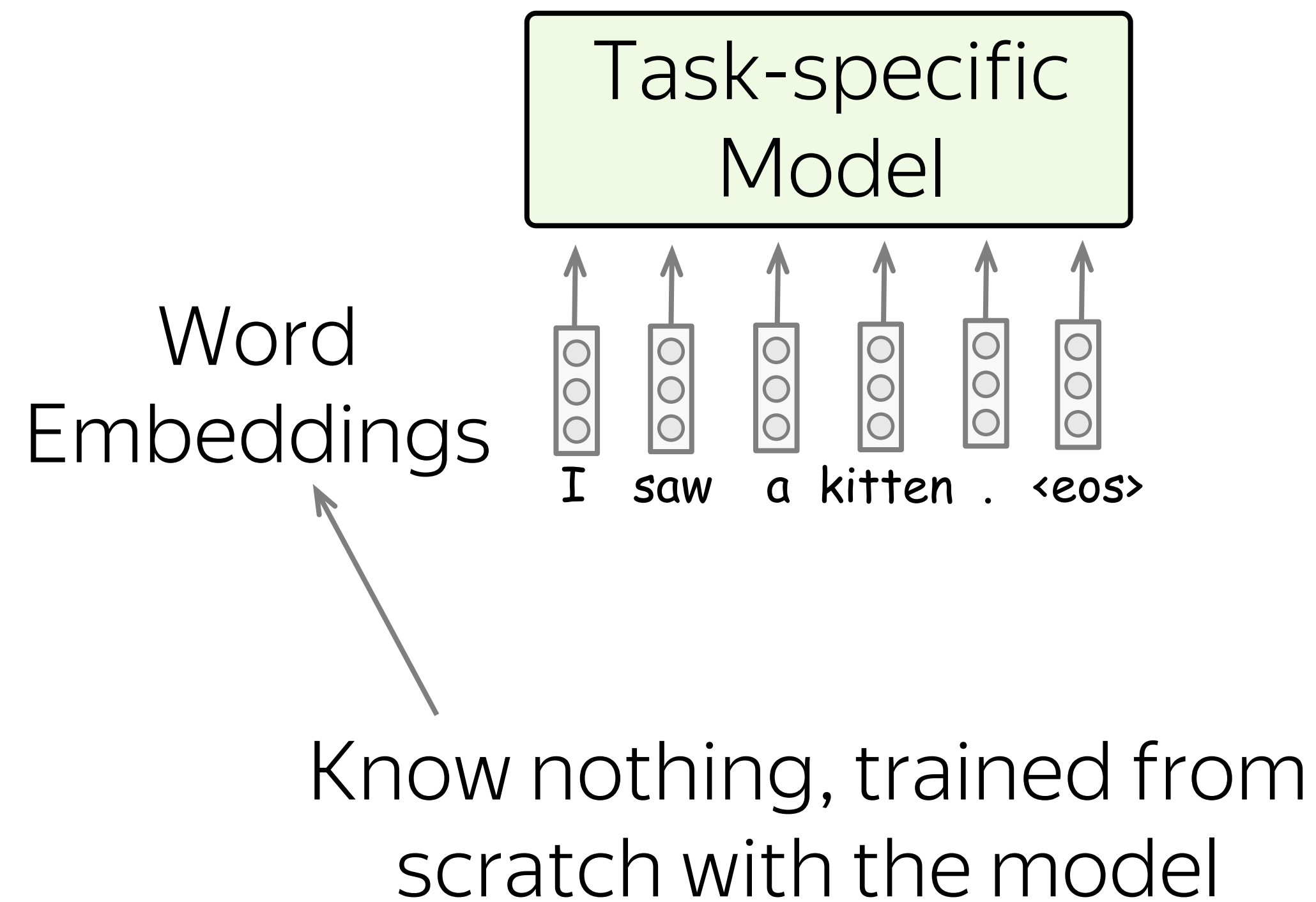
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Simplest (recap once again): Word Embeddings (Word2Vec, GloVe)



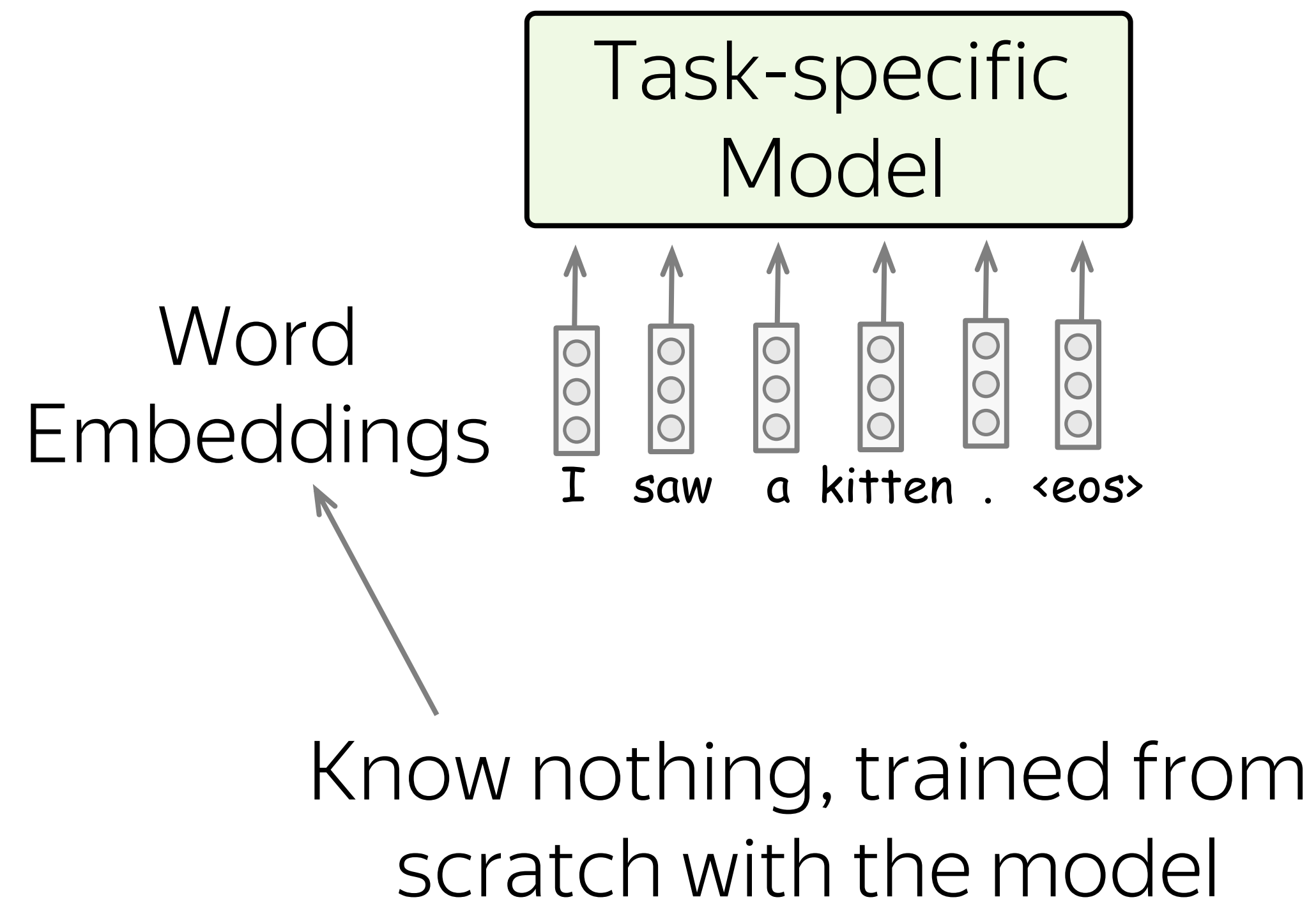
Transfer Through Word Embedding

Before

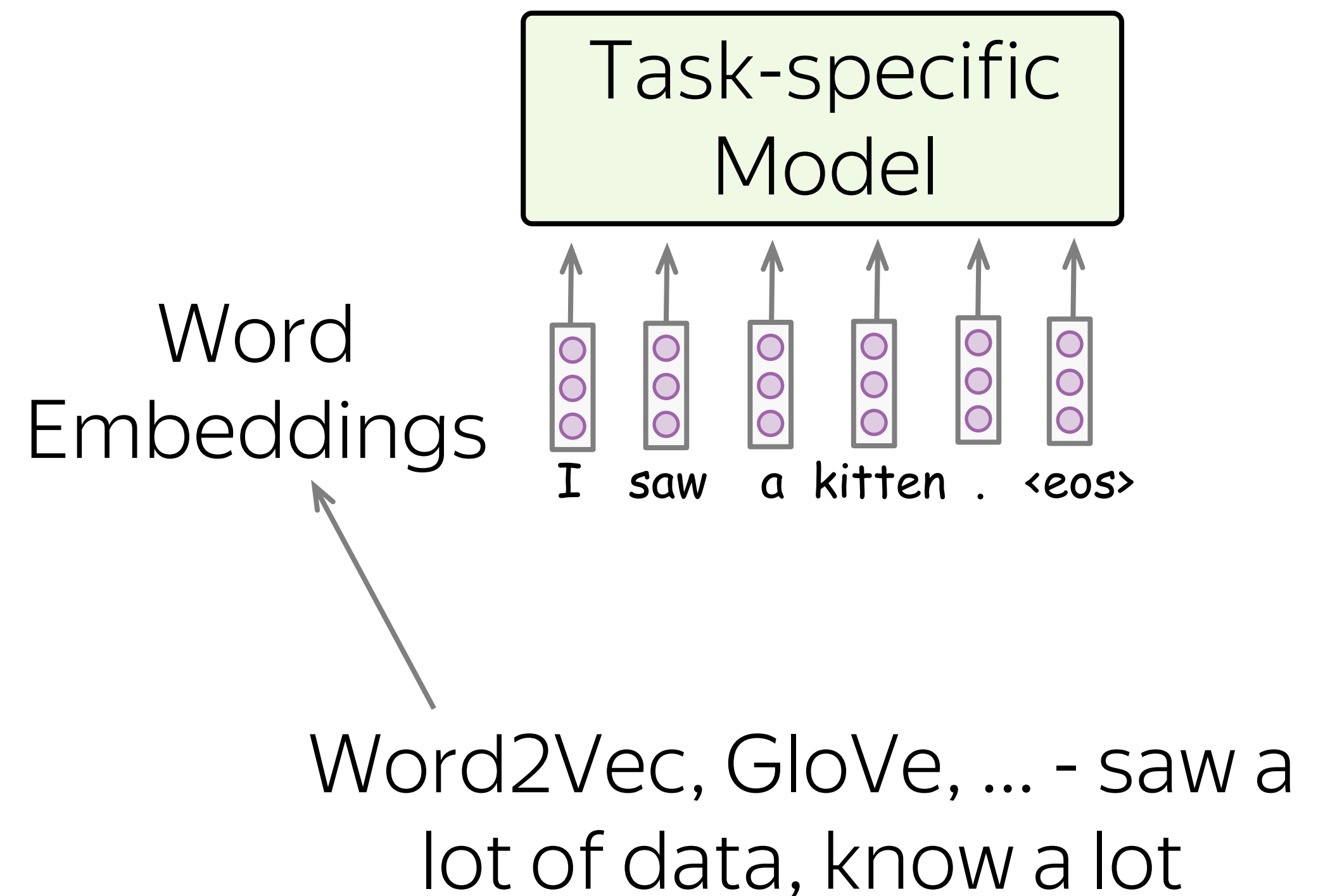


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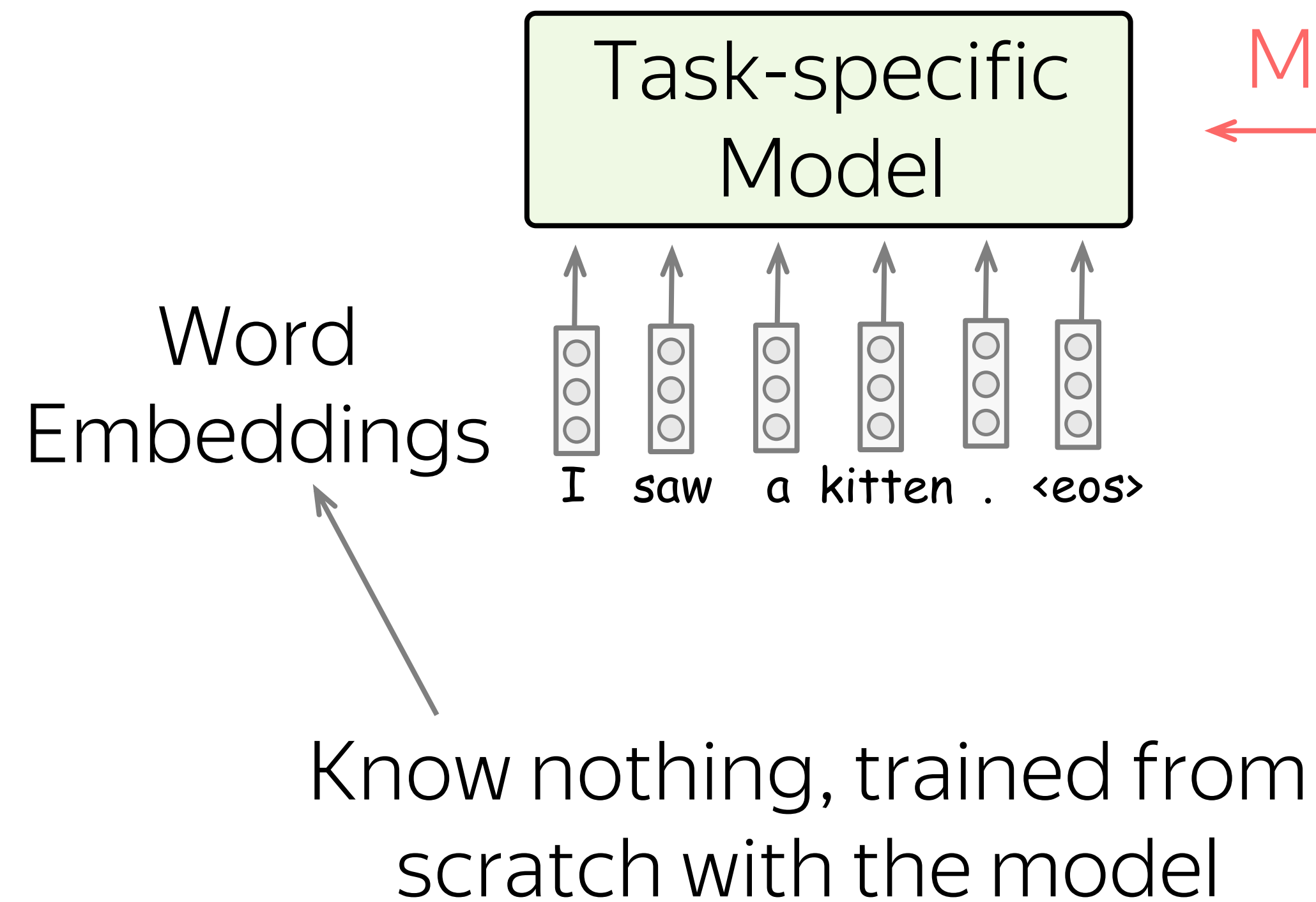


After

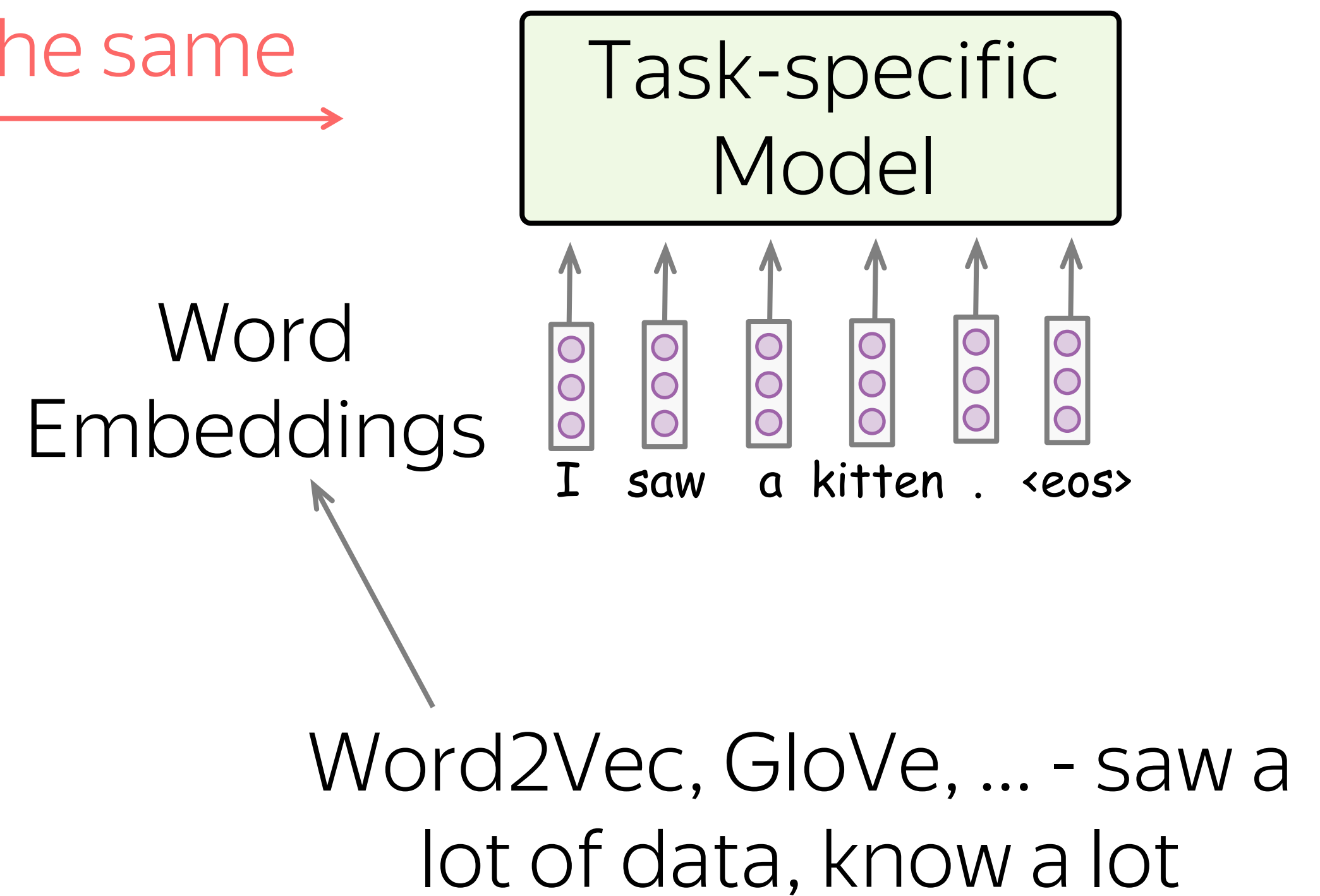


Transfer Through Word Embedding

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After



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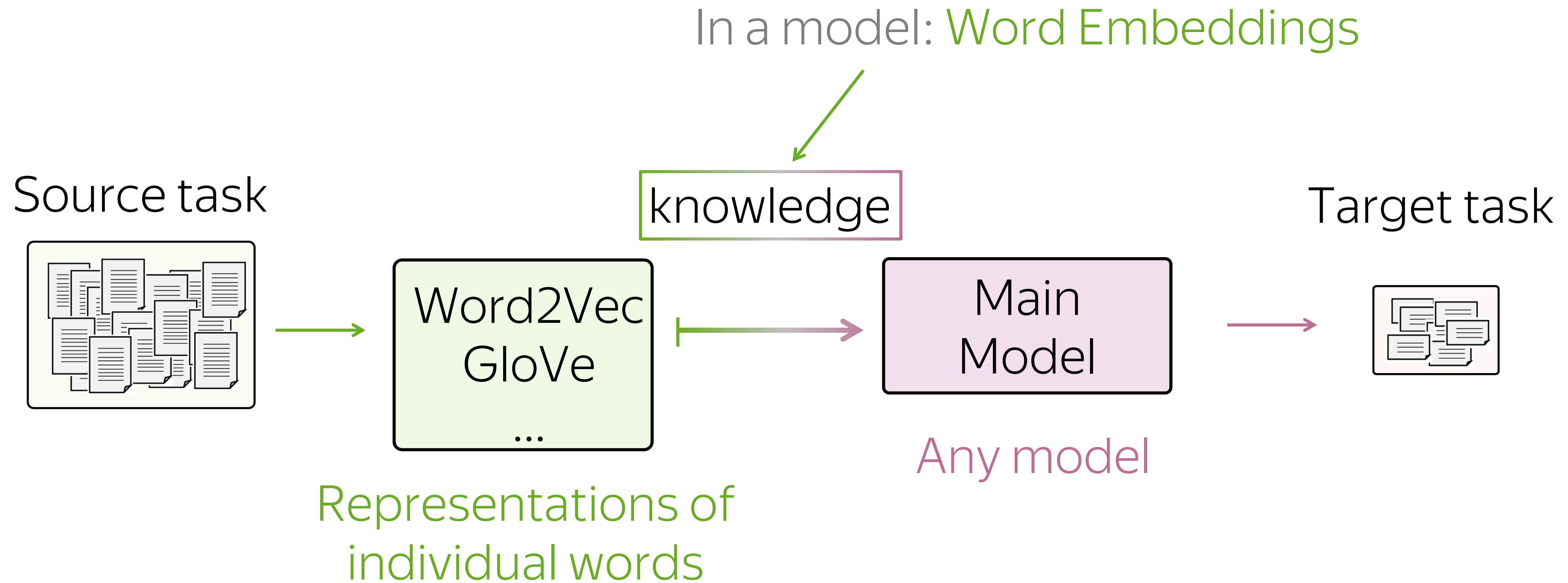
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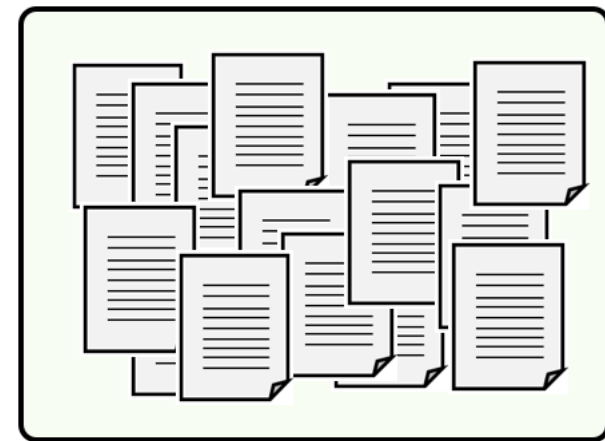
Idea: Through the embeddings, we "transfer" the knowledge of their training data to our task-specific model.



Pretrained Models

Idea: Through `_insert_your_model_`, we "transfer" the knowledge of their training data to our task-specific model.

Source task



Model

knowledge

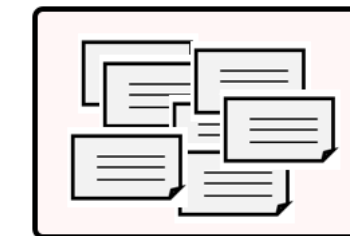


Main
Model

Any model



Target task



The Two Great Ideas

GloVe, Word2Vec

What is encoded

Words

How it is used for
downstream tasks

Input for task-
specific models

The Two Great Ideas

GloVe, Word2Vec  CoVe, ELMo

What is encoded

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Words in context

How it is used for
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The Two Great Ideas

GloVe, Word2Vec $\longrightarrow$ CoVe, ELMo

Great idea 1

What is encoded

Words $\longrightarrow$ Words in context

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The Two Great Ideas

GloVe, Word2Vec → CoVe, ELMo → BERT, GPT

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Words → Words in context

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→ Instead of task-
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What is encoded

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Great idea 2

How it is used for
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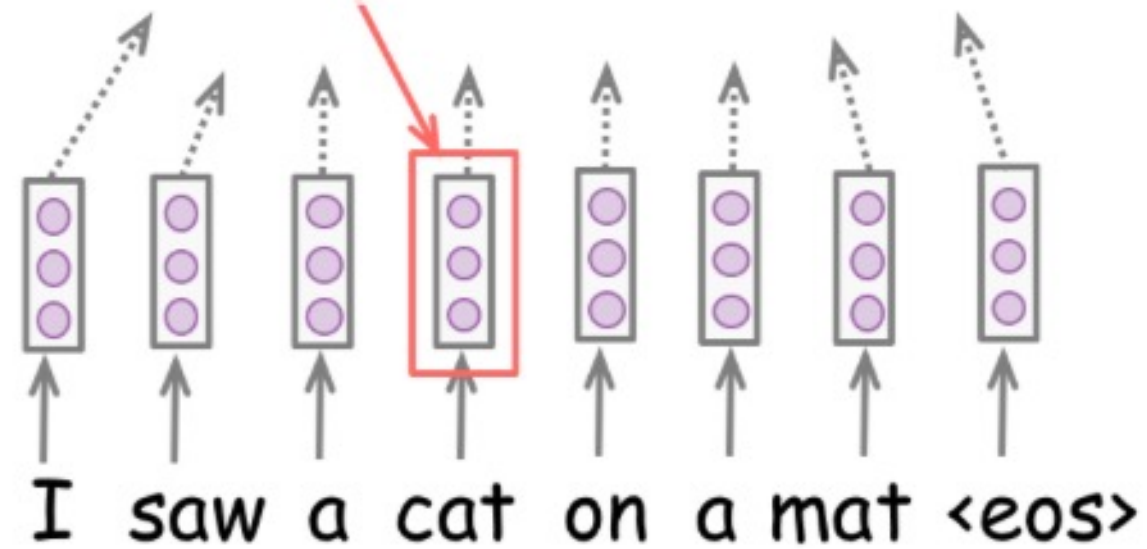
Great Idea 1: From Words to Words-in-Context

Before

“cat”. Just “cat”



use for your task



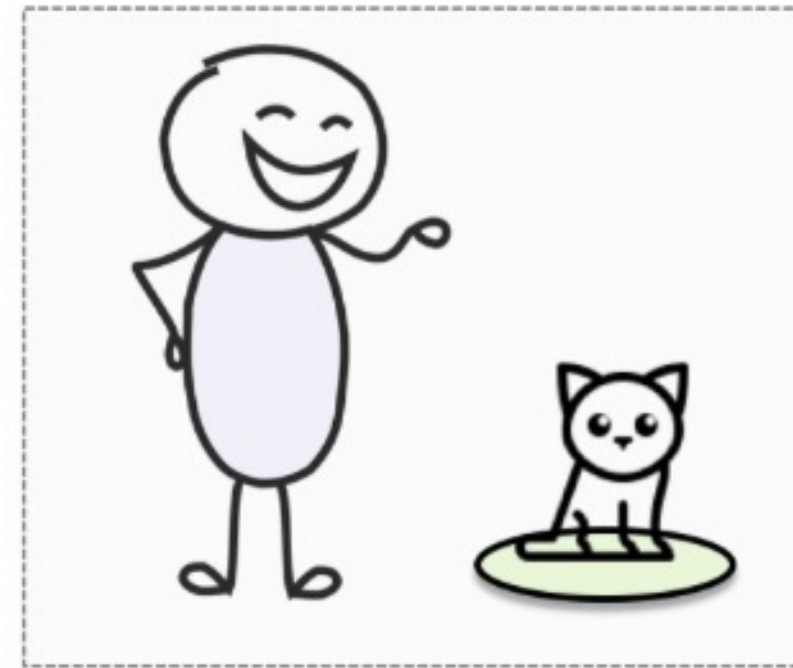
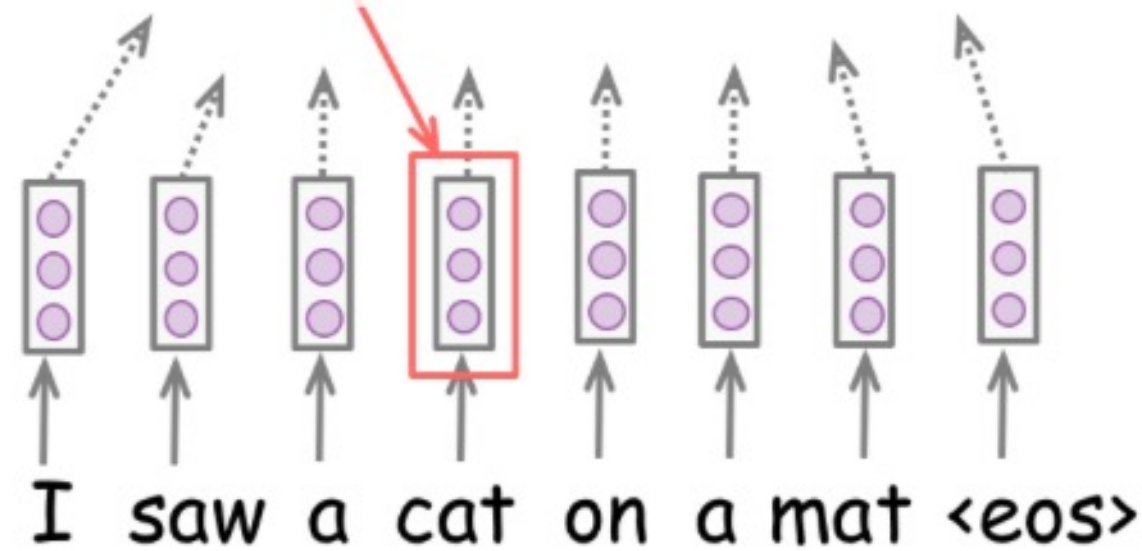
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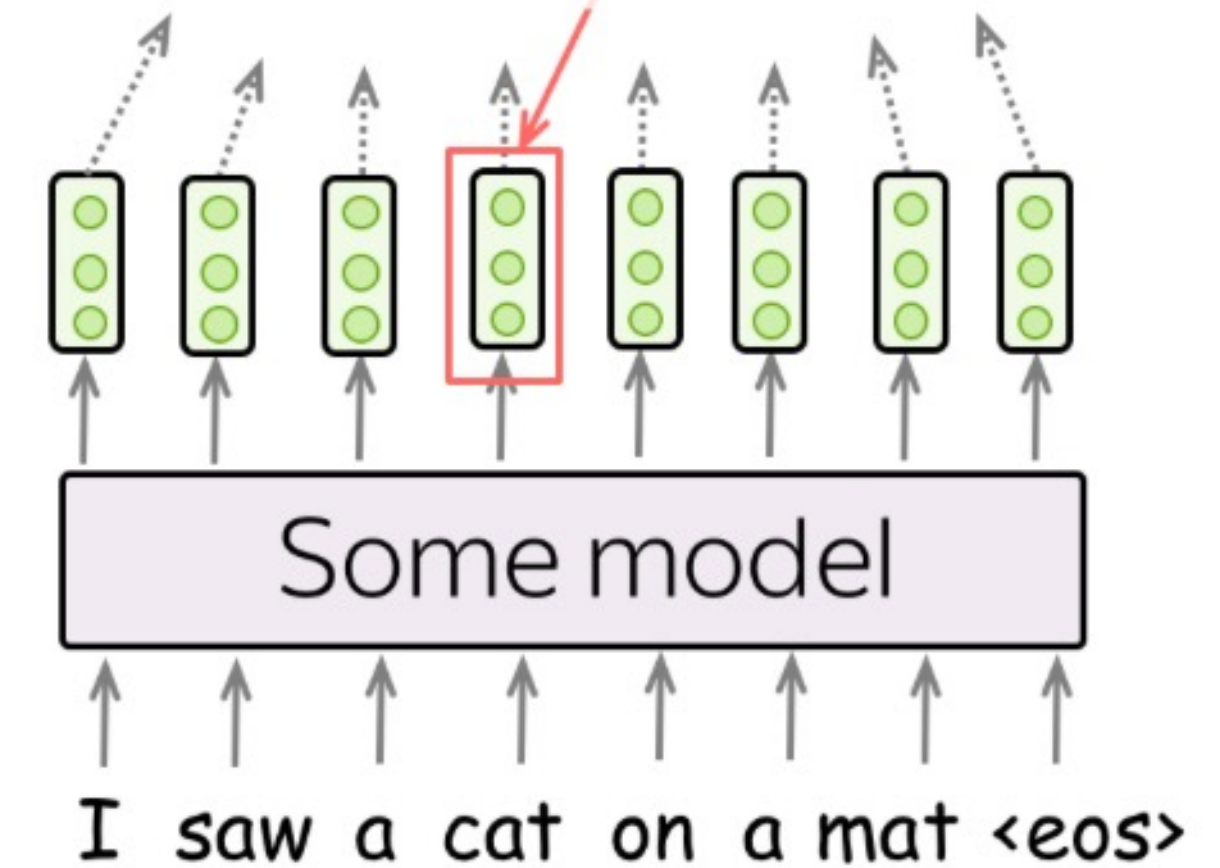
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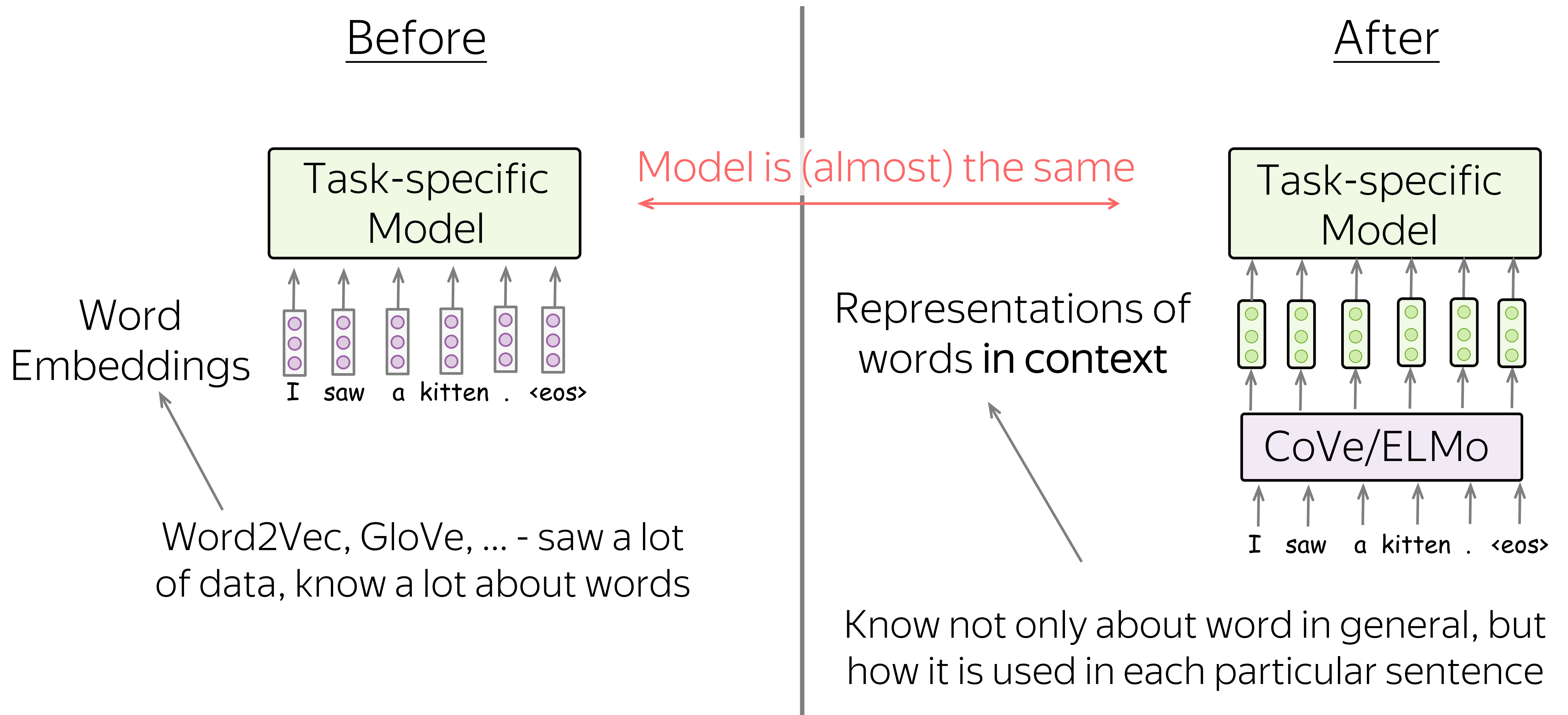
After

Not just any “cat”!
This is the **cat** that **I** saw, the
one who **sat on the mat**

use for your task



Great Idea 1: From Words to Words-in-Context



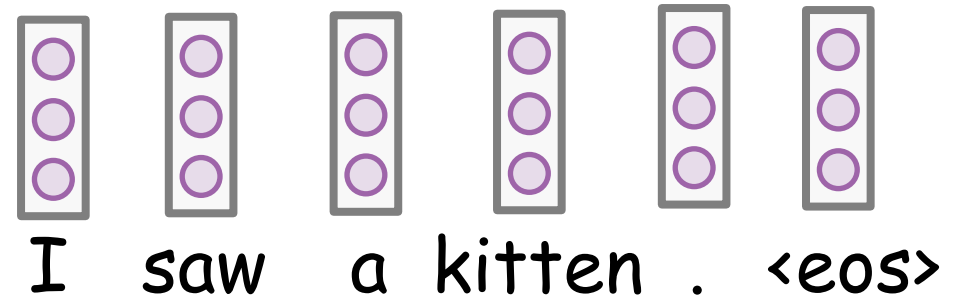
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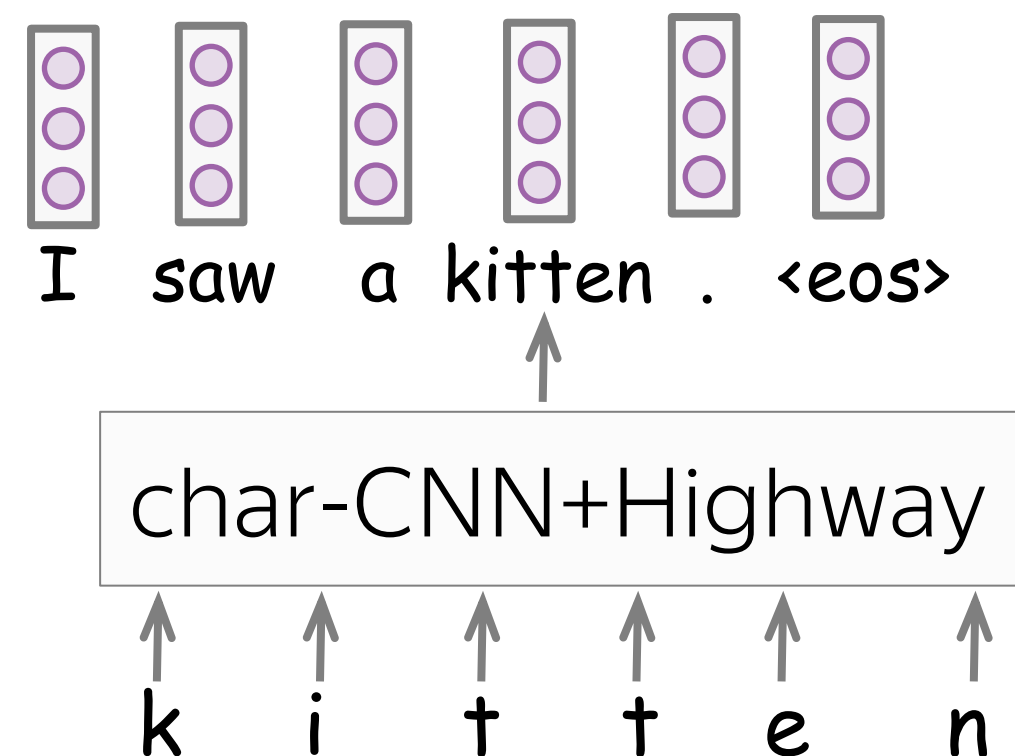
ELMo: From Words to Words-in-Context



The diagram illustrates the transition from individual words to words-in-context. It shows a sequence of six vertical rectangles, each containing three purple circles, representing word embeddings. Below each rectangle is a corresponding token from the sentence "I saw a kitten ." followed by an end-of-sentence token "<eos>".

I saw a kitten . <eos>

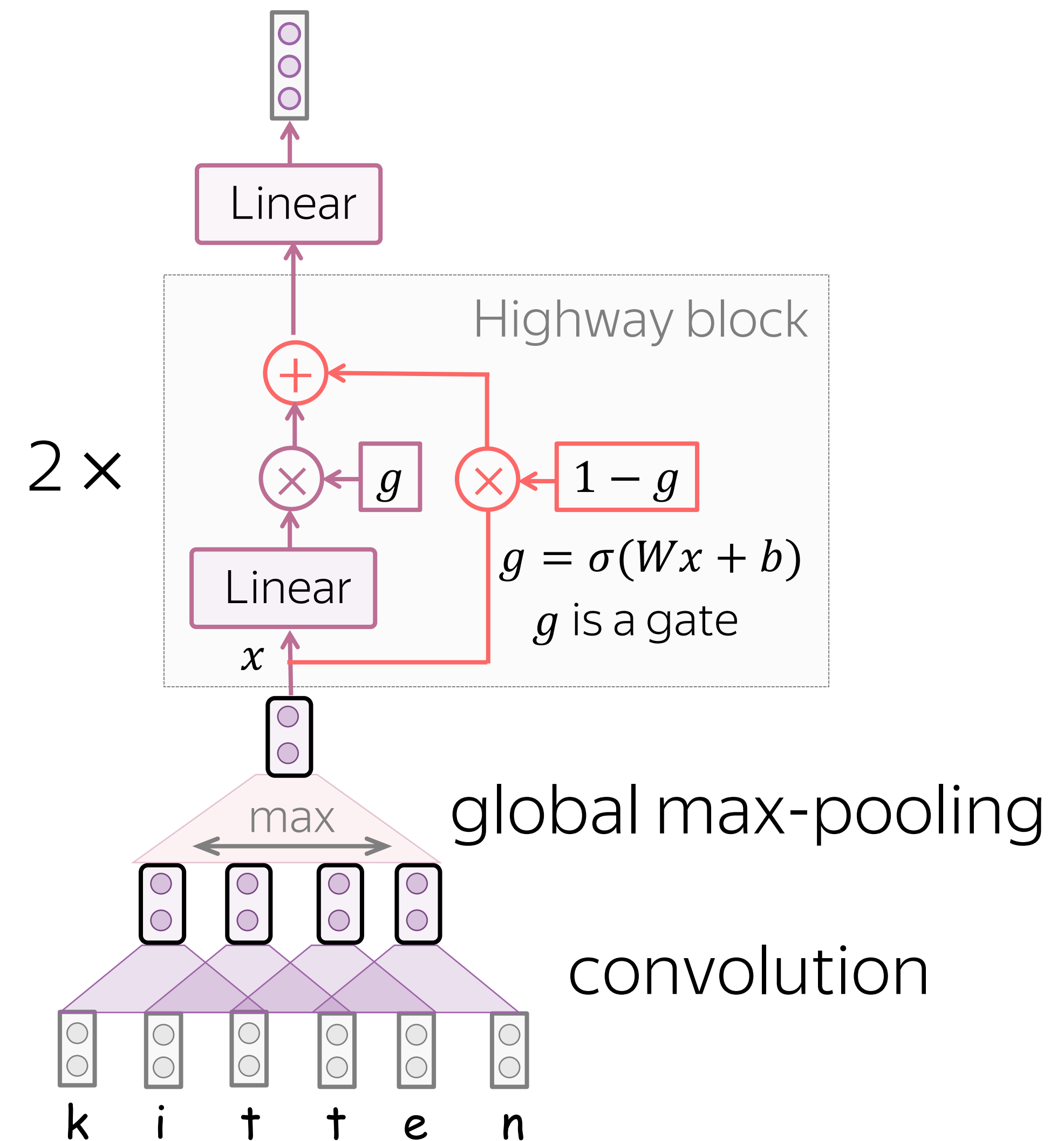
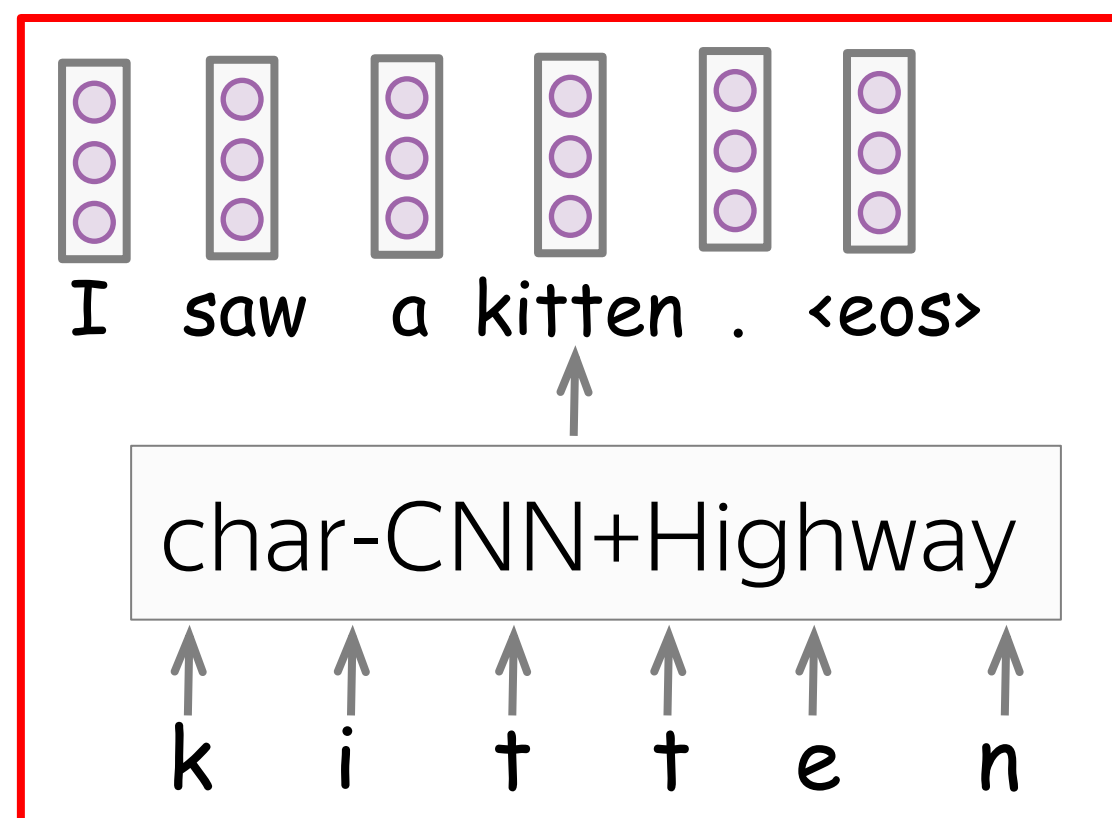
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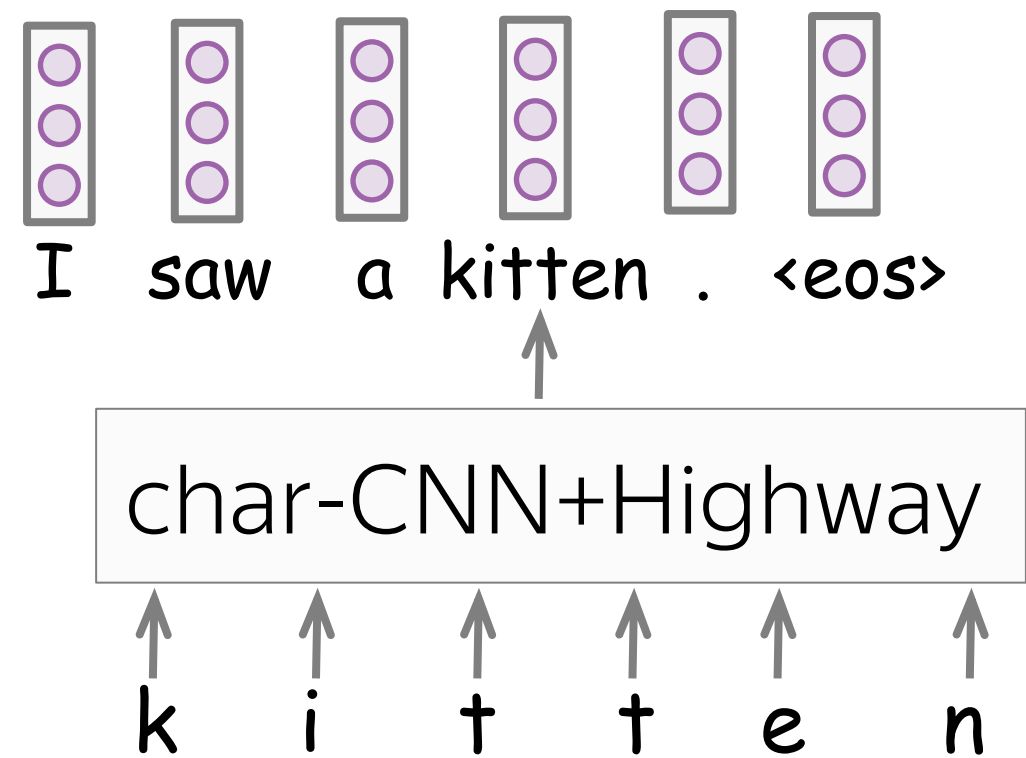
Character-level CNN:

- makes it possible to represent even unknown words
- tells a network which words are written similarly

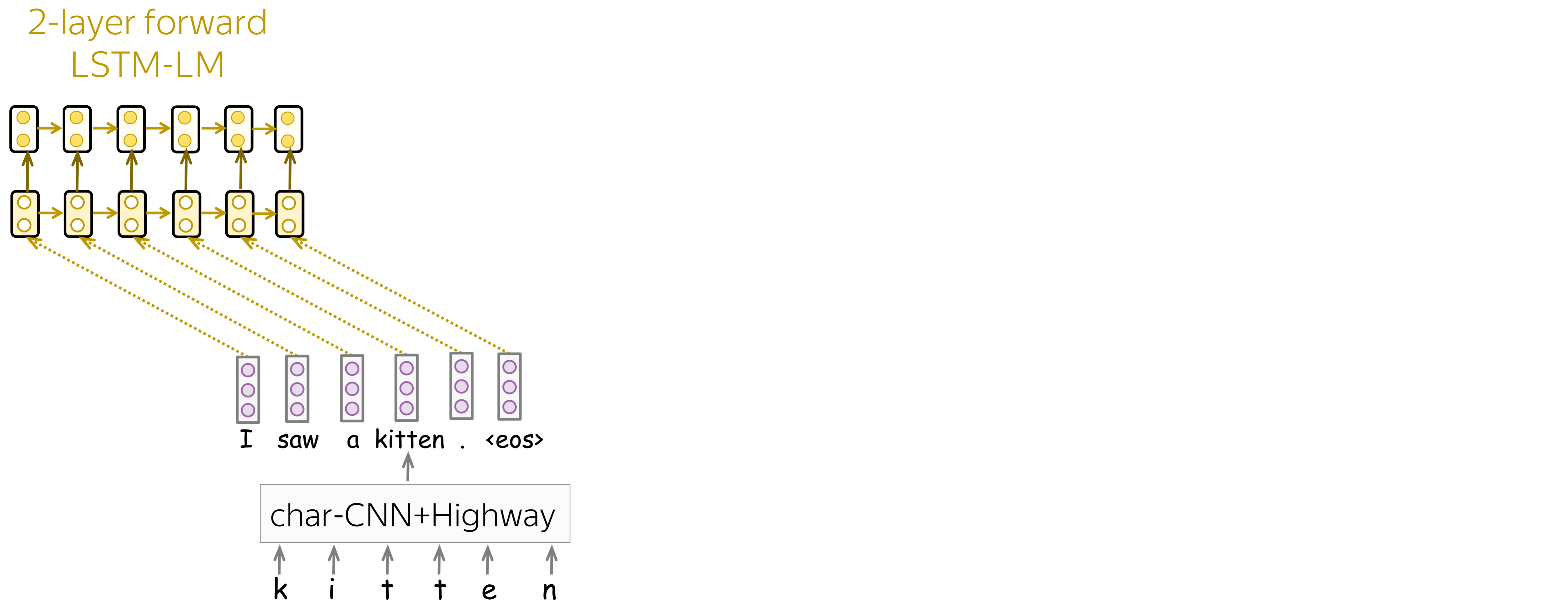
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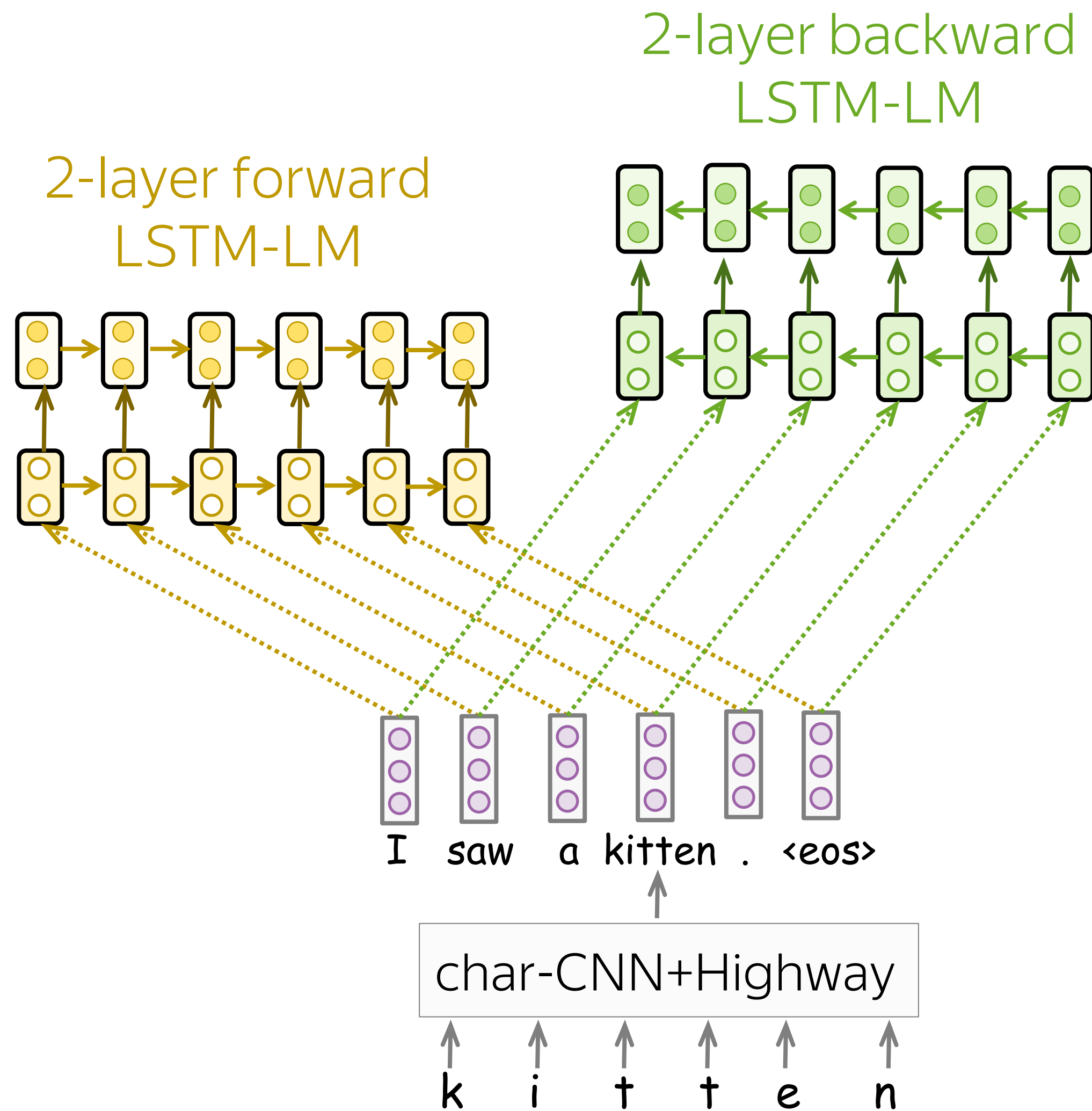
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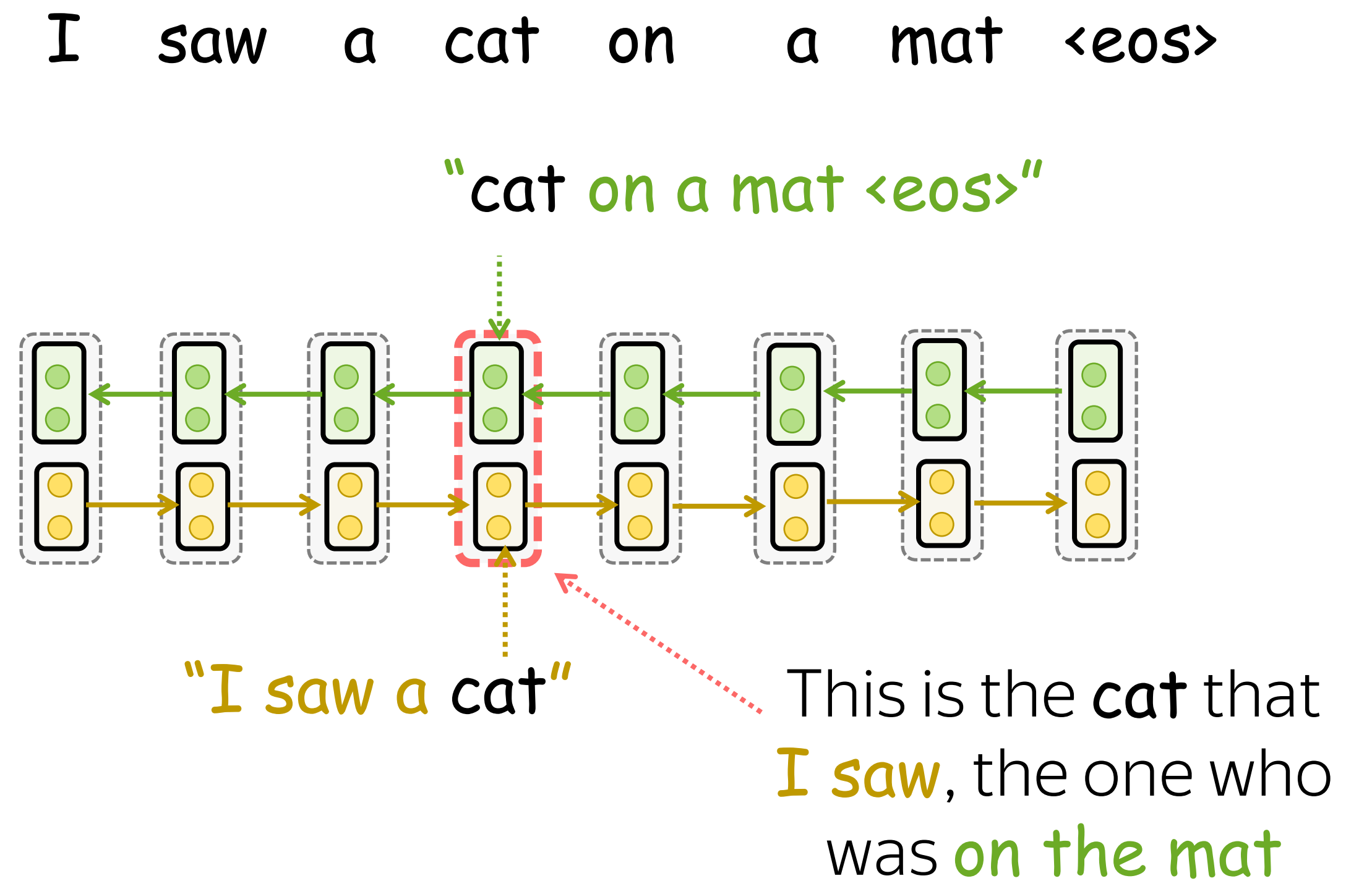
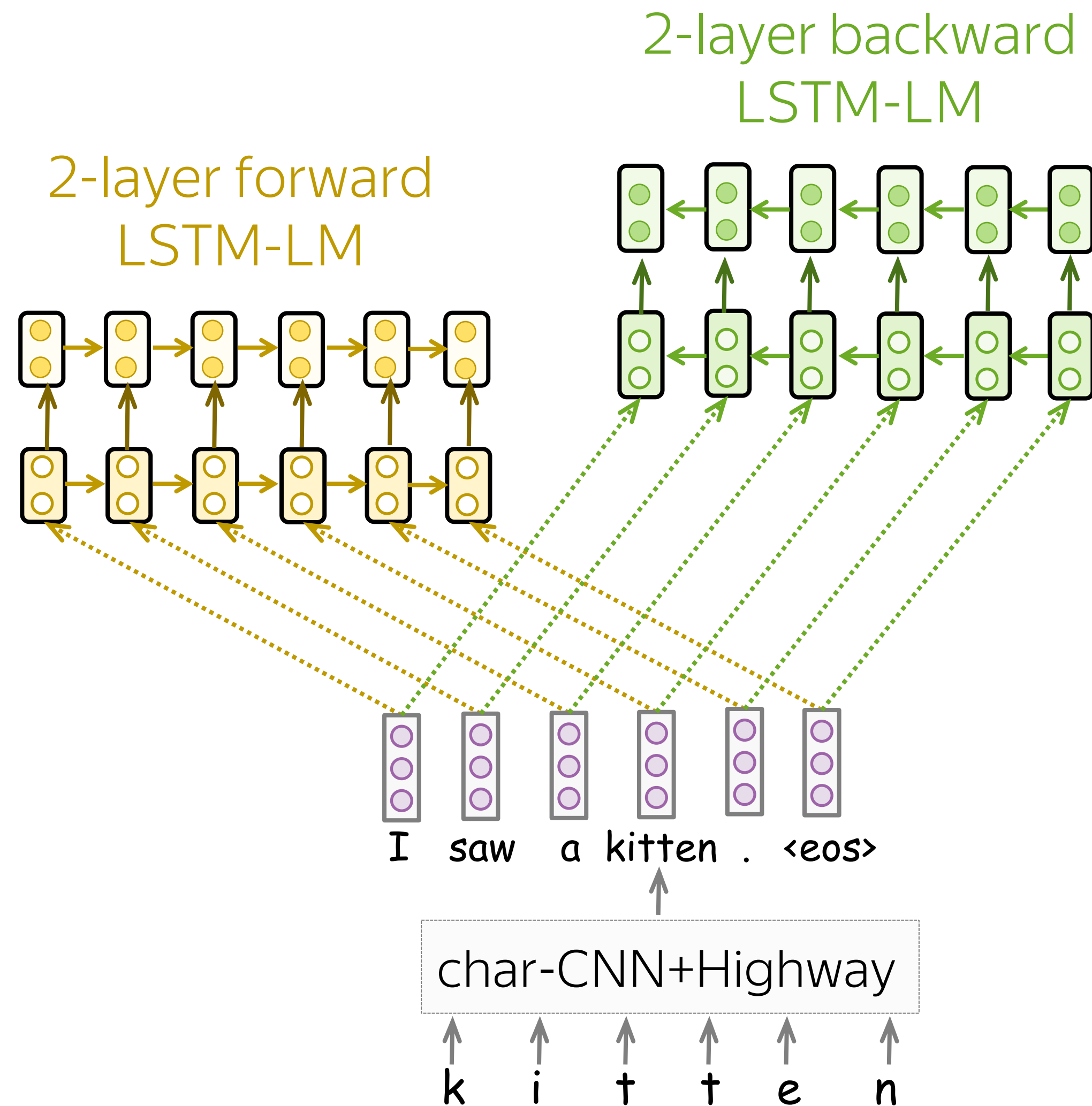
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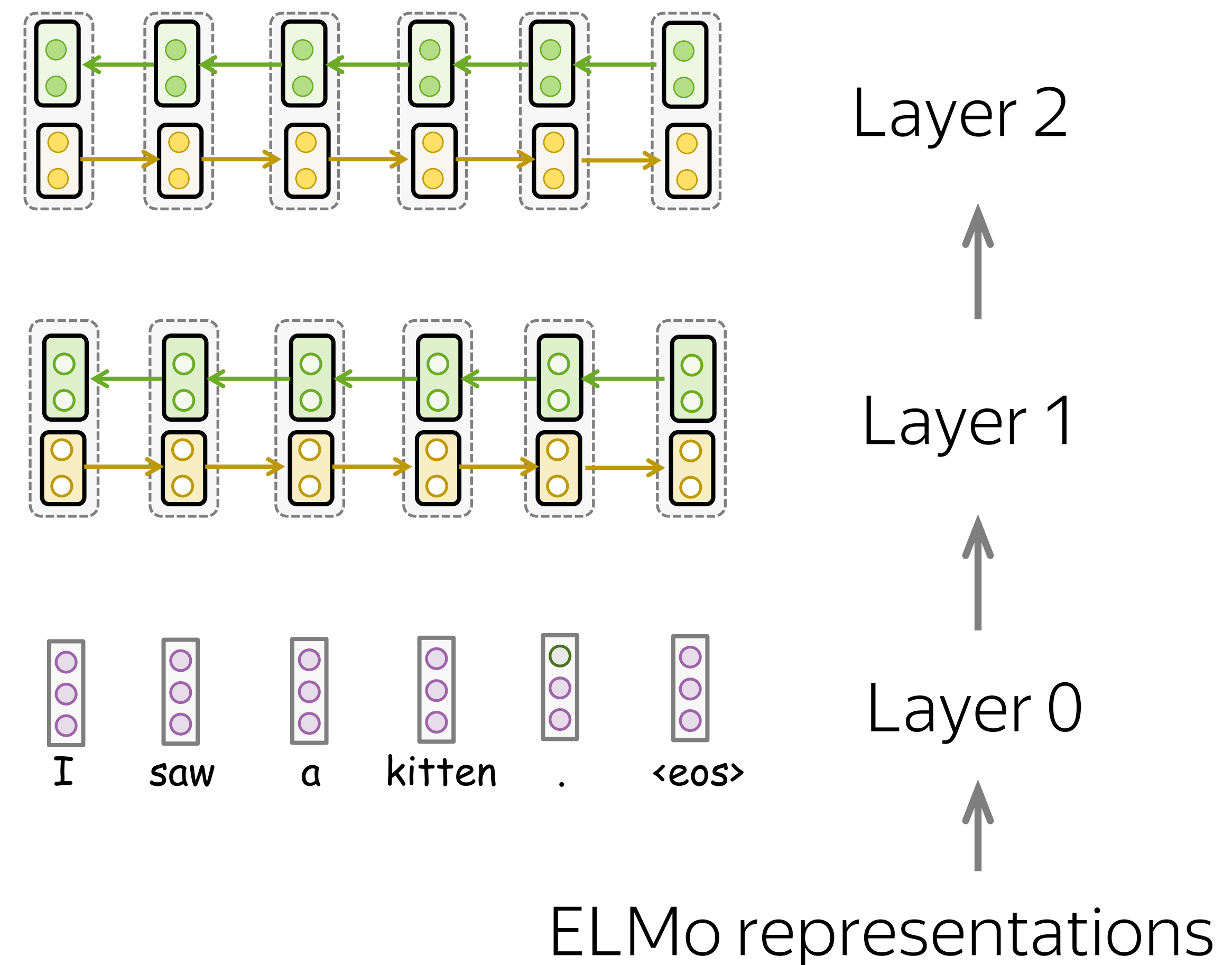
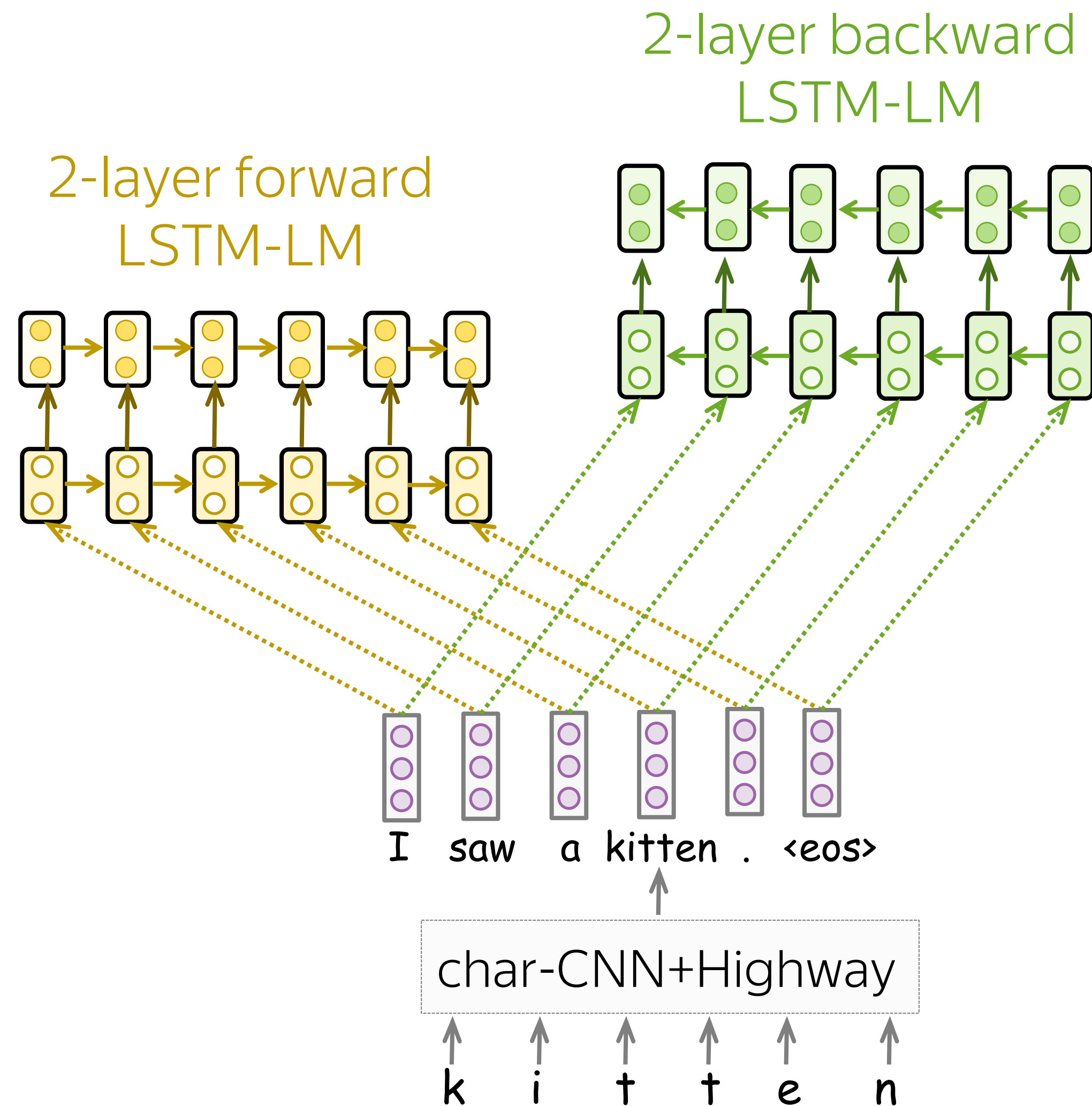
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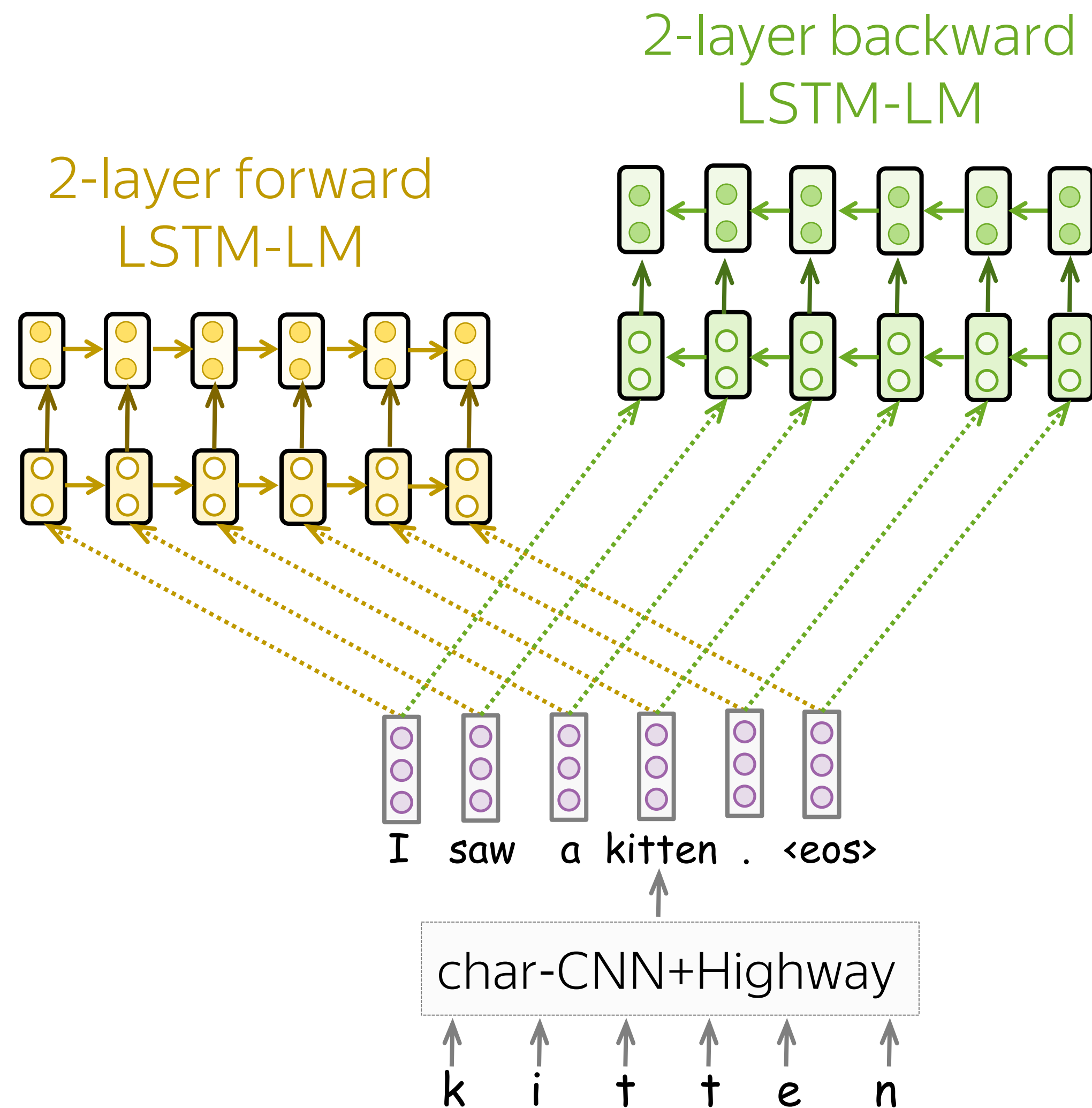
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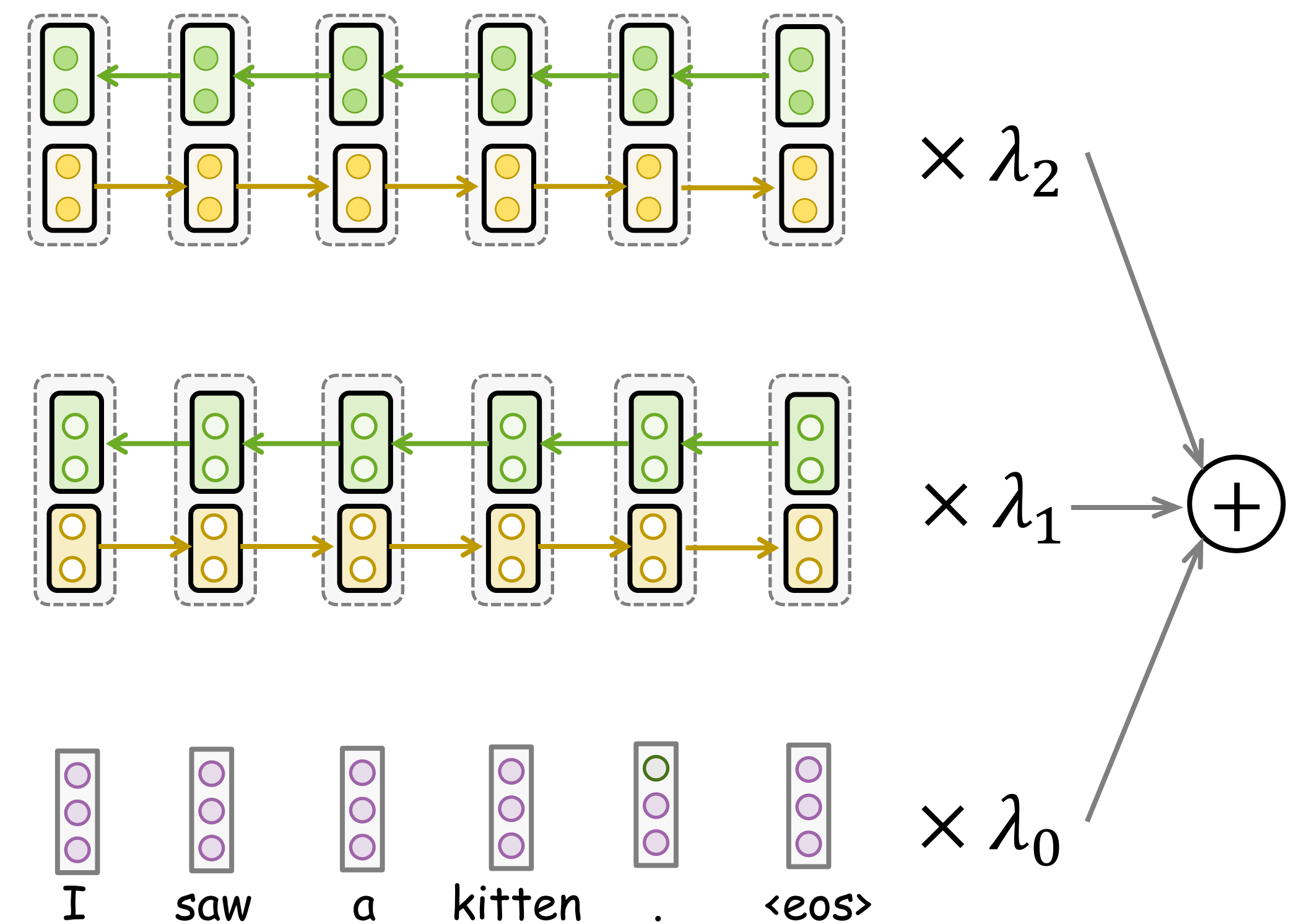
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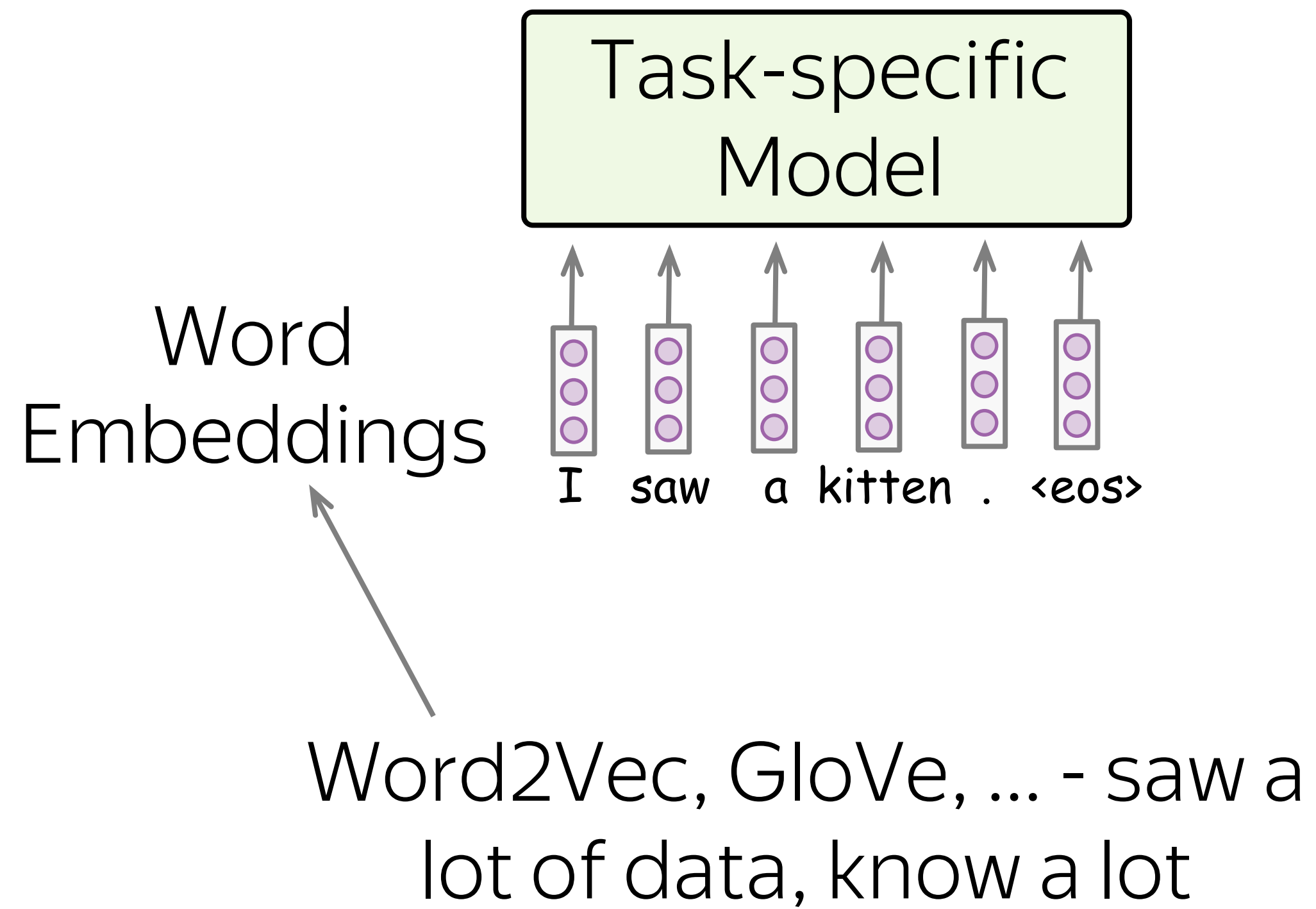


Learn specific $\lambda_0, \lambda_1, \lambda_2$ for each task



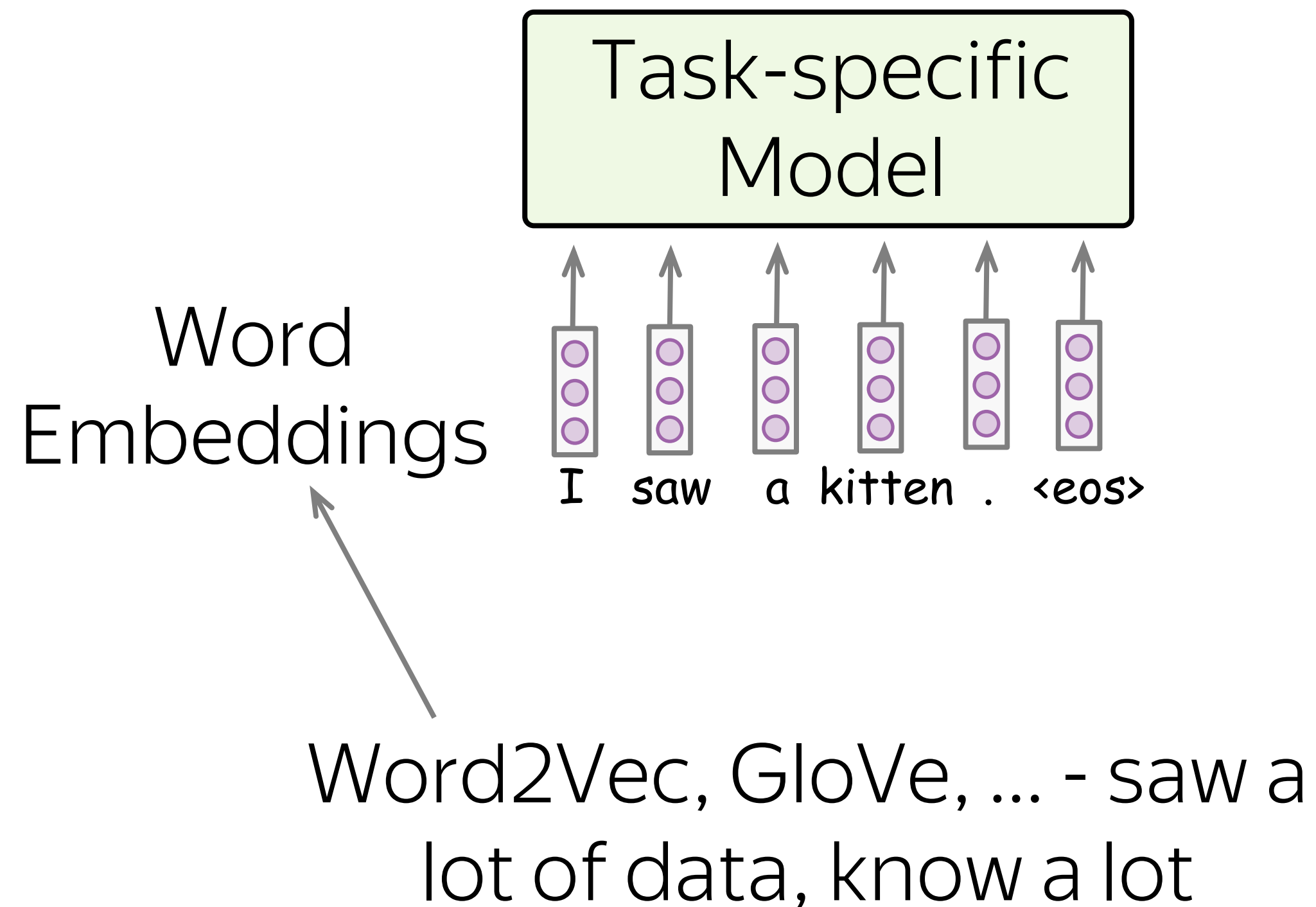
ELMo: How to Use?

Before



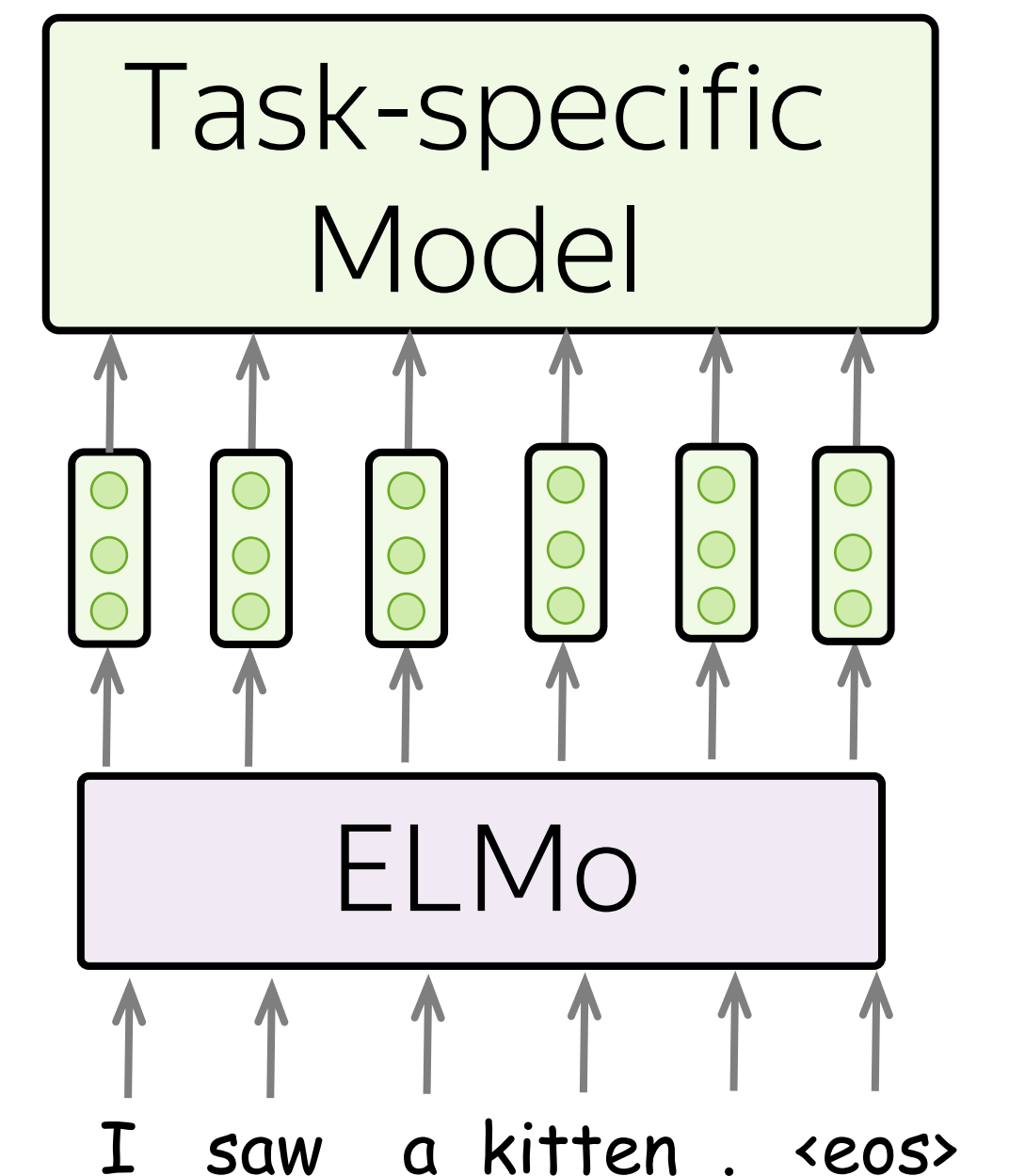
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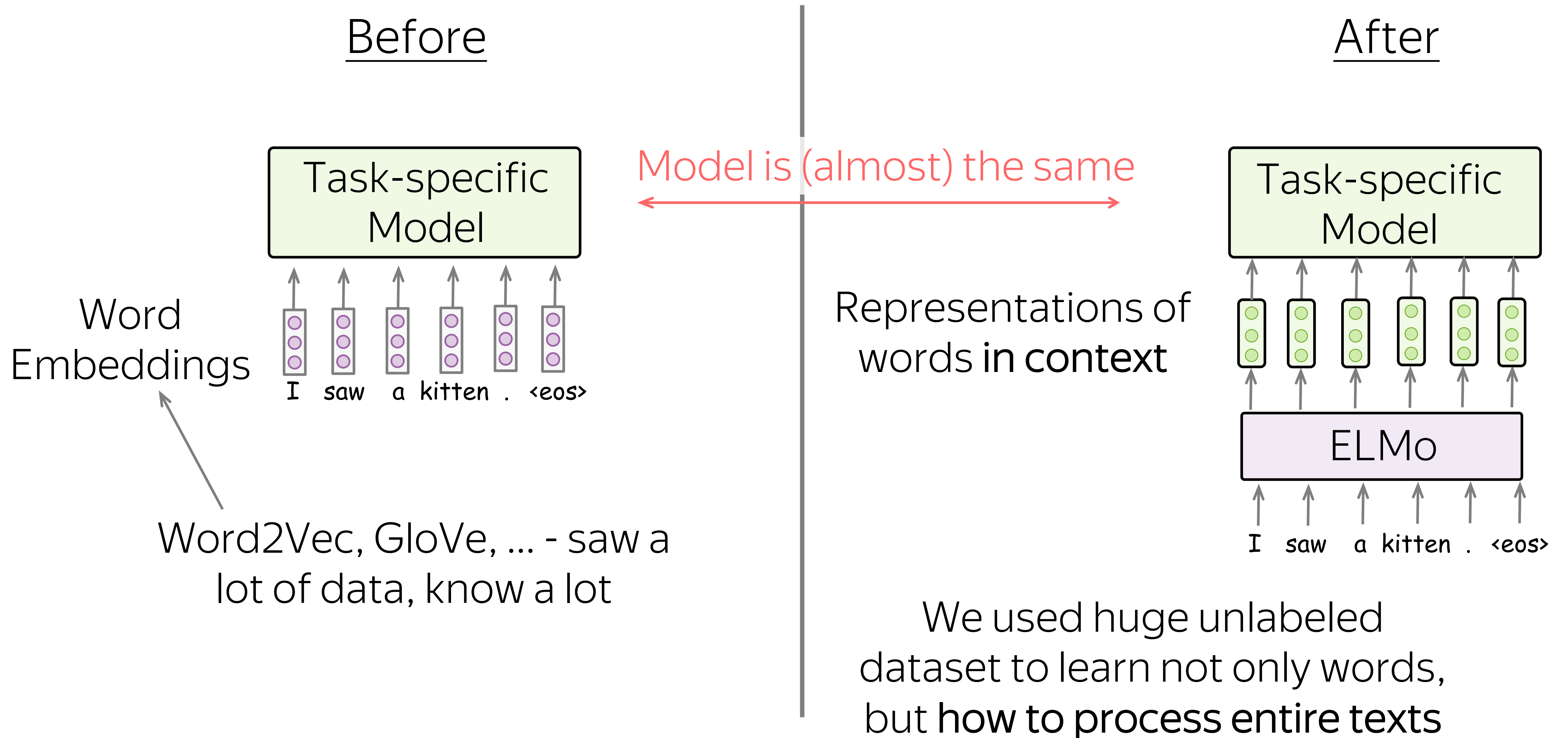
After

Representations of words in context



We used huge unlabeled dataset to learn not only words, but how to process entire texts

ELMo: How to Use?



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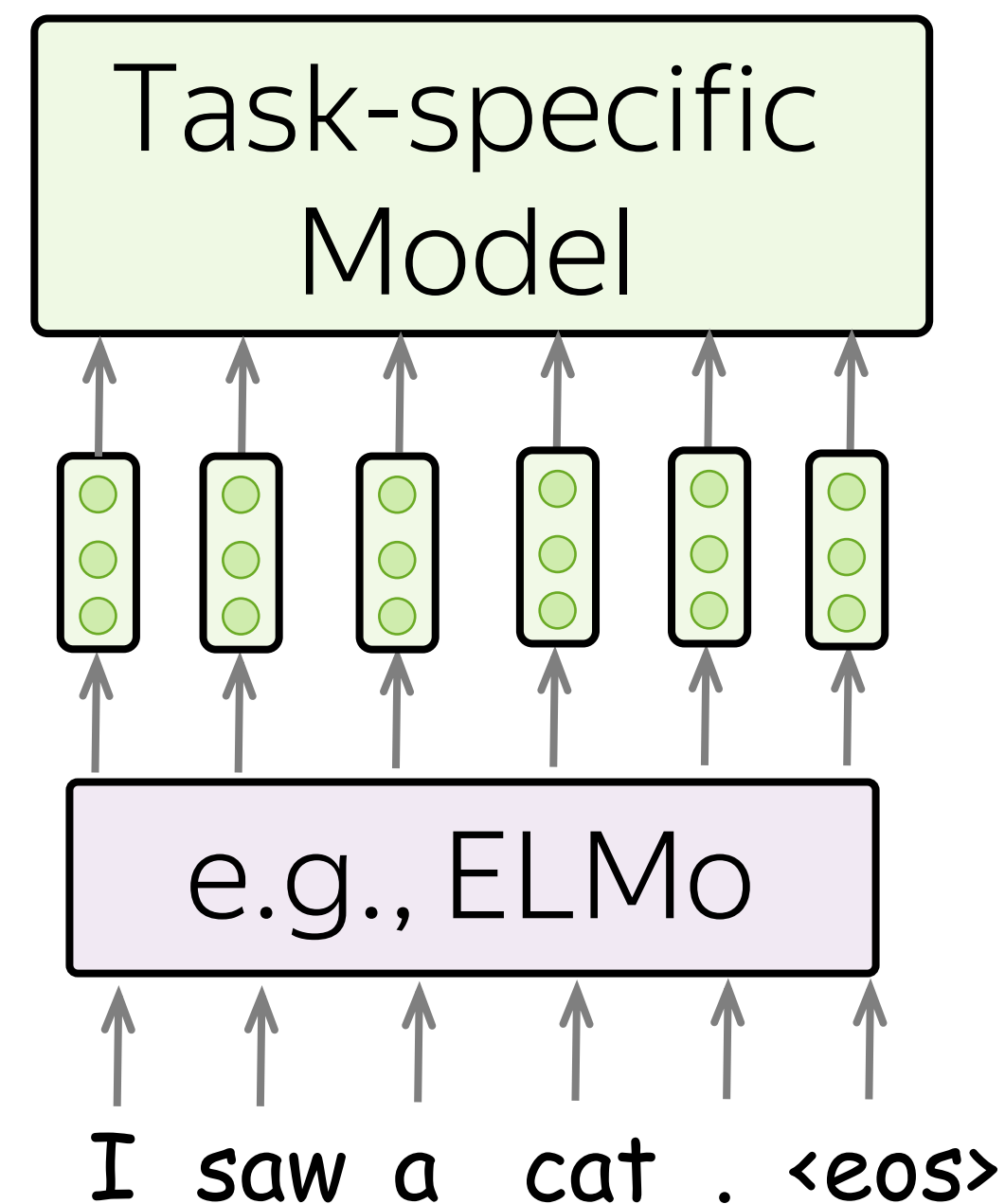
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Great Idea 2: From task-Specific Models to Unified

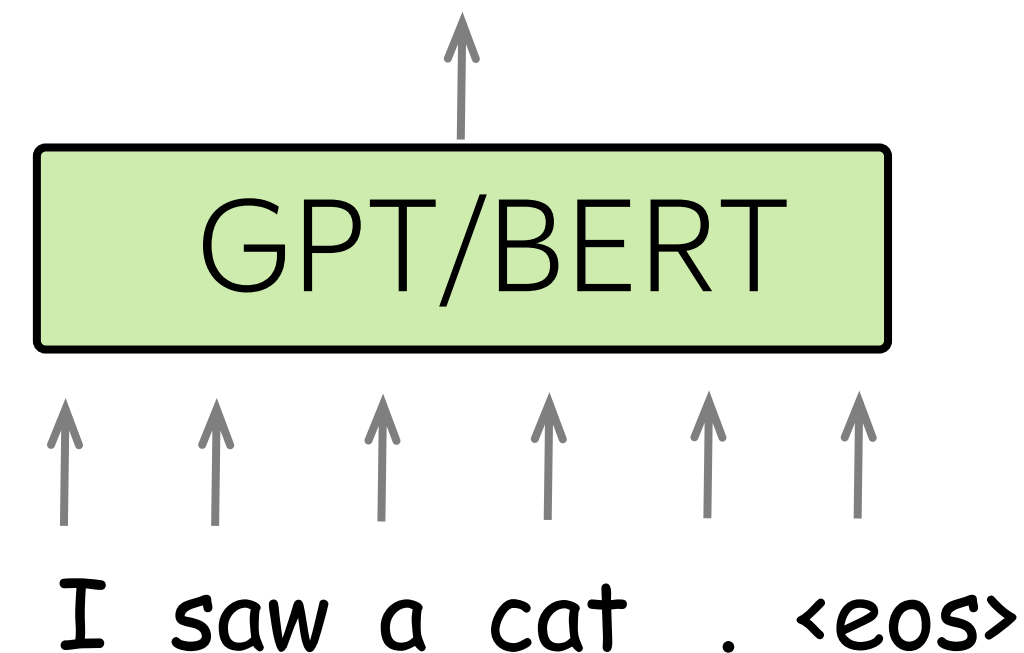
Idea: Instead of replacing word embeddings, replace entire models.

Before



After

No task-specific models!



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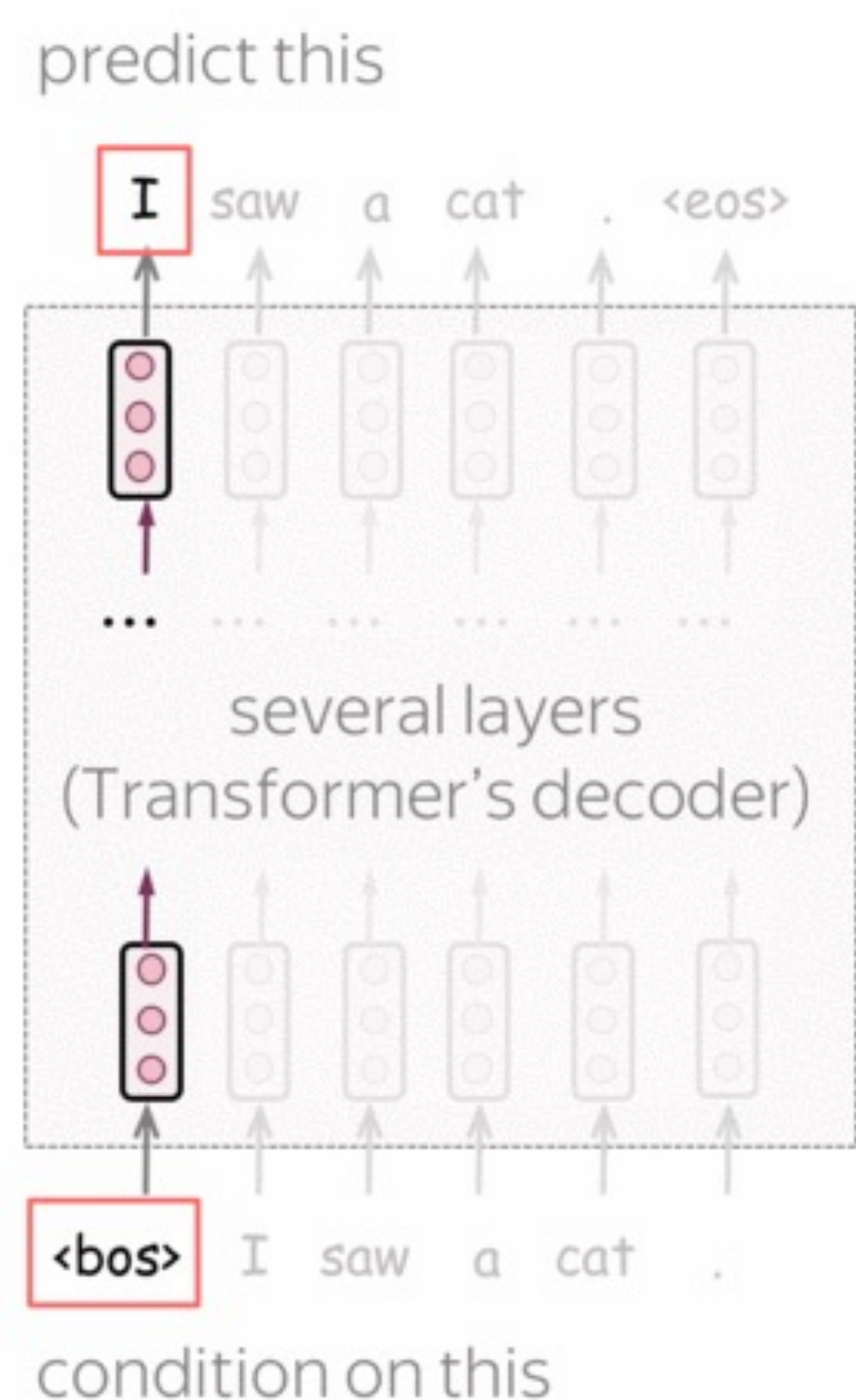
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- GPT
- BERT

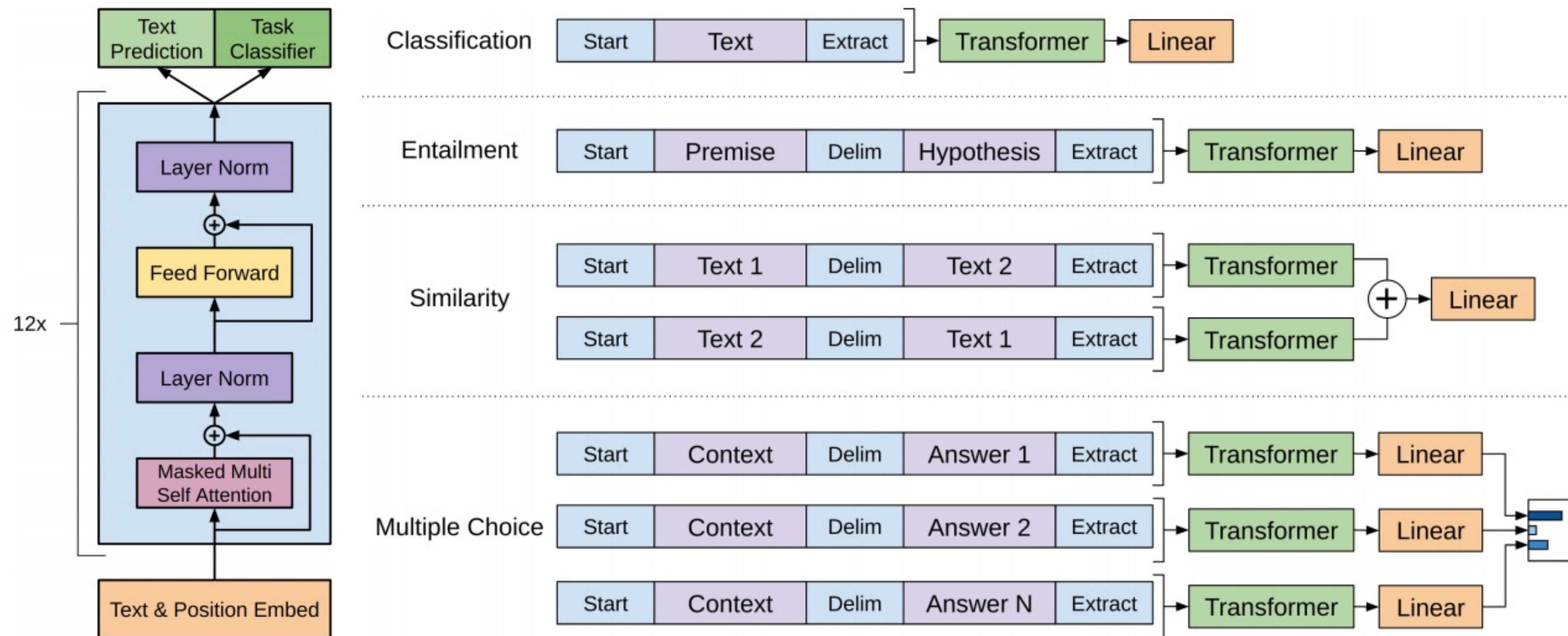
GPT(1-2-3): Transformer Decoder

- Transformer left-to-right LM
- GPT-1: Decoder with 12 layers



Fine-Tuning: Using GPT for Downstream Tasks

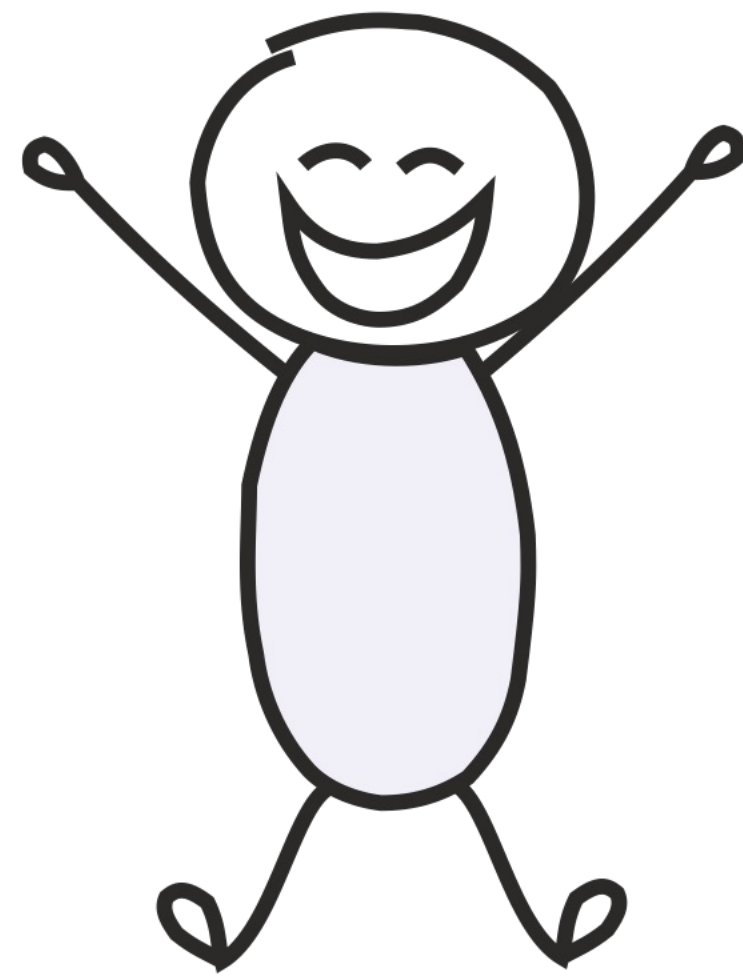
Fine-tuning loss: $L = L_{xent} + \lambda \cdot L_{task}$



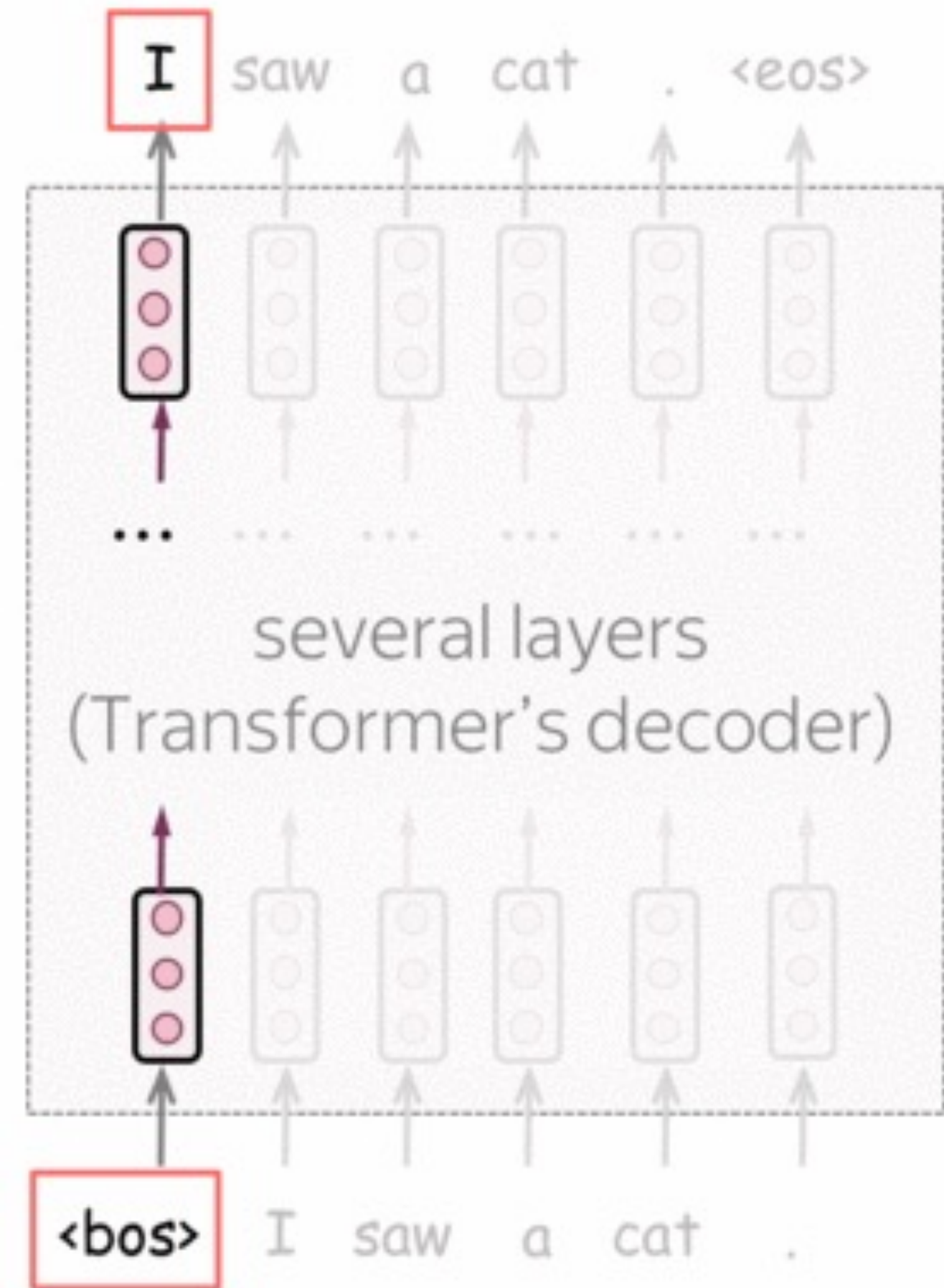
The figure is from the paper Improving Language Understanding by Generative Pretraining

GPT(1-2-3): More In the Next Lecture!

More on the later models
in the next lecture!



predict this



condition on this

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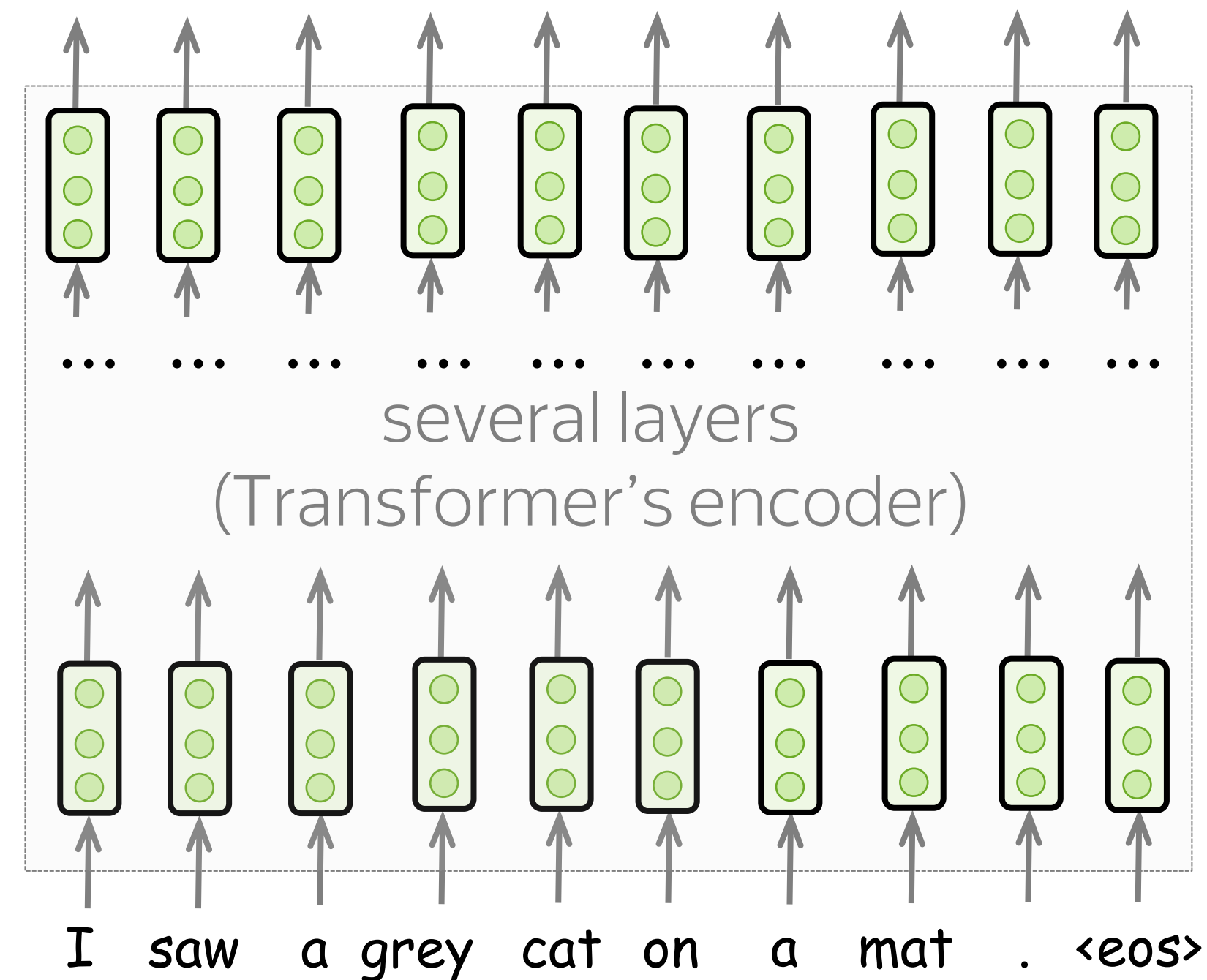
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BERT: Transformer Encoder with Fancy Training



Model architecture:

- Transformer encoder

What is special about it:

- Training objectives
 - MLM: Masked language modeling
 - NSP: Next sentence prediction
- Lots of data

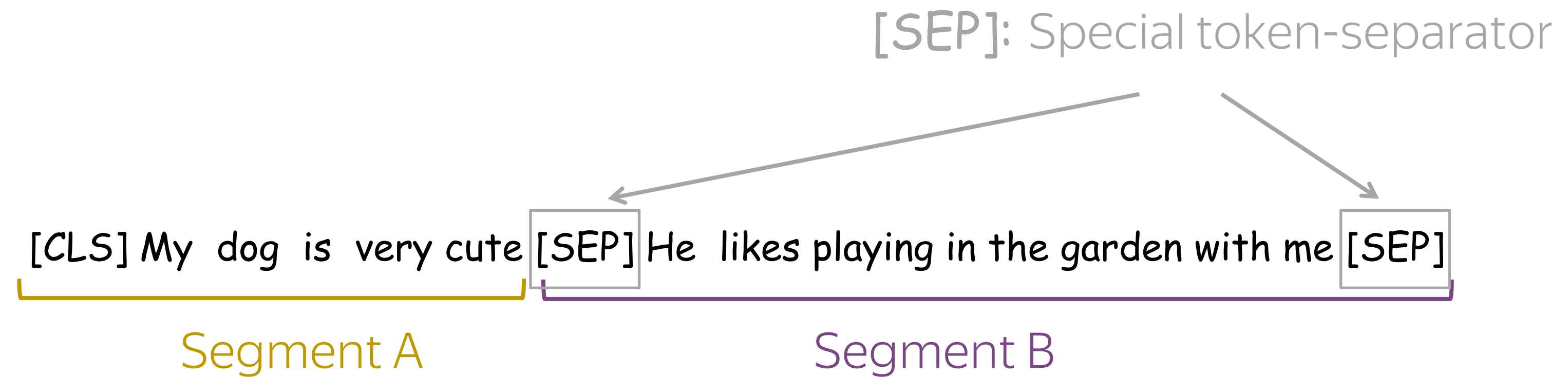
BERT Input

[CLS] My dog is very cute [SEP] He likes playing in the garden with me [SEP]

Segment A Segment B

Training on pairs of sentences: either
consecutive or random (50%/50%)

BERT Input

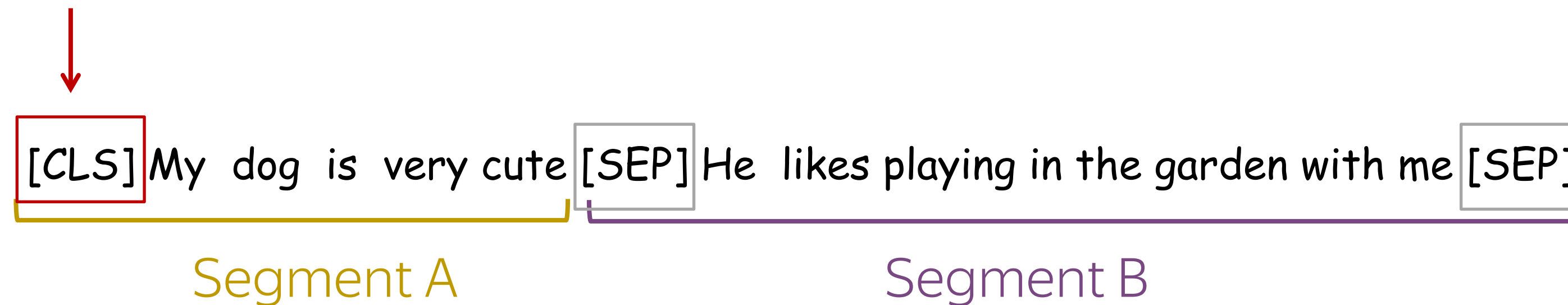


Training on pairs of sentences: either
consecutive or random (50%/50%)

BERT Input

[CLS]: Special token

- Training time: predict if sentences are consecutive or not (Next Sentence Prediction /NSP objective)
- Test time: downstream tasks (e.g., classification)



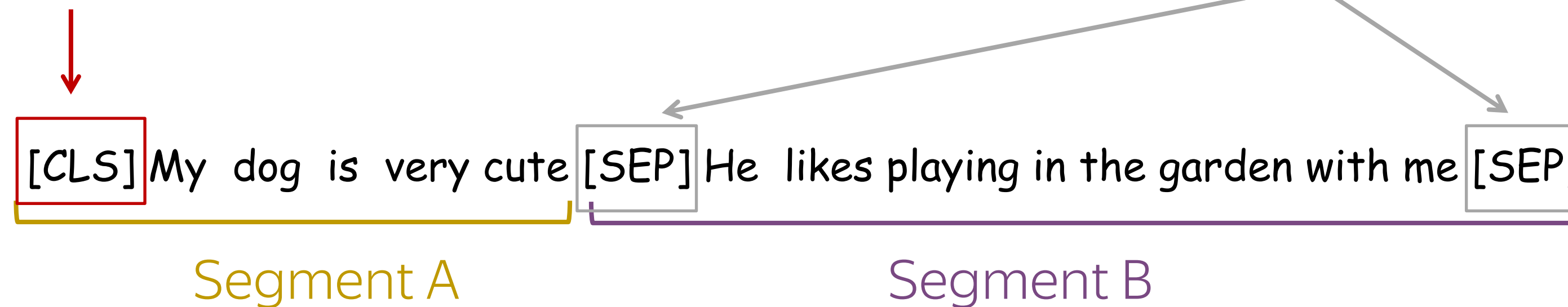
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BERT Input

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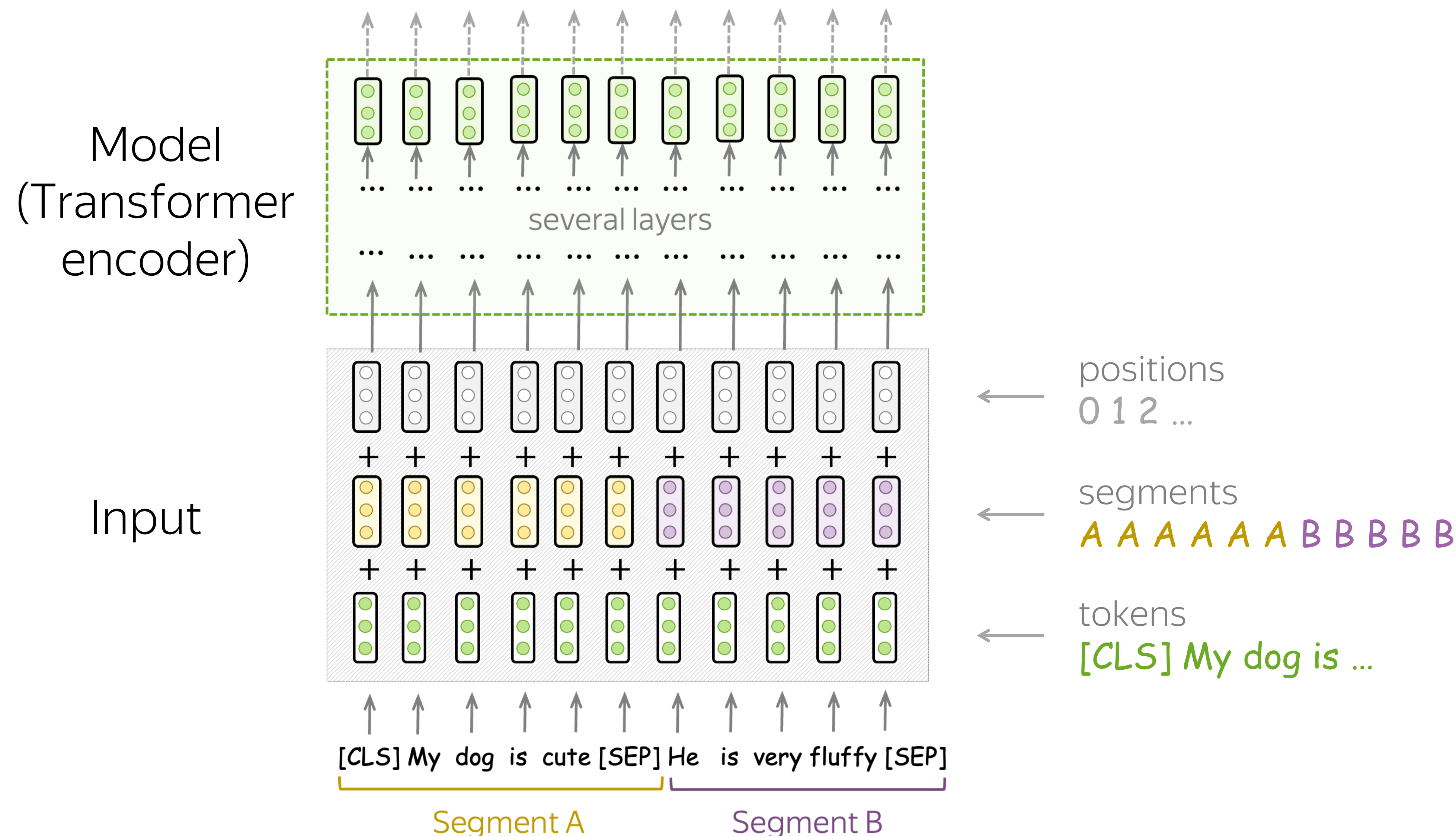
- Training time: predict if sentences are consecutive or not (Next Sentence Prediction /NSP objective)
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[SEP]: Special token-separator

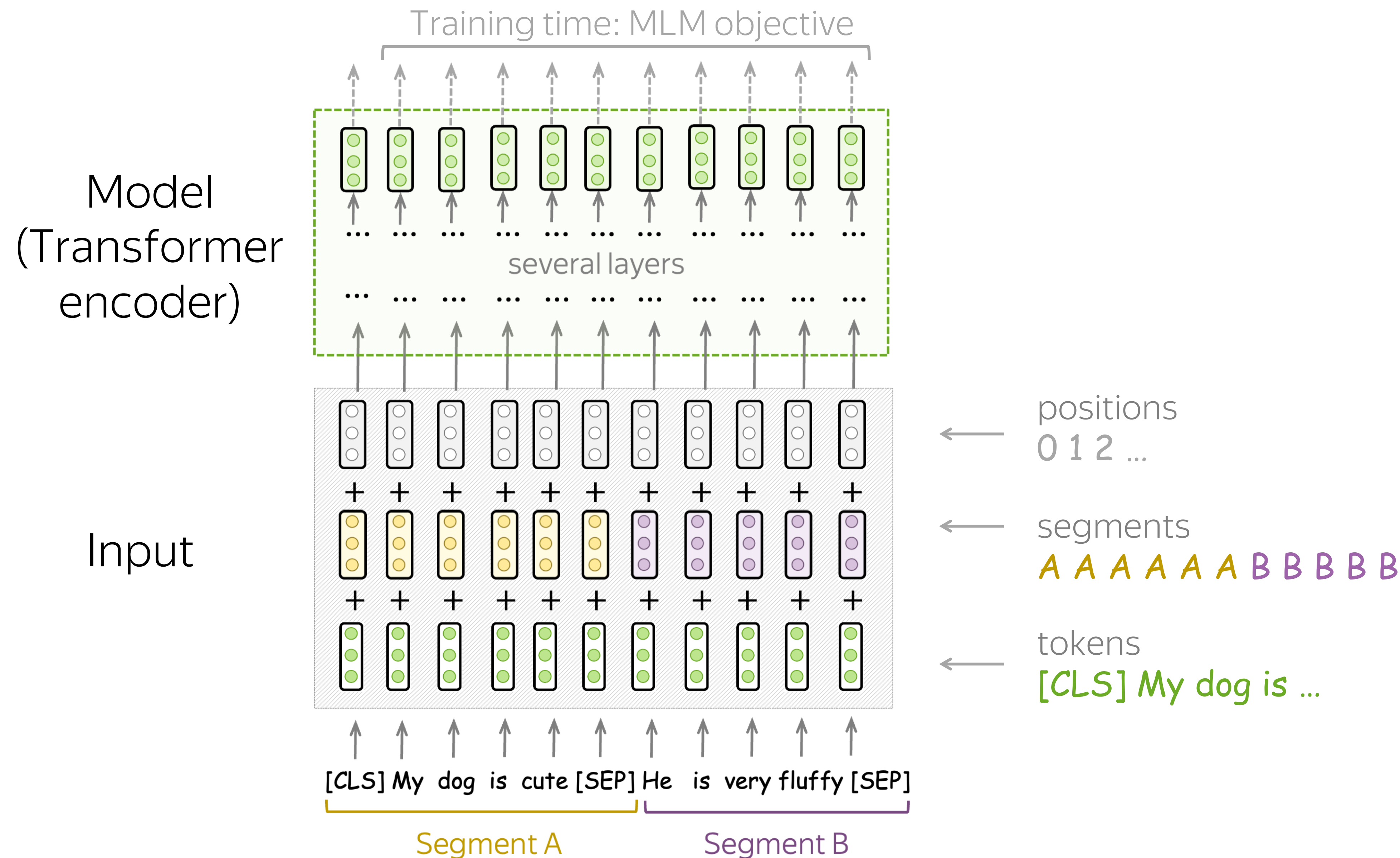


Training on pairs of sentences: either consecutive or random (50%/50%)

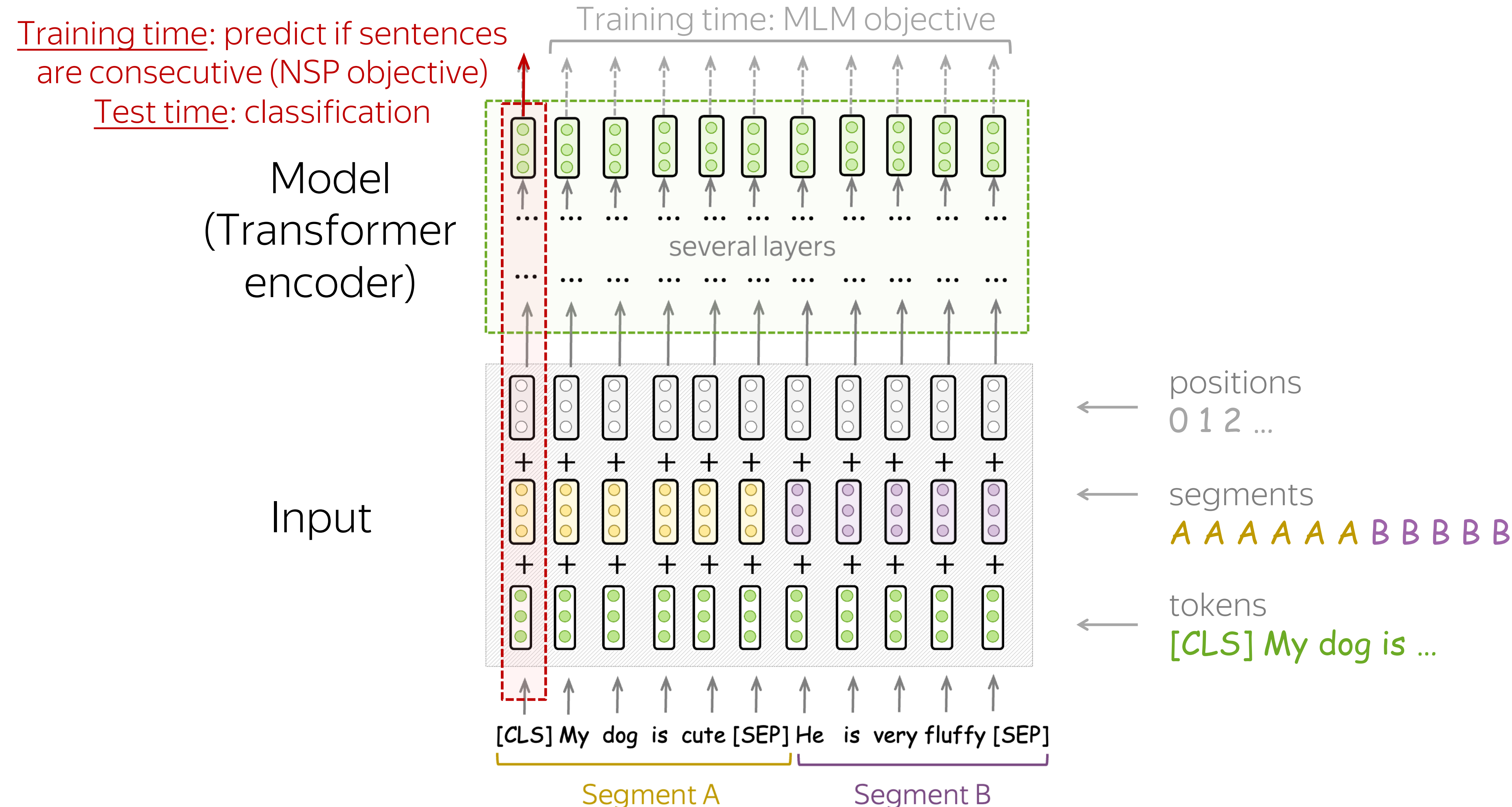
BERT Input: Full Pipeline



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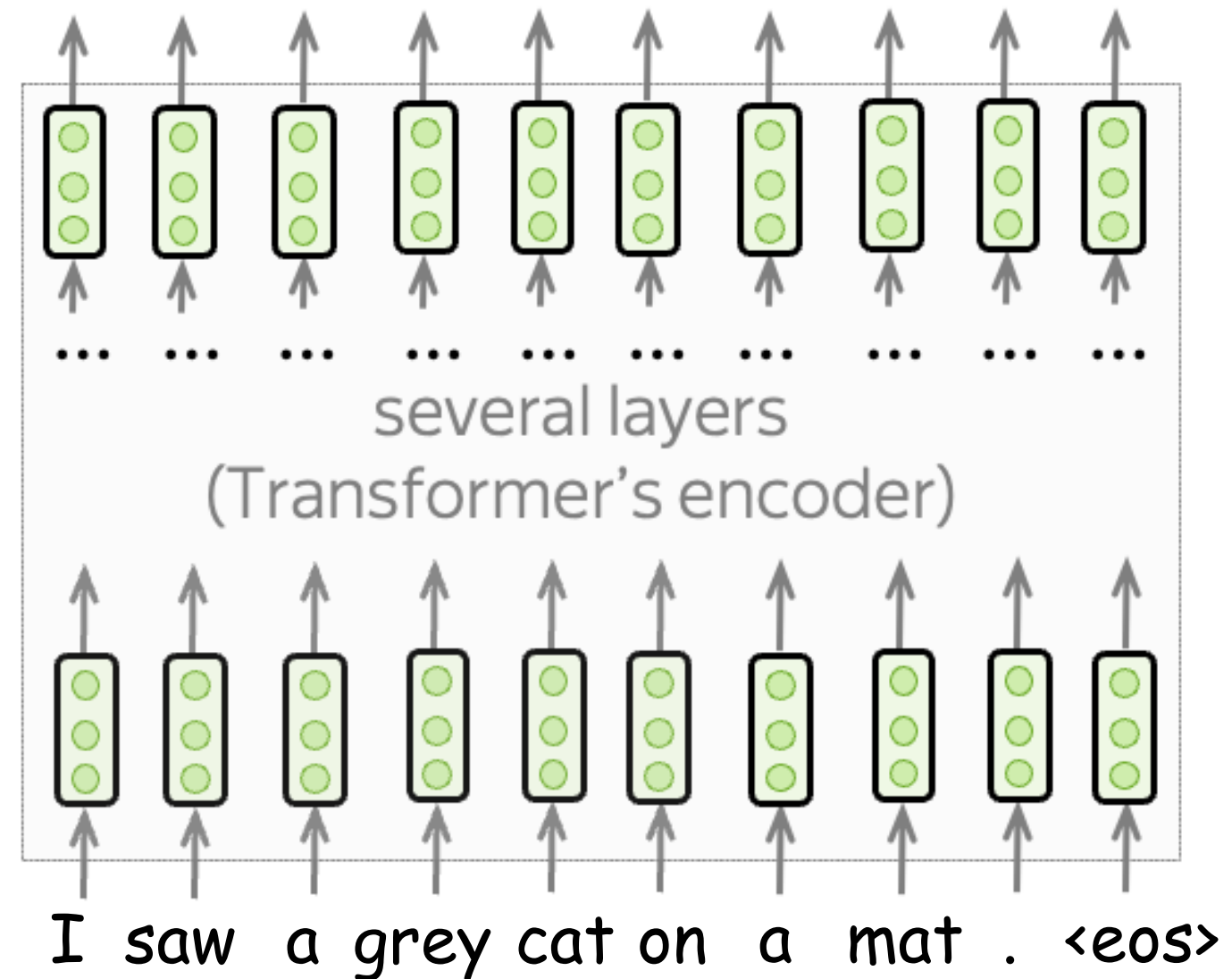


BERT Input: Full Pipeline



BERT: Masked Language Modeling Objective

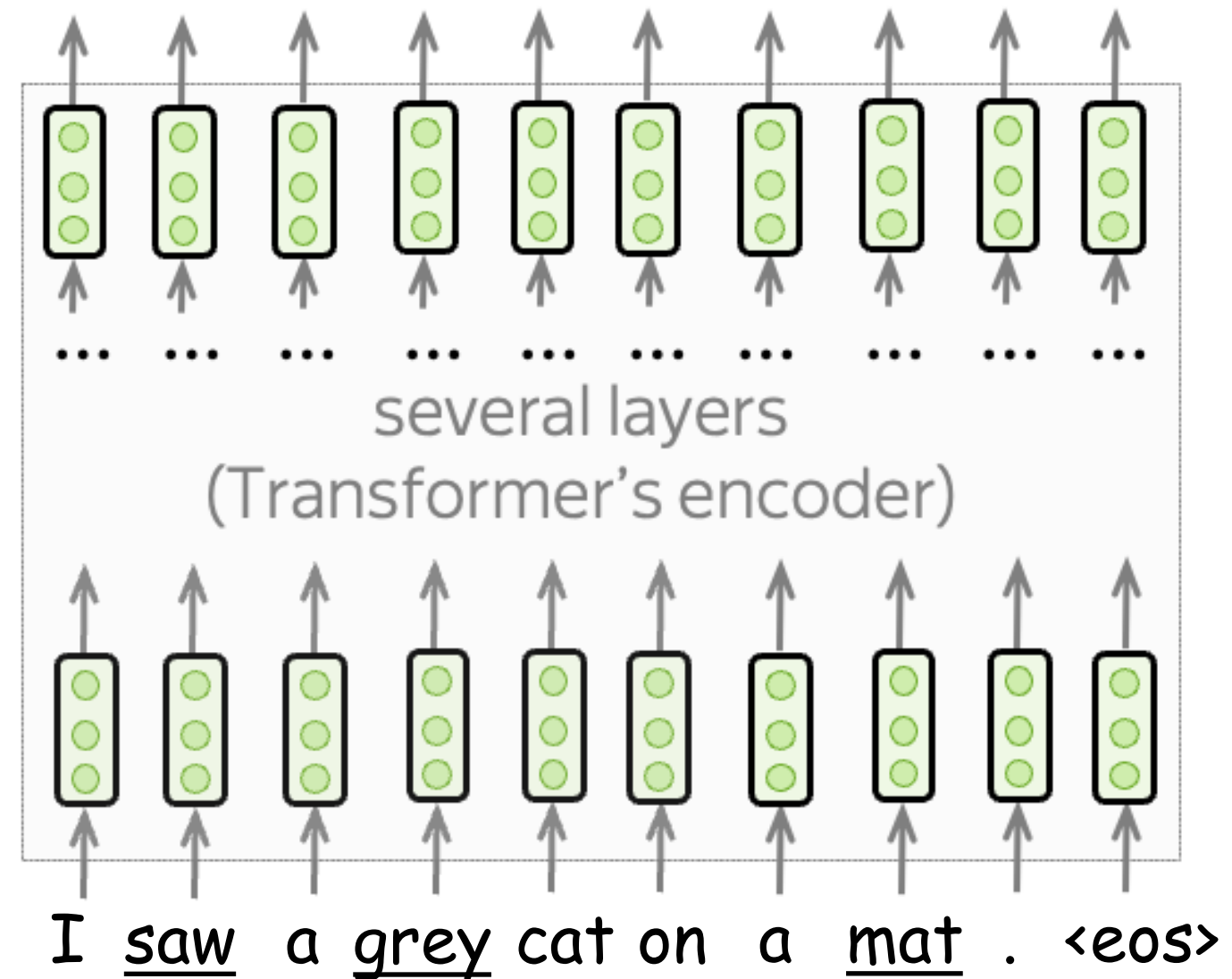
At each training step:



BERT: Masked Language Modeling Objective

At each training step:

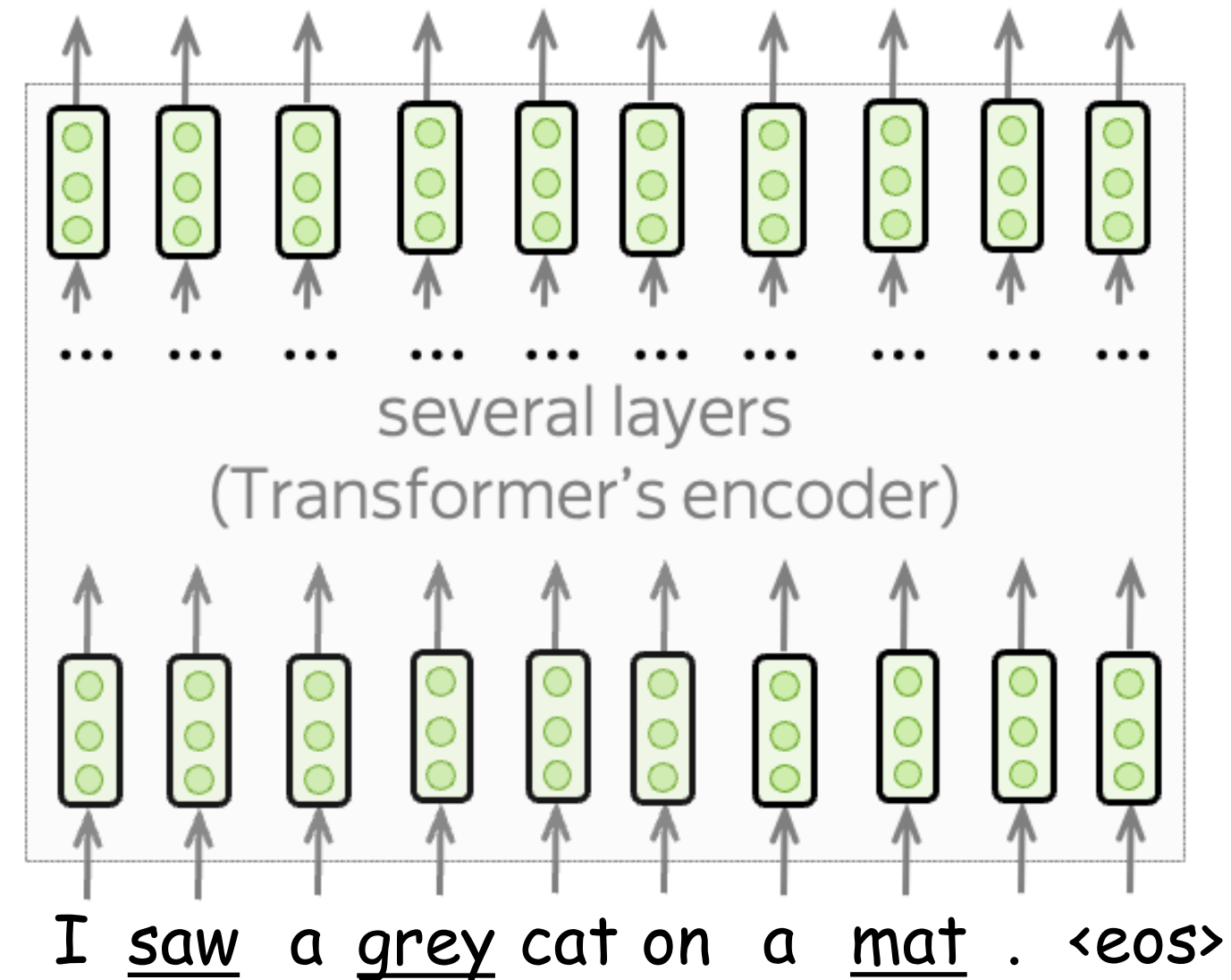
- pick randomly 15% of tokens



BERT: Masked Language Modeling Objective

At each training step:

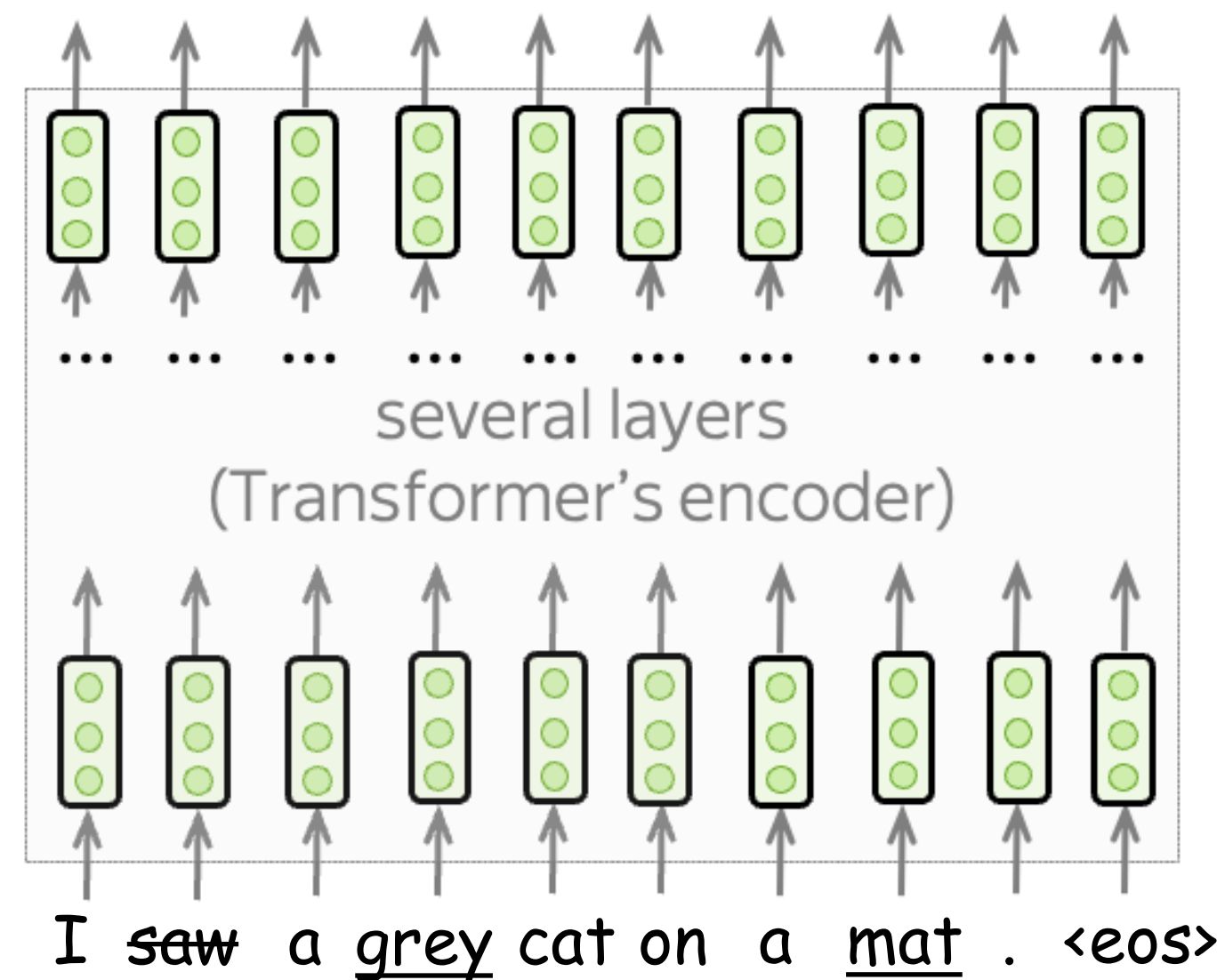
- pick randomly 15% of tokens
- replace each of the chosen tokens with something



BERT: Masked Language Modeling Objective

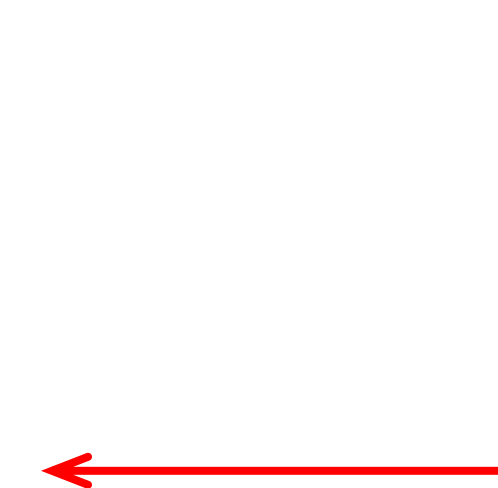
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[MASK]

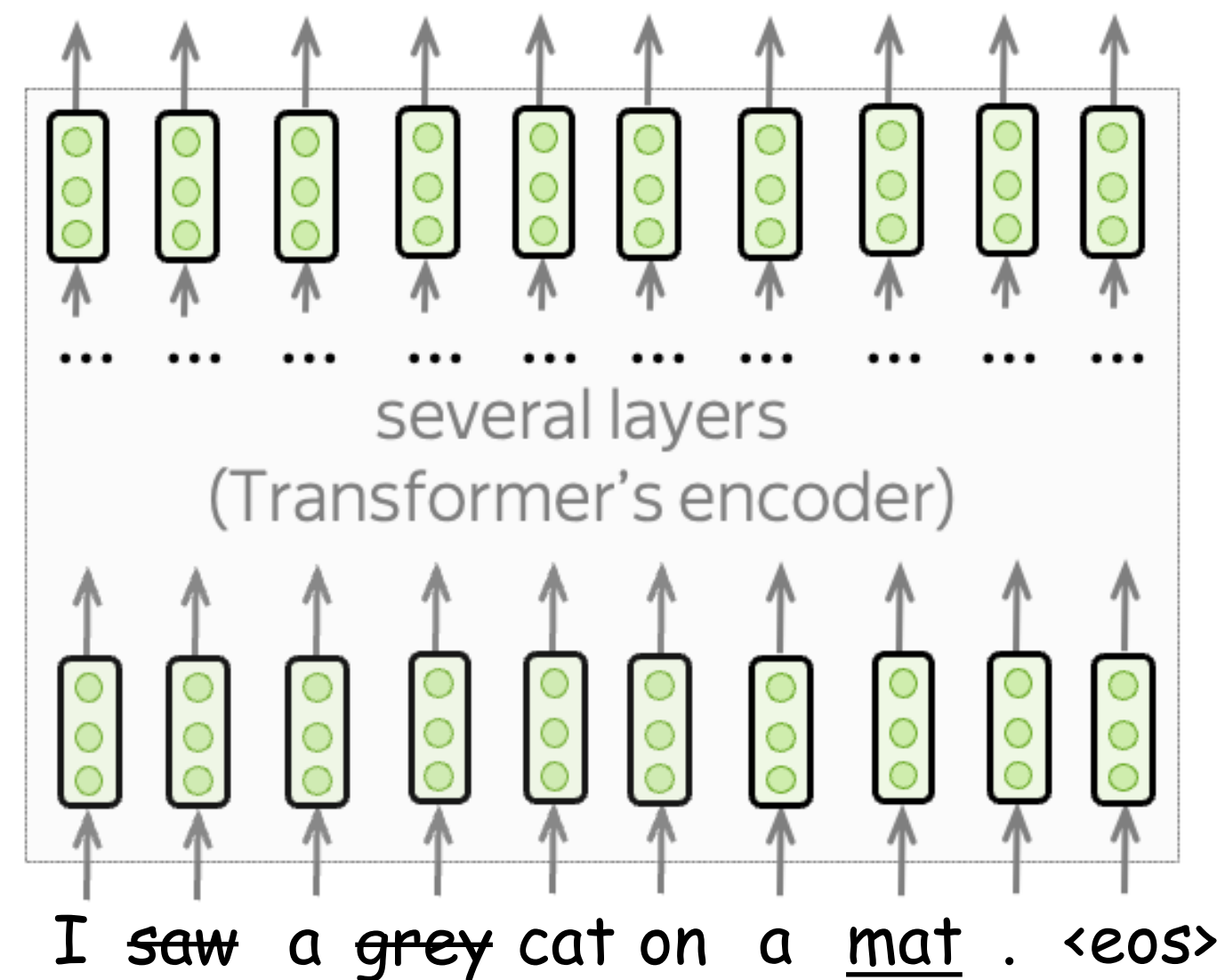
- [MASK],
with $p = 80\%$



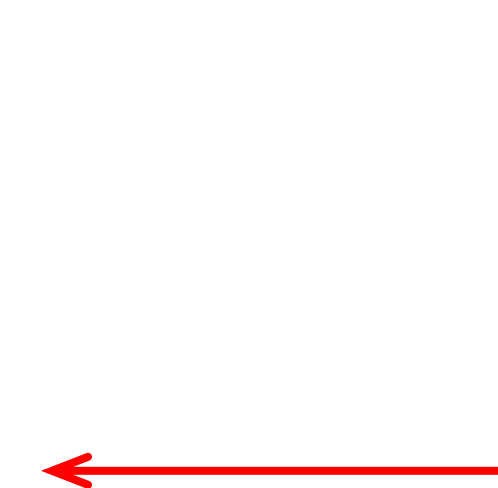
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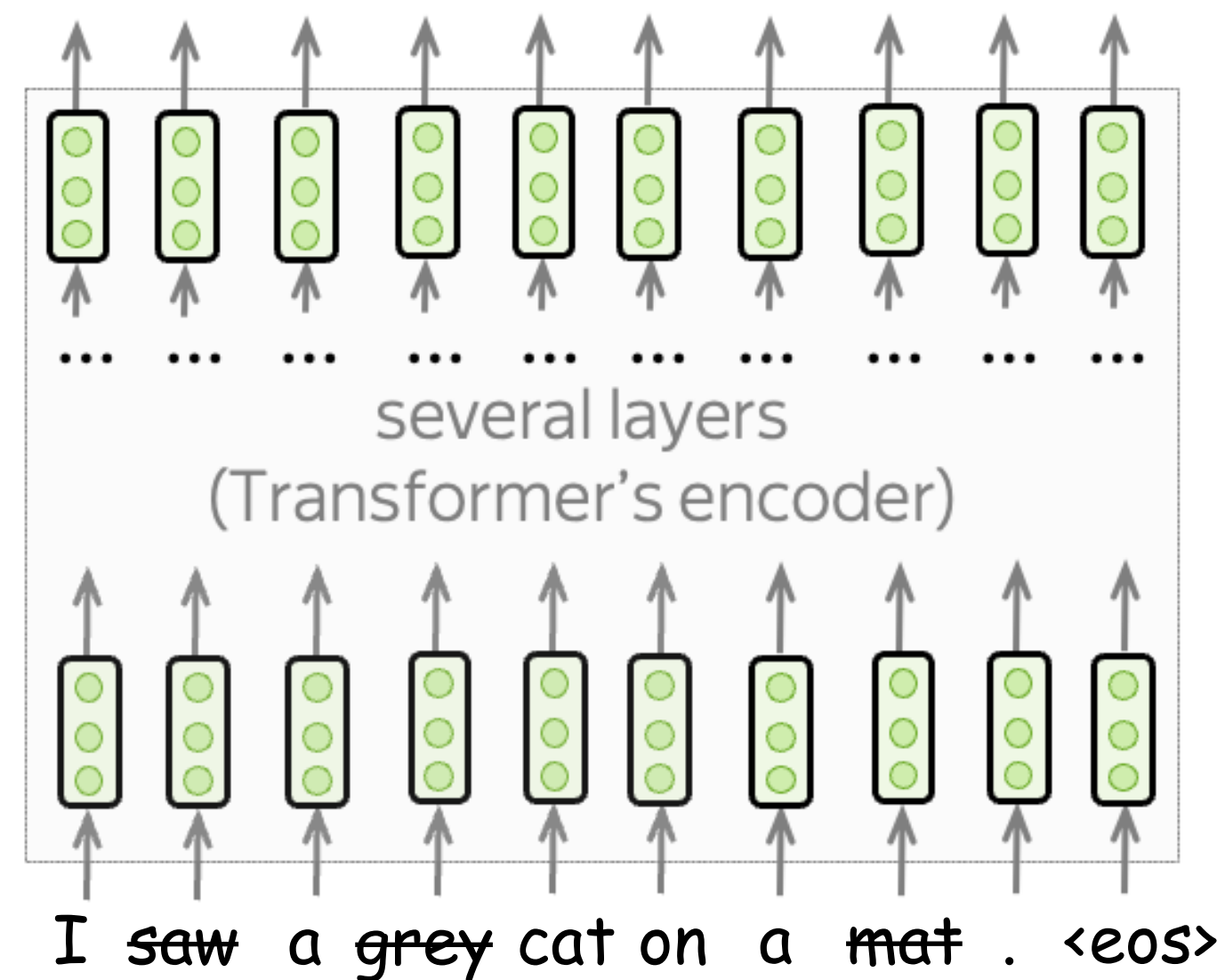
- **[MASK]**,
with $p = 80\%$
- Random token,
with $p = 10\%$



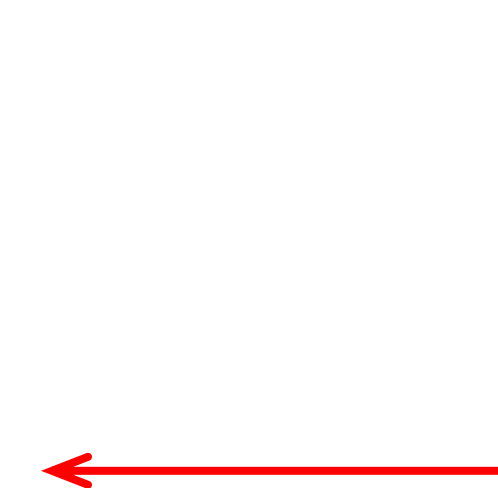
BERT: Masked Language Modeling Objective

At each training step:

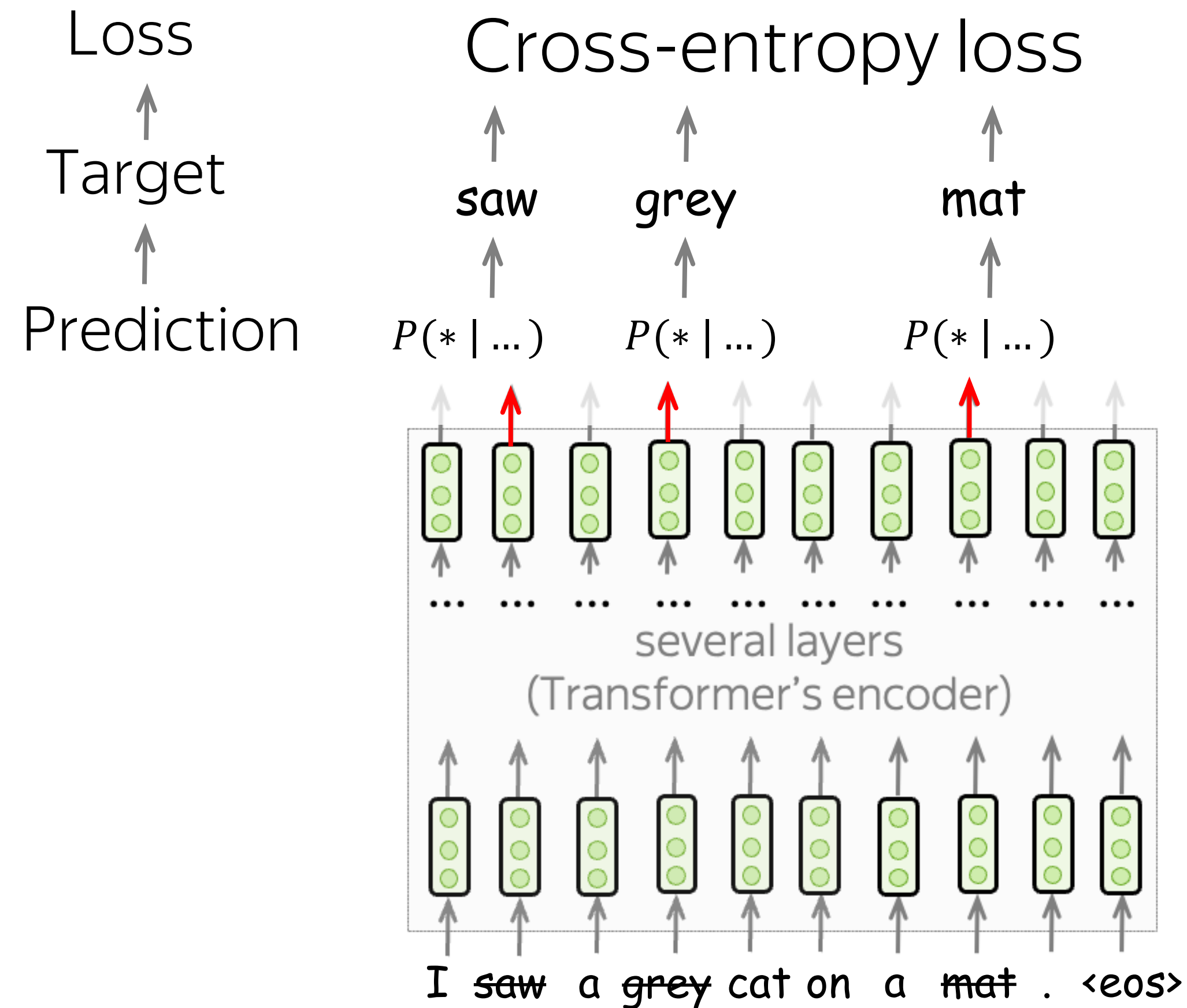
- pick randomly 15% of tokens
- replace each of the chosen tokens with something



- **[MASK]**,
with $p = 80\%$
- Random token,
with $p = 10\%$
- Original token,
with $p = 10\%$



BERT: Masked Language Modeling Objective



At each training step:

- pick randomly 15% of tokens
- replace each of the chosen tokens with something
- predict original chosen tokens

- **[MASK]**, with $p = 80\%$
- Random token, with $p = 10\%$
- Original token, with $p = 10\%$

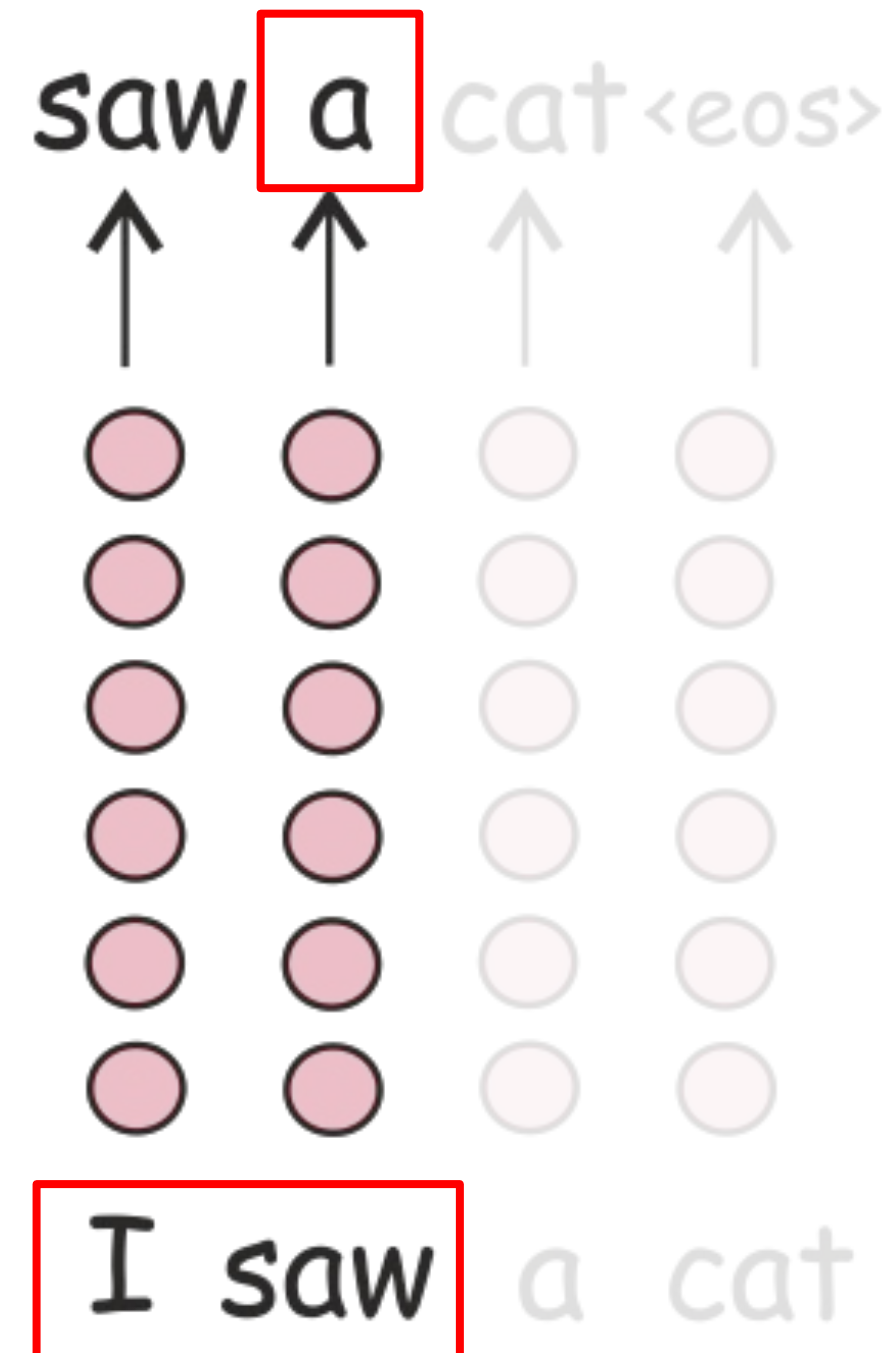
Training Objectives: LM vs MLM

Language Modeling

Target: next token

Prediction: $P(* | \text{I saw})$

left-to-right, does
not see future



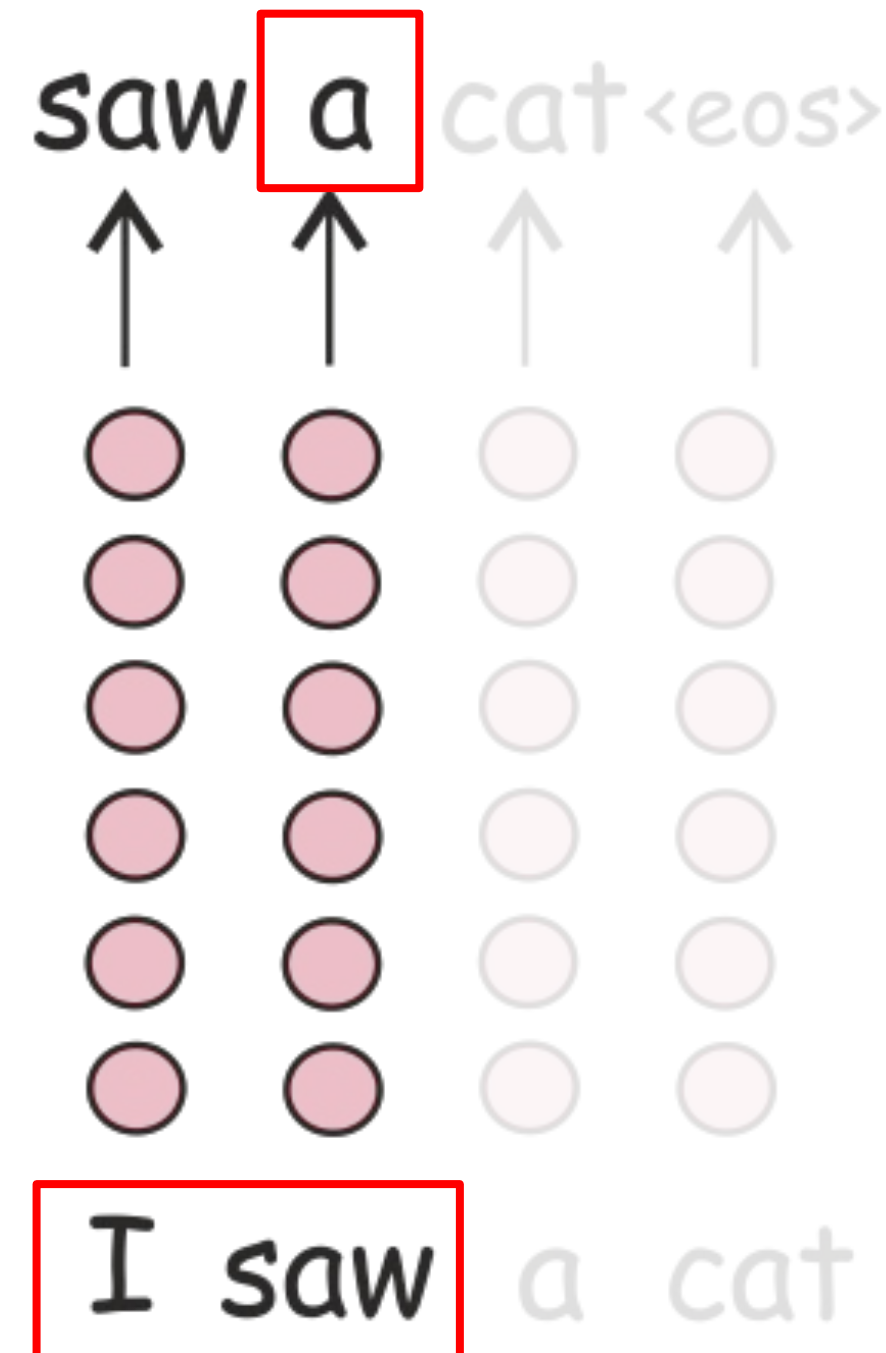
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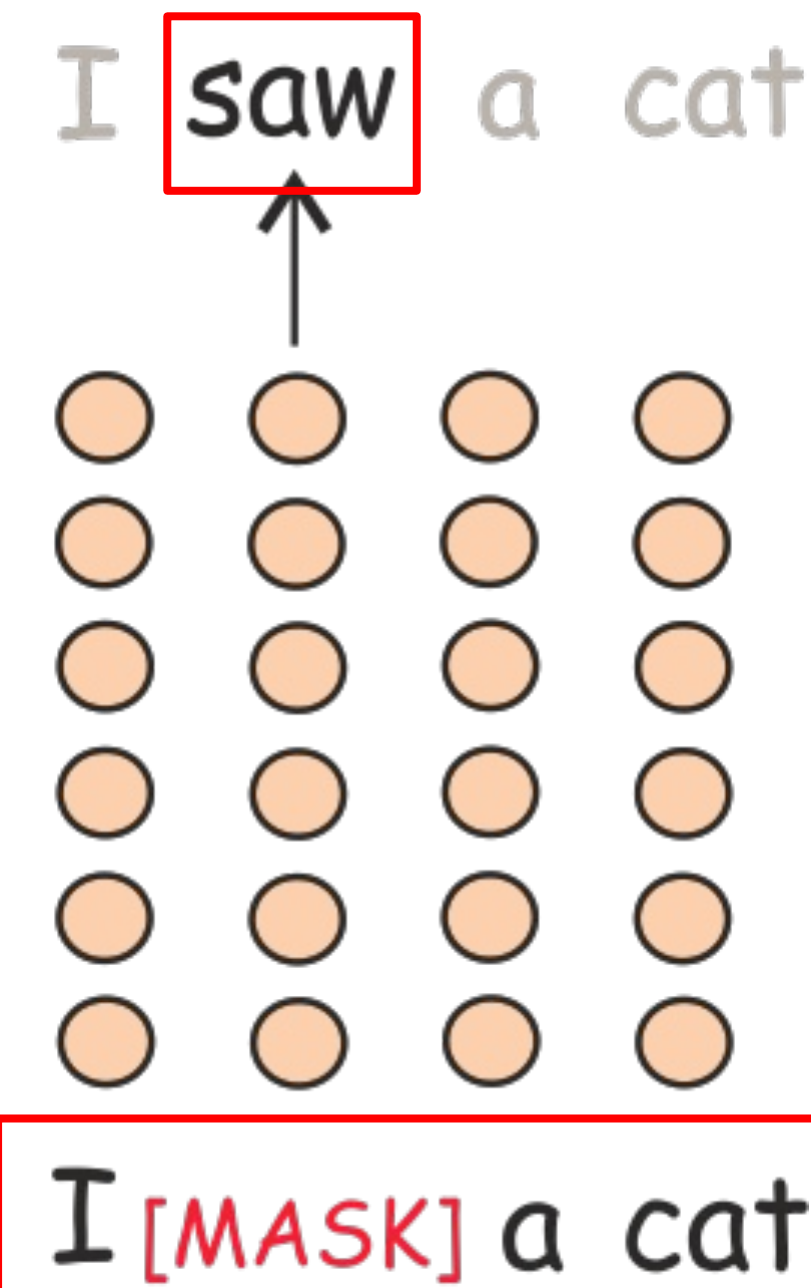
left-to-right, does
not see future



Masked Language Modeling

Target: current token (the true one)

Prediction: $P(* | \text{I [MASK] a cat})$

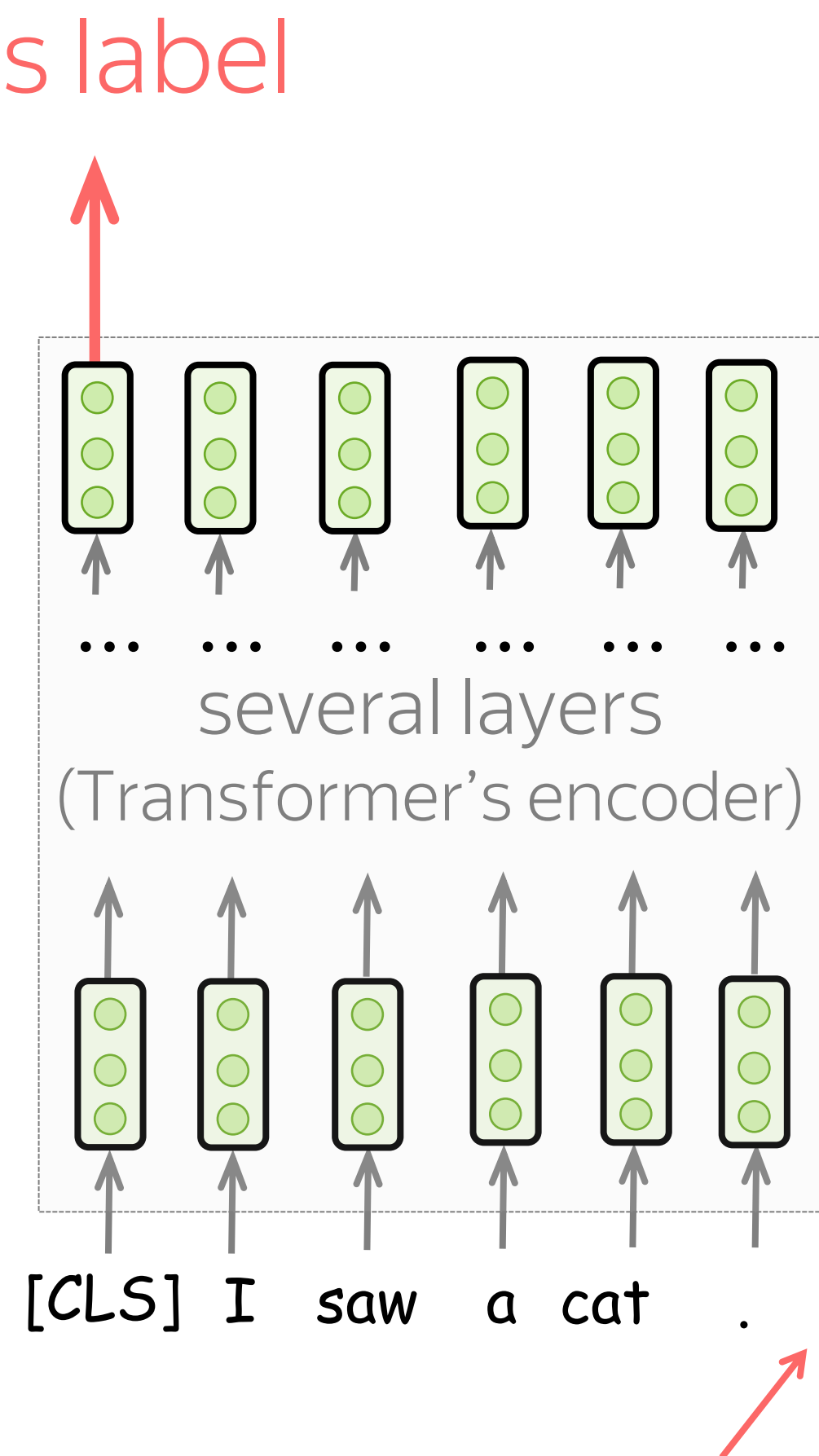


sees the whole text, but
something is corrupted

Finetuning BERT: Single-Sentence Classification

Examples of tasks:

- SST-2 – binary sentiment classification (we saw it in the text classification lecture)
- CoLA (Corpus of Linguistic Acceptability) – say whether a sentence is linguistically acceptable



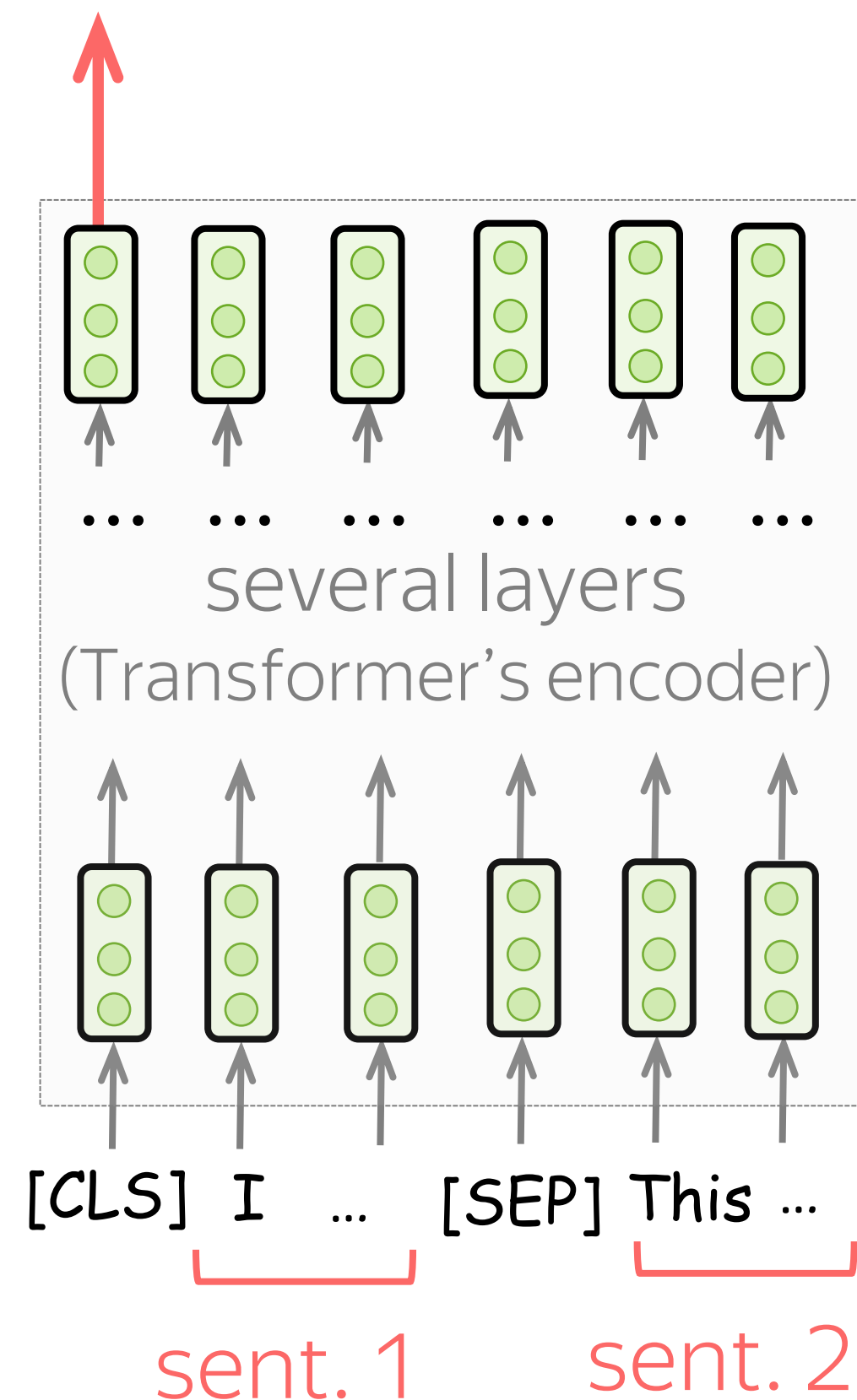
No second sentence!

Finetuning BERT: Sentence Pair Classification

Examples of tasks:

- MLNI – entailment classification. Given a pair of sentences, say if the second is an **entailment**, **contradiction** or **neutral**
- QQP (Quora Question Pairs) – given two questions say if they are semantically equivalent
- STS-B – given two sentences return a similarity score from 1 to 5

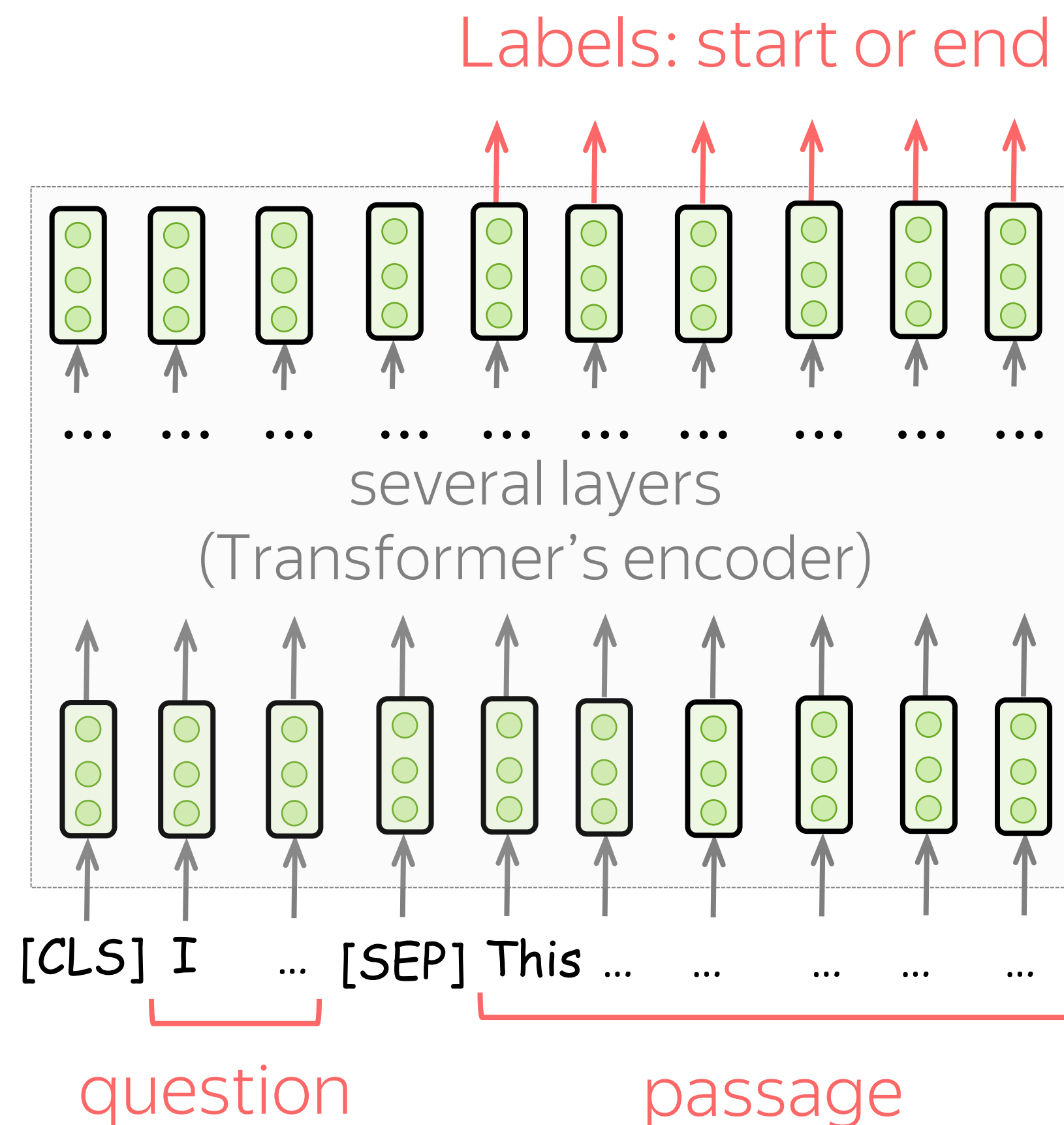
class label



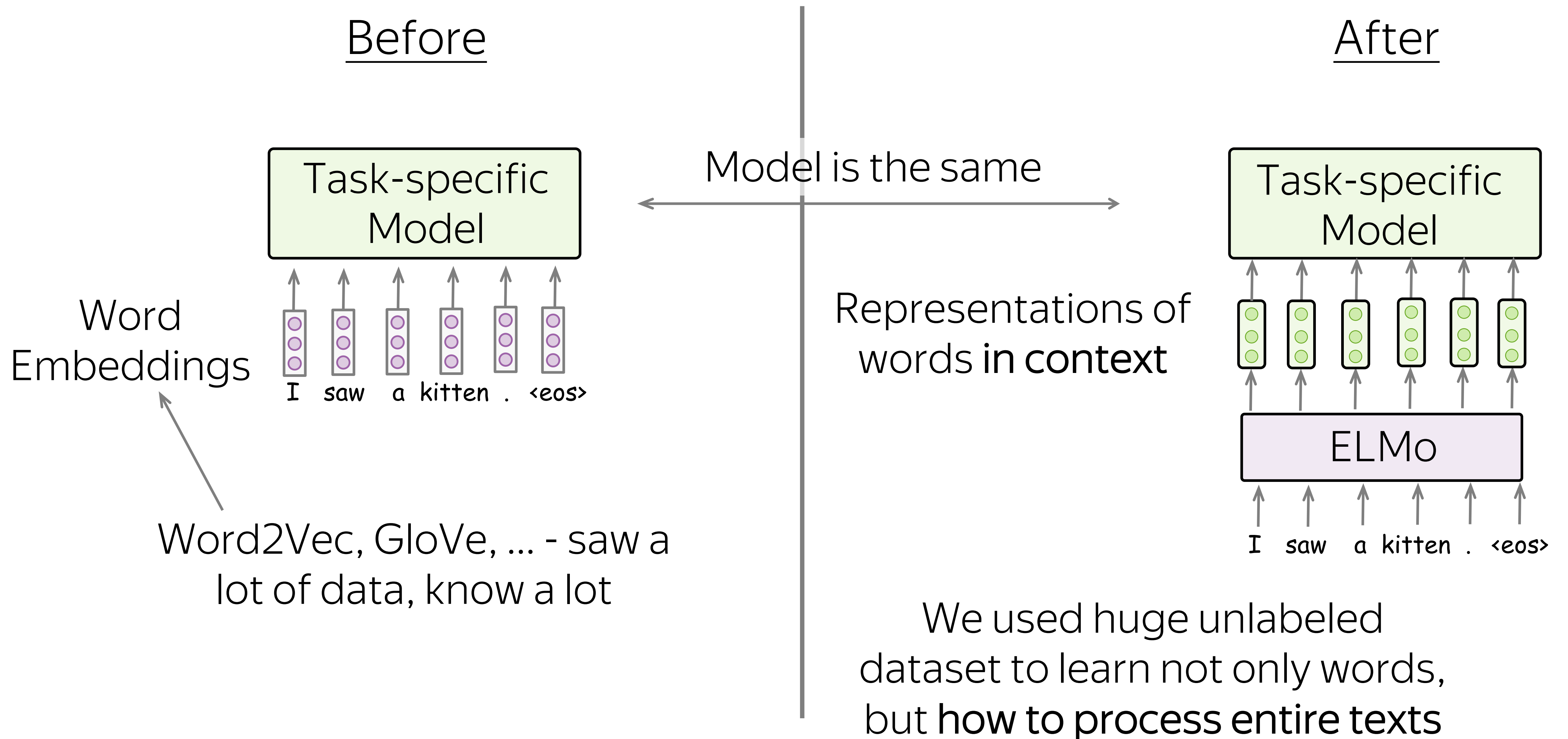
Finetuning BERT: Question Answering

Examples of tasks:

- SQUAD – dataset with pairs of question-passage; the passage contains the answer – need to indicate where



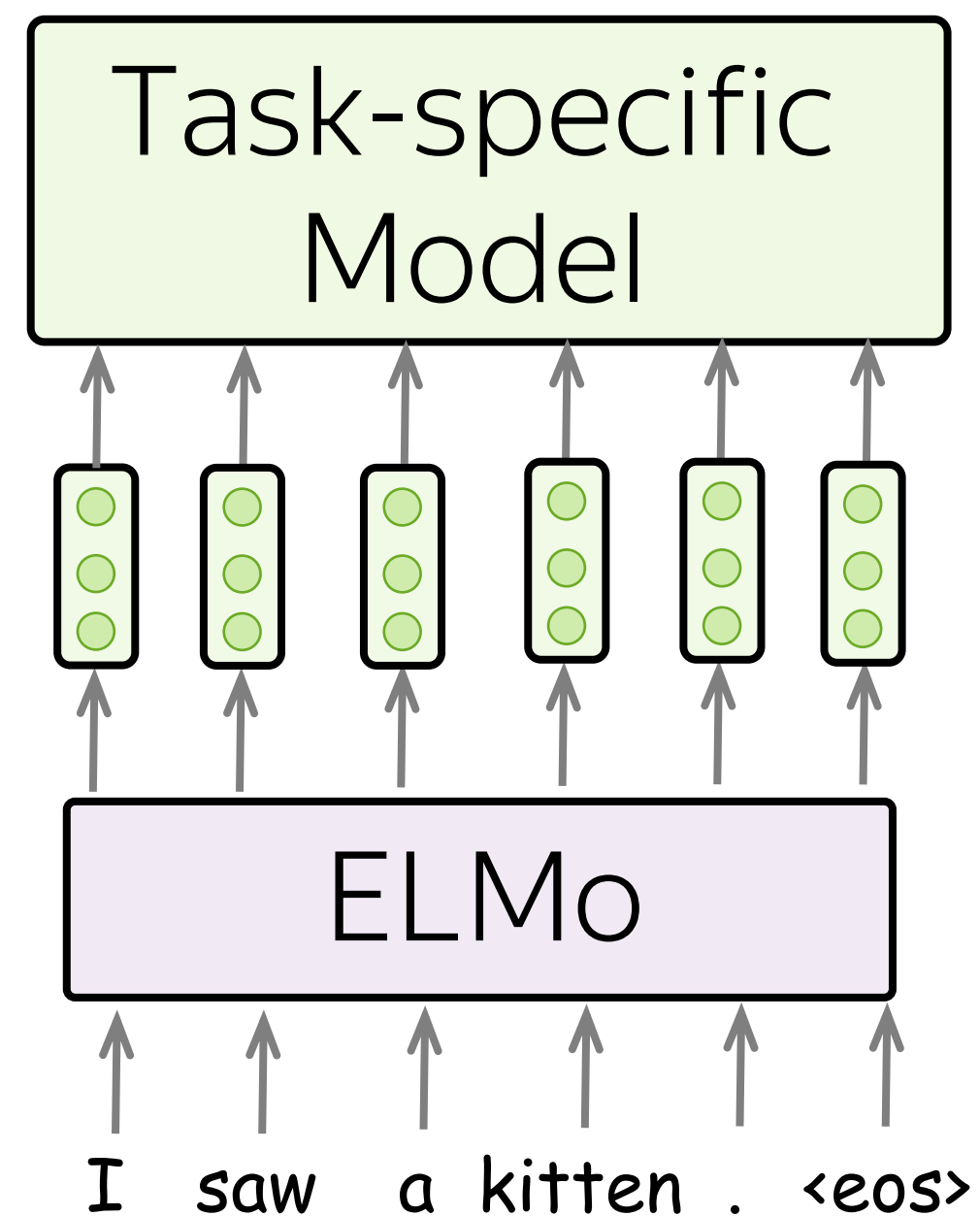
ELMo: What's changed?



BERT: What's changed?

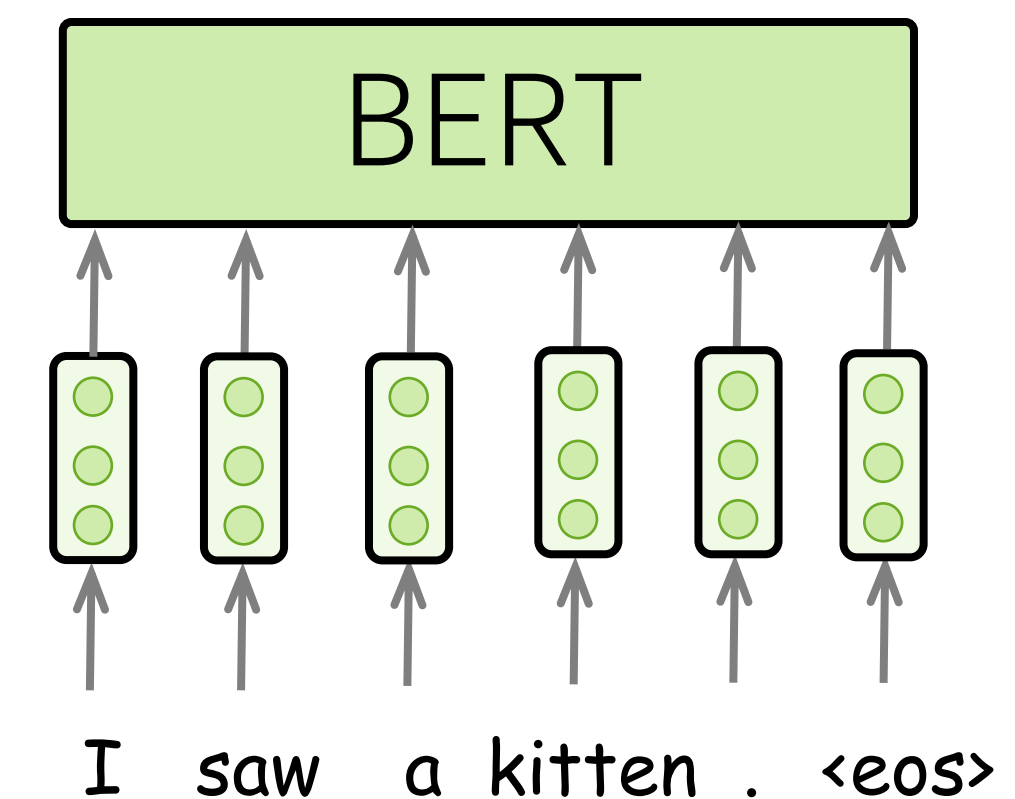
Before

Representations of
words in context



After

No task-specific model at all!



What is going to happen:

- Transfer Learning Idea
 - (recap) Word Embeddings
- Pretrained Models
 - Great Idea 1: Words $\mapsto$ Words in Context
 - Great Idea 2: Task-Specific $\mapsto$ Unified
-  Analysis and Interpretability



- GPT
- BERT

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Analysis Methods

The methods we used previously:

- (model-specific) looking at model components

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- Heads in Multi-Head Attention (BERT)
 - FFNs as Key-Value Memories

Analysis Methods

The methods we used previously:

- (model-specific) looking at model components
 - (model-agnostic) probing for linguistic structure
- 
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Analysis Methods

The methods we used previously:

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| | | BERT and the classical NLP pipeline |

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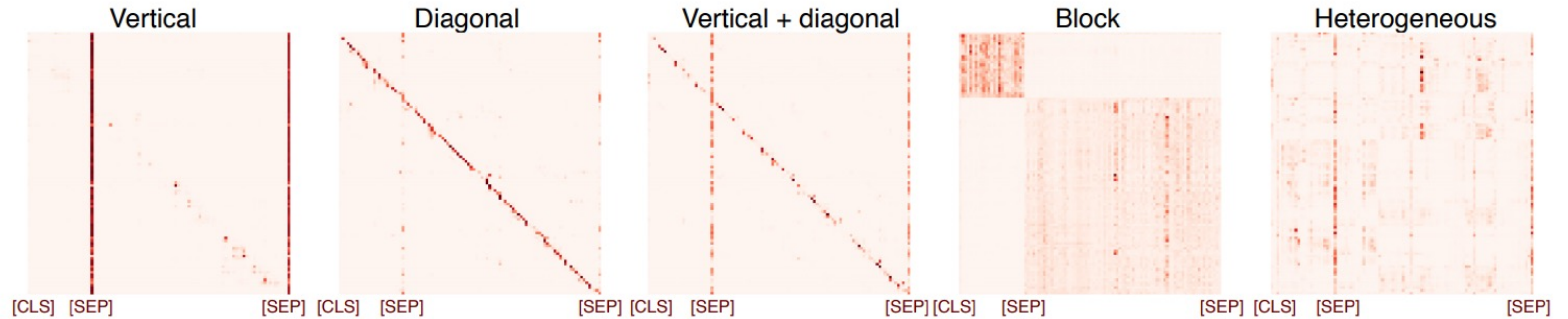
What we will see in this lecture:

- Heads in Multi-Head Attention (BERT)
 - FFNs as Key-Value Memories
- BERT and the classical NLP pipeline
- BERT as knowledge base

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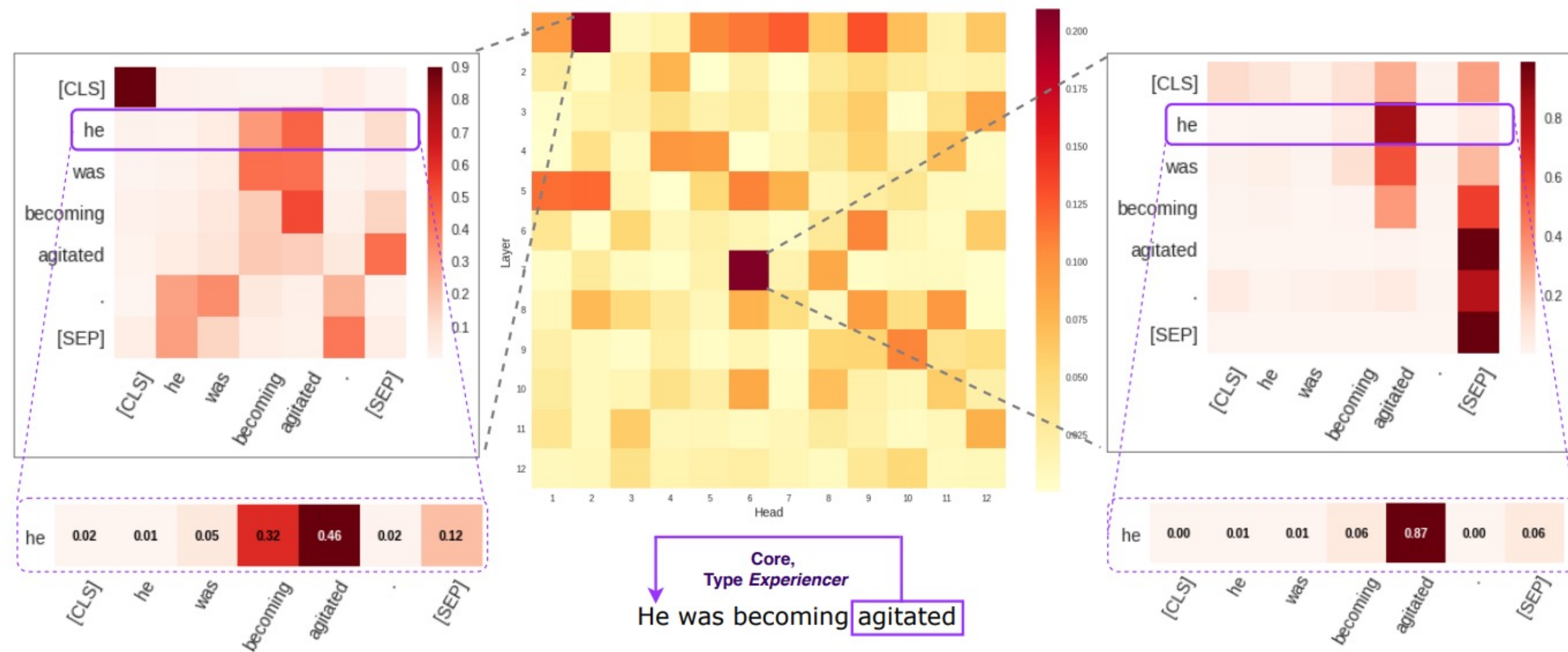
BERT Self-Attention Heads



Typical self-attention patterns

The figure is from the paper [Revealing the Dark Secrets of BERT](#)

BERT Self-Attention Heads



Heads that encode information correlated to semantic links in the input text

The figure is from the paper Revealing the Dark Secrets of BERT

Analysis Methods

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Analysis Methods

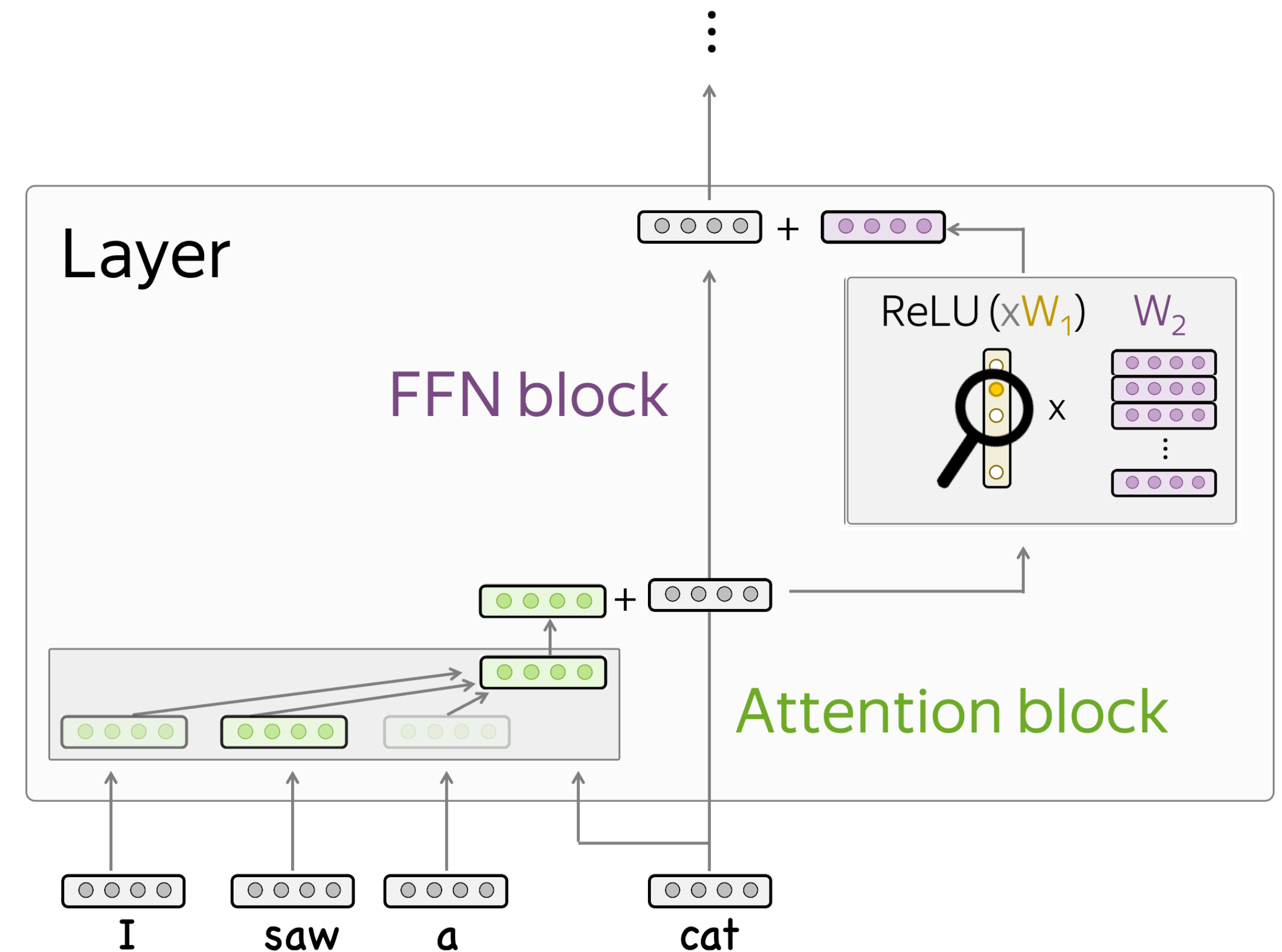
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Transformer Feed-Forward Layers are Key-Value Memories

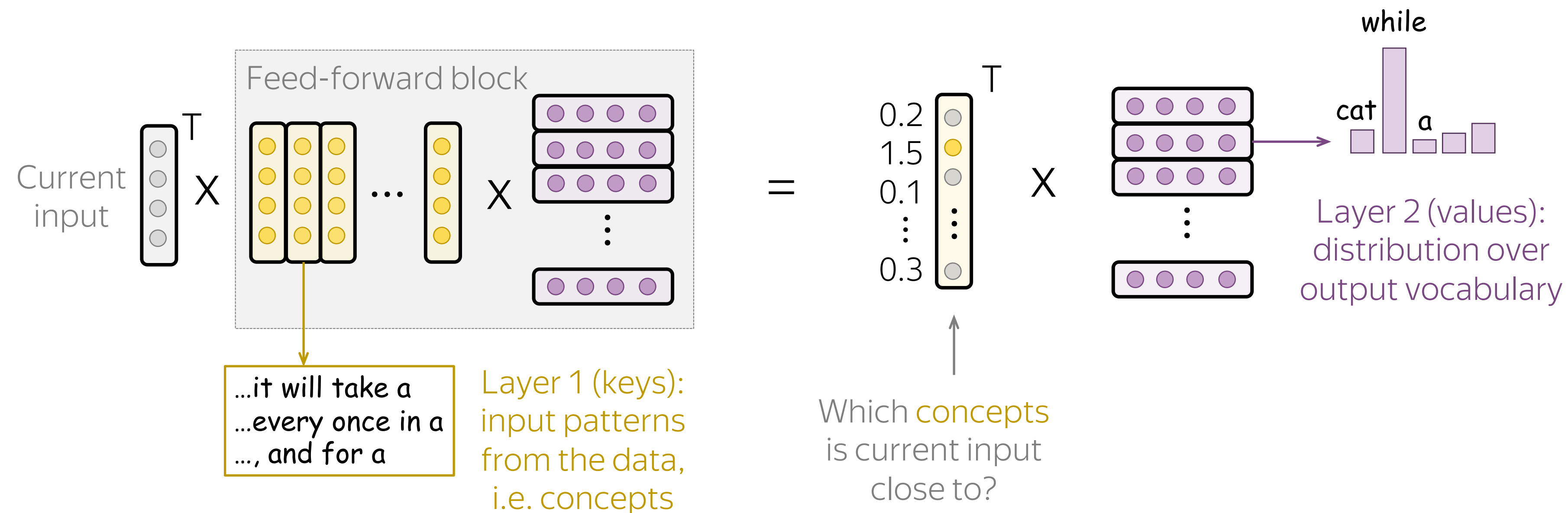
Transformer: Attn -> FFN -> Attn -> FFN -> ...

- Token representation evolves from input token embedding to the final prediction
- Attention and FFN update this representation
- Stack of representations for the same token is called **residual stream**



Transformer Feed-Forward Layers are Key-Value Memories

Transformer: Attn -> FFN -> Attn -> FFN -> ...



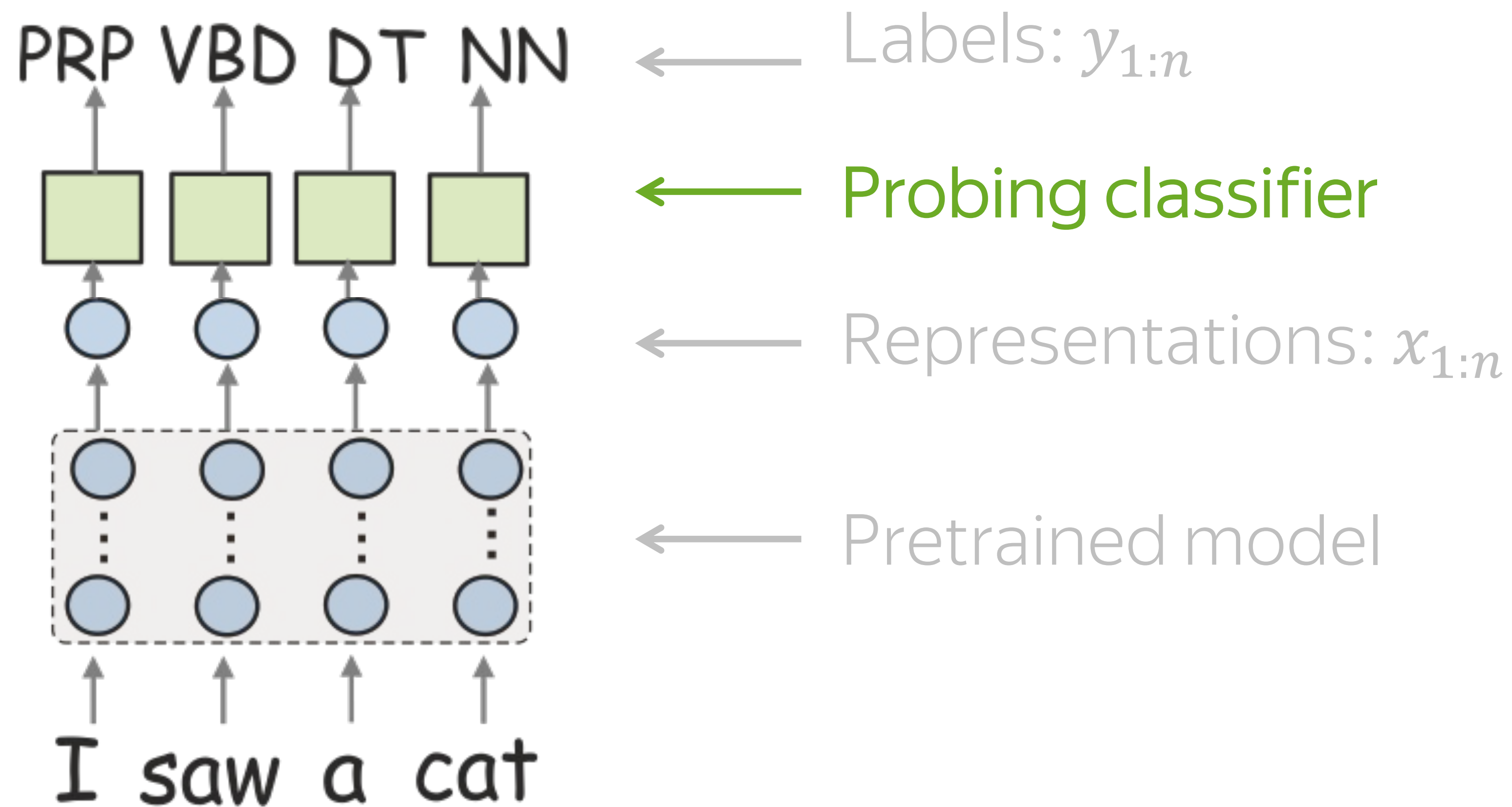
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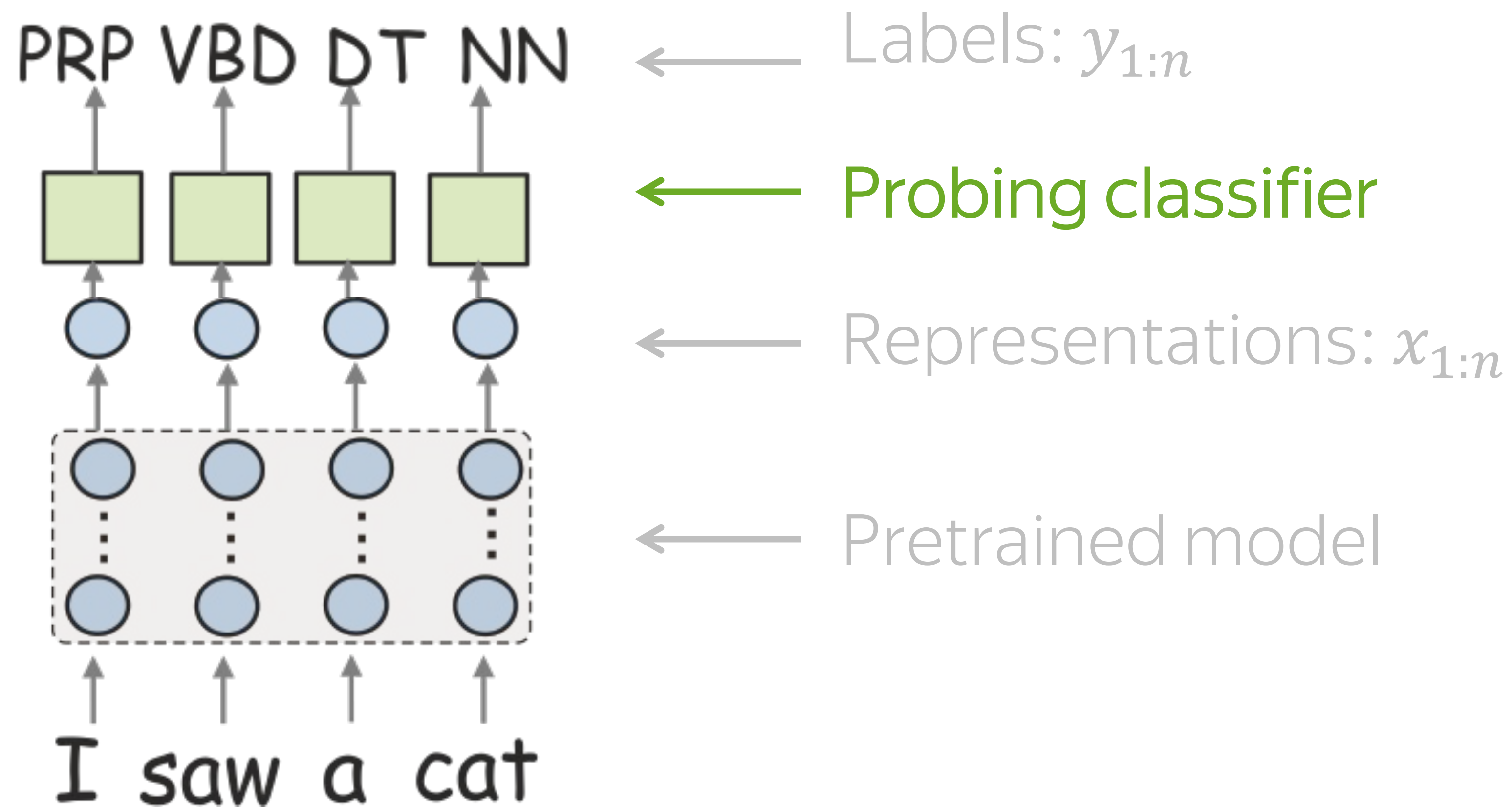
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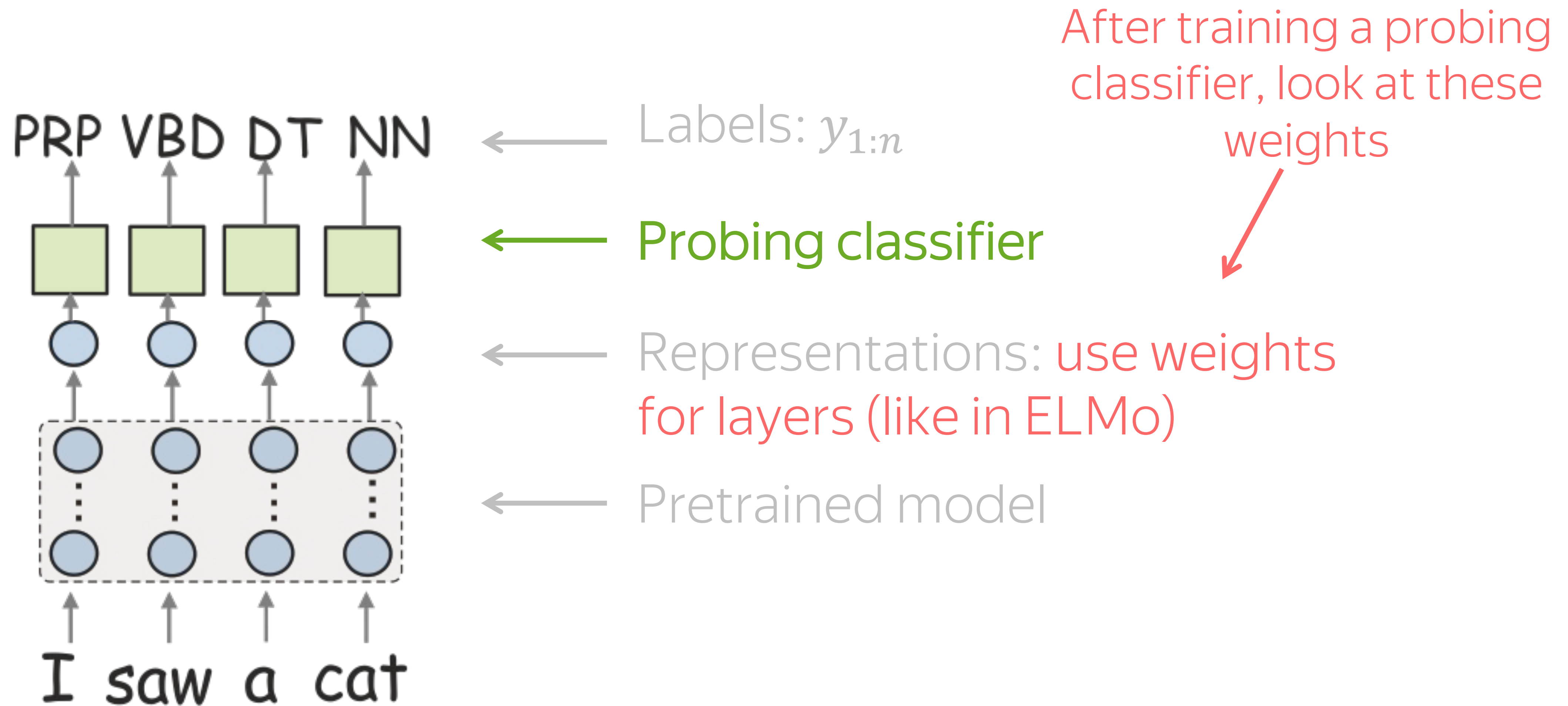
RECAP: Probing for linguistic structure



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BERT Rediscovered the Classical NLP Pipeline

Part of speech: I want to find more , [something] bigger or deeper . → NN (Noun)

The figure is from the paper [BERT Rediscovered the Classical NLP Pipeline](#)

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Semantic Role Labeling: I want to [find]₁ [more , something bigger or deeper]₂ . → Agr1 (Agent)

Coreference: So [the followers]₁ waited to say anything about what [they]₂ saw . → True

The figure is from the paper [BERT Rediscovered the Classical NLP Pipeline](#)

BERT Rediscovered the Classical NLP Pipeline

Part of speech:

Constituents:

Dependencies:

Entities:

Semantic Role
Labeling:

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In classical NLP, to
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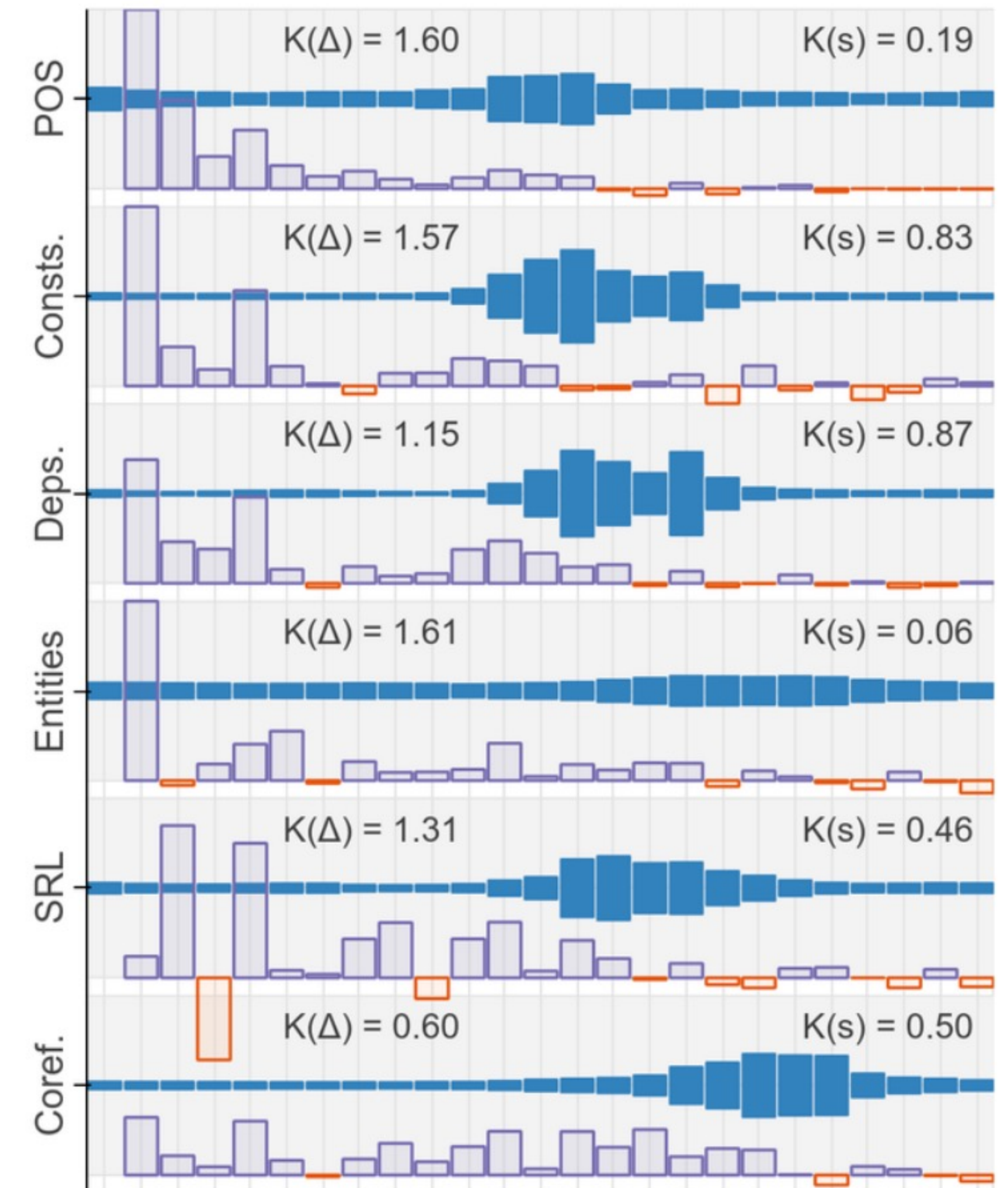
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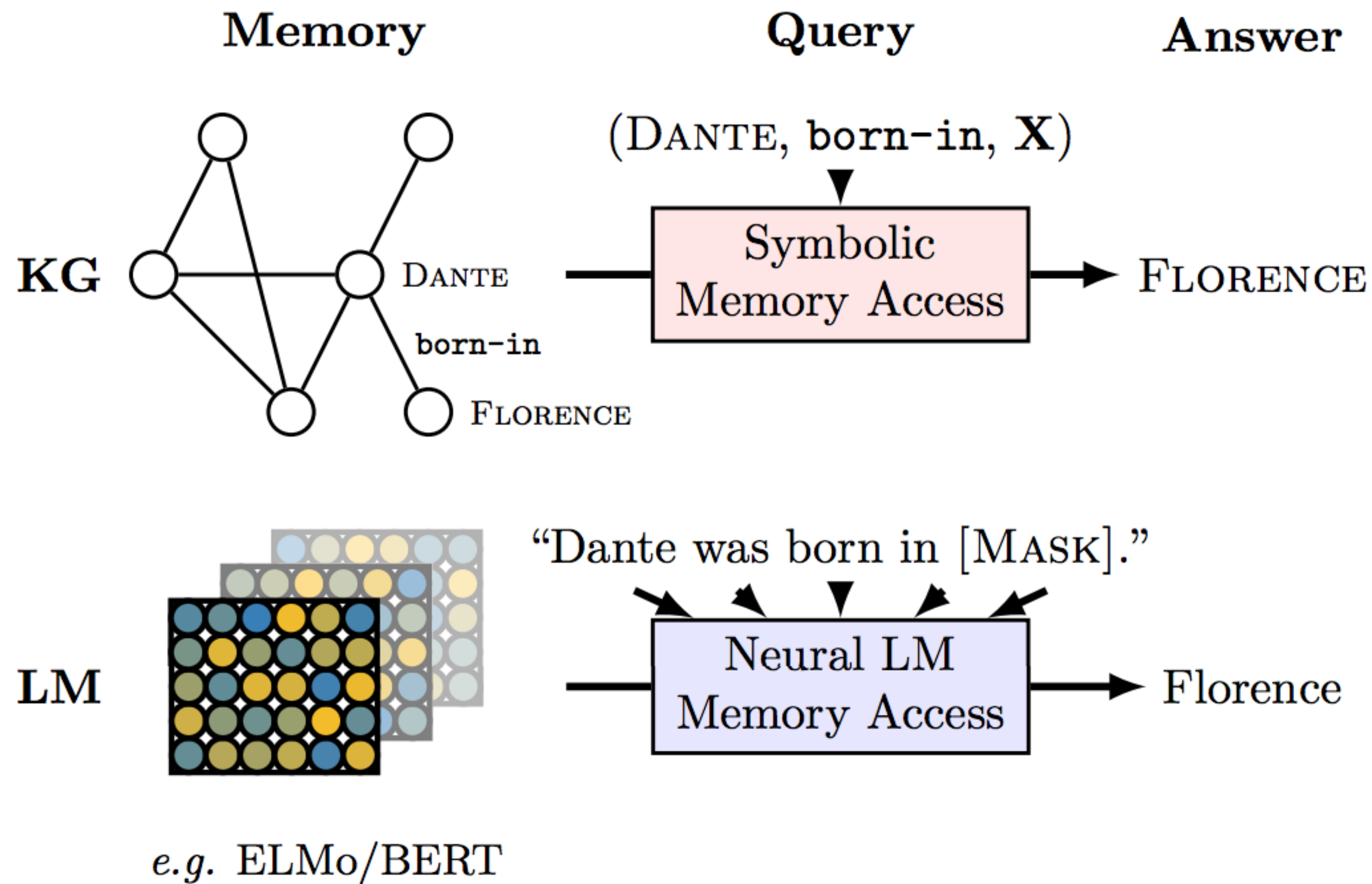
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Language Models as Knowledge Bases?



The figure is from the paper Language Models as Knowledge Bases?

Language Models as Knowledge Bases?

	Relation	Query	Answer	Generation
T-Rex	P19	Francesco Bartolomeo Conti was born in ____.	Florence	Rome [-1.8] , Florence [-1.8] , Naples [-1.9] , Milan [-2.4] , Bologna [-2.5]
	P20	Adolphe Adam died in ____.	Paris	Paris [-0.5] , London [-3.5] , Vienna [-3.6] , Berlin [-3.8] , Brussels [-4.0]
	P279	English bulldog is a subclass of ____.	dog	dogs [-0.3] , breeds [-2.2] , dog [-2.4] , cattle [-4.3] , sheep [-4.5]
	P37	The official language of Mauritius is ____.	English	English [-0.6] , French [-0.9] , Arabic [-6.2] , Tamil [-6.7] , Malayalam [-7.0]
	P413	Patrick Oboya plays in ____ position.	midfielder	centre [-2.0] , center [-2.2] , midfielder [-2.4] , forward [-2.4] , midfield [-2.7]
	P138	Hamburg Airport is named after ____.	Hamburg	Hess [-7.0] , Hermann [-7.1] , Schmidt [-7.1] , Hamburg [-7.5] , Ludwig [-7.5]
	P364	The original language of Mon oncle Benjamin is ____.	French	French [-0.2] , Breton [-3.3] , English [-3.8] , Dutch [-4.2] , German [-4.9]
	P54	Dani Alves plays with ____ .	Barcelona	Santos [-2.4] , Porto [-2.5] , Sporting [-3.1] , Brazil [-3.3] , Portugal [-3.7]
	P106	Paul Toungui is a ____ by profession .	politician	lawyer [-1.1] , journalist [-2.4] , teacher [-2.7] , doctor [-3.0] , physician [-3.7]
	P527	Sodium sulfide consists of ____.	sodium	water [-1.2] , sulfur [-1.7] , sodium [-2.5] , zinc [-2.8] , salt [-2.9]
	P102	Gordon Scholes is a member of the ____ political party.	Labor	Labour [-1.3] , Conservative [-1.6] , Green [-2.4] , Liberal [-2.9] , Labor [-2.9]
	P530	Kenya maintains diplomatic relations with ____.	Uganda	India [-3.0] , Uganda [-3.2] , Tanzania [-3.5] , China [-3.6] , Pakistan [-3.6]
	P176	iPod Touch is produced by ____.	Apple	Apple [-1.6] , Nokia [-1.7] , Sony [-2.0] , Samsung [-2.6] , Intel [-3.1]
	P30	Bailey Peninsula is located in ____.	Antarctica	Antarctica [-1.4] , Bermuda [-2.2] , Newfoundland [-2.5] , Alaska [-2.7] , Canada [-3.1]
	P178	JDK is developed by ____.	Oracle	IBM [-2.0] , Intel [-2.3] , Microsoft [-2.5] , HP [-3.4] , Nokia [-3.5]
	P1412	Carl III used to communicate in ____.	Swedish	German [-1.6] , Latin [-1.9] , French [-2.4] , English [-3.0] , Spanish [-3.0]
	P17	Sunshine Coast, British Columbia is located in ____.	Canada	Canada [-1.2] , Alberta [-2.8] , Yukon [-2.9] , Labrador [-3.4] , Victoria [-3.4]
	P39	Pope Clement VII has the position of ____ .	pope	cardinal [-2.4] , Pope [-2.5] , pope [-2.6] , President [-3.1] , Chancellor [-3.2]
	P264	Joe Cocker is represented by music label ____.	Capitol	EMI [-2.6] , BMG [-2.6] , Universal [-2.8] , Capitol [-3.2] , Columbia [-3.3]
	P276	London Jazz Festival is located in ____.	London	London [-0.3] , Greenwich [-3.2] , Chelsea [-4.0] , Camden [-4.6] , Stratford [-4.8]
	P127	Border TV is owned by ____.	ITV	Sky [-3.1] , ITV [-3.3] , Global [-3.4] , Frontier [-4.1] , Disney [-4.3]
	P103	The native language of Mammootty is ____.	Malayalam	Malayalam [-0.2] , Tamil [-2.1] , Telugu [-4.8] , English [-5.2] , Hindi [-5.6]
	P495	The Sharon Cuneta Show was created in ____.	Philippines	Manila [-3.2] , Philippines [-3.6] , February [-3.7] , December [-3.8] , Argentina [-4.0]

The figure is from the paper Language Models as Knowledge Bases?

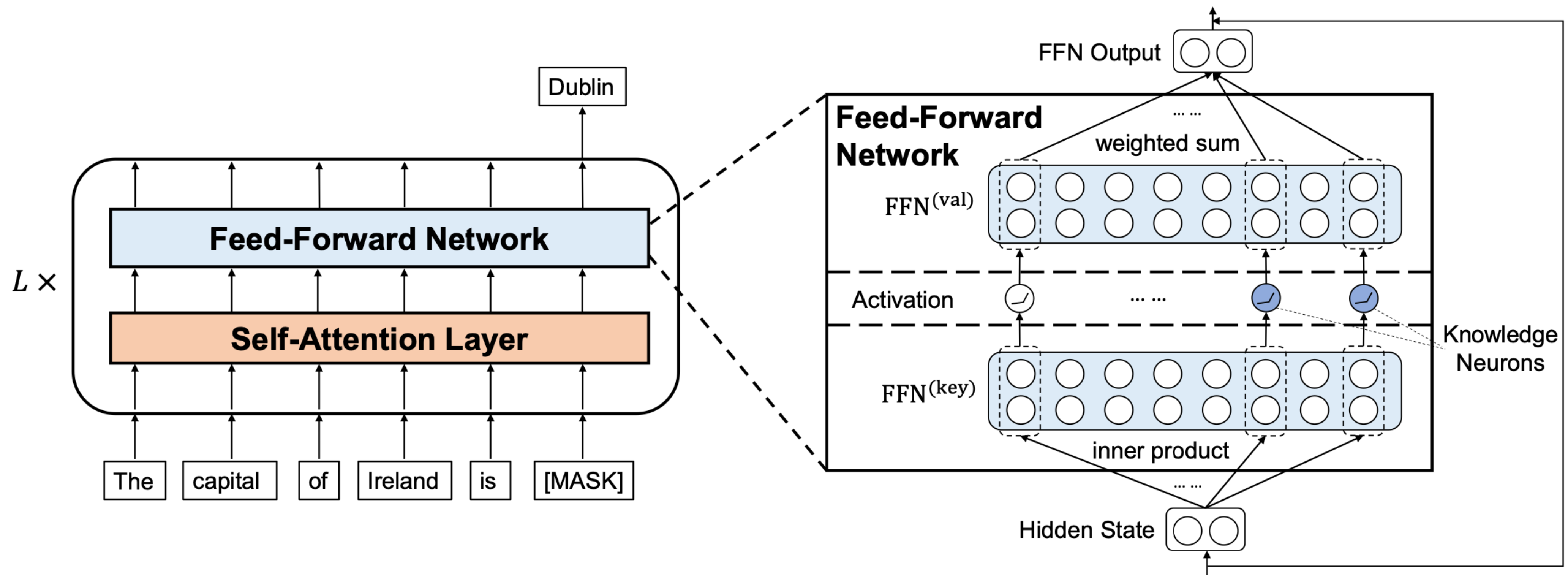
Language Models as Knowledge Bases?

ConceptNet	AtLocation	You are likely to find a overflow in a ____.	drain	sewer [-3.1] , canal [-3.2] , toilet [-3.3] , stream [-3.6] , drain [-3.6]
	CapableOf	Ravens can ____.	fly	fly [-1.5] , fight [-1.8] , kill [-2.2] , die [-3.2] , hunt [-3.4]
	CausesDesire	Joke would make you want to ____.	laugh	cry [-1.7] , die [-1.7] , laugh [-2.0] , vomit [-2.6] , scream [-2.6]
	Causes	Sometimes virus causes ____.	infection	disease [-1.2] , cancer [-2.0] , infection [-2.6] , plague [-3.3] , fever [-3.4]
	HasA	Birds have ____.	feathers	wings [-1.8] , nests [-3.1] , feathers [-3.2] , died [-3.7] , eggs [-3.9]
	HasPrerequisite	Typing requires ____.	speed	patience [-3.5] , precision [-3.6] , registration [-3.8] , accuracy [-4.0] , speed [-4.1]
	HasProperty	Time is ____.	finite	short [-1.7] , passing [-1.8] , precious [-2.9] , irrelevant [-3.2] , gone [-4.0]
	MotivatedByGoal	You would celebrate because you are ____.	alive	happy [-2.4] , human [-3.3] , alive [-3.3] , young [-3.6] , free [-3.9]
	ReceivesAction	Skills can be ____.	taught	acquired [-2.5] , useful [-2.5] , learned [-2.8] , combined [-3.9] , varied [-3.9]
	UsedFor	A pond is for ____.	fish	swimming [-1.3] , fishing [-1.4] , bathing [-2.0] , fish [-2.8] , recreation [-3.1]

The figure is from the paper Language Models as Knowledge Bases?

Back to the Key-Value View: Knowledge Neurons

Data: (subject, relation) $\rightarrow$ object triplets



The figure is from the paper Knowledge Neurons in Pretrained Transformers

What is going to happen:

- Transfer Learning Idea
 - (recap) Word Embeddings
- Pretrained Models
 - Great Idea 1: Words $\mapsto$ Words in Context
 - Great Idea 2: Task-Specific $\mapsto$ Unified
-  Analysis and Interpretability

Thank you!

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